CAN THO UNIVERSITY

COLLEGE OF INFORMATION AND COMMUNICATION TECHNOLOGY

GROUP PROJECT

COURSE: INTRODUCTION TO SOFTWARE ENGINEERING

COURSE CLASS: CT114H

VIEBUS -
BUS TRACKING SOFTWARE

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Instructor:

Ph.D. Phan Phuong Lan

Group Members:

Can Tho, 04/2024

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• Revision History

Name	Date	Reason For Changes	Version

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6	AEmail	AEMAIL	varchar(20)	20			

• References

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https://www.figma.com/file/lslAMCWt6DCoopXAponaNj/Mobile-App-Onboarding--SignIn-SignUp-Forgot-Password-(Community)?type=design&t=3mzH0UykaGlPvLUJ-6 (Last accessed: 4/2024)

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● Acronyms and Abbreviations

#	Acronyms and Abbreviations	Explanation

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1. Plan

1.1 Group Organization

#	Student Code	Full Name	Role
1			Team Leader
2			Vice of Team Leader
3			Member
4			Member
5			Member

1.2 Plan

Week	Task	Member (Name)	Deadline	Level of completion
Software Requirement Specification				
6	Product Perspective		1/31	100%
	Product Function		1/31	100%
	User Classes and Characteristics		2/28	100%
	Operating Environment		2/28	100%
	Design and Implementation Constraints		2/28	100%
7	User Interfaces		2/28	100%
	Hardware Interfaces		2/28	100%
	Software Interfaces		2/28	100%
	Communication Interfaces		2/28	100%
8	Nonfunctional Requirement		2/28	100%

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9	Function 1: Real-time Bus Routes Tracking		2/28	100%
	Function 2: Account Registration		2/28	100%
	Function 3: Bus Routes Editing		2/28	100%
	Function 4: Bus Routes Searching		2/28	100%
	Function 5: Online Ticket Booking		2/28	100%
15				
Software Design				
10	**Data Design**			
	Application Architecture Diagram	All	3/13	100%
	Application Architecture Description	All	3/13	100%
	Data Description	All	3/13	100%
	Conceptual Data Model	All	3/13	100%
	Logical Data Model	All	3/13	100%
	Physical Data Model	All	3/13	100%
	Data Dictionary	All	3/13	100%
11	**Detail Design**			
	Function 1: Real-Time Bus Routes Tracking		4/10	
	Function 2: Account Registration		4/10	100%
	Function 3: Bus Routes Editing		4/10	100%

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12	Function 4: Bus Routes Searching		4/10	100%
	Function 5: Online Ticket Booking		4/10	100%
	References to Requirements		4/10	100%
13	Function 1: Bus Routes Tracking		4/24	
	Function 2: Account Registration		4/24	100%
	Function 3: Bus Routes Editing		4/24	100%
14	Function 4: Routes Searching		4/24	100%
	Function 5: Online Ticket Booking		4/24	100%
Preparation (for presentation and report)				
15	Report Slides		5/1	
16	Report Summary		5/1	75%
17	Revision & Summary		5/1	

1.3 Rules on the group

- Contact Channel:

○ Platform: Zalo

○ Group Name: Nhóm Nhập môn công nghệ phần mềm HK2 2023-2024

- Group Meeting:

○ Time: Every Wednesday

○ Venue: In the classroom

- Group data storage:

○ Report document : Google Drive

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File: Report.docx, idea.docx
Folder: Requirements and Template
○ Use case & Flow chart, graph designer: StarUML, Visual Paradigm
○ Interface design: Figma Group Project

1.4 Software Requirement Specification

1.5 Overall Description

1.5.1 Product Perspective

The Bus Tracking System represents a groundbreaking approach to addressing the escalating demand for streamlined public transportation management across the country. As a novel and independent software solution, it stands poised to revolutionise how commuters access real-time information about bus routes. Unlike mere replacements for existing systems, this pioneering platform is meticulously crafted to significantly bolster the accessibility and dependability of bus services. By offering a dedicated package tailored to each province or major city, the system ensures seamless integration nationwide, catering to diverse user needs with unparalleled efficiency.

1.5.2 Product Functions

Main functions of this product include:
1. Real-time Bus Route Tracking: The system provides live tracking of bus routes, allowing drivers and users to monitor bus locations in real-time.
2. Account Registration: Users can create accounts to access personalized features and services.
3. Bus Routes Editing: Authorized personnel can edit and update bus routes as needed to ensure accuracy and efficiency.
4. Bus Routes Searching: Users can search for specific bus routes based on their origin, destination, or other criteria.
5. Online Ticket Booking: The platform enables users to book bus tickets online, streamlining the ticketing process for commuters.

1.5.3 User Classes and Characteristics

- Passengers (Most Important)
+ Characteristics:

- Frequent users (daily or weekly)
- May have varying technical expertise (from novice to tech-savvy)
- Primarily interested in trip planning and real-time information
- Need a user-friendly interface for ease of use on the go

+ Important Requirements:

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- Real-time bus tracking with estimated arrival times
- Route planning and trip visualization
- Fare information and potential purchase options (if applicable)
- Service alerts and notifications for delays or disruptions
- Ability to report issues or provide feedback

- Administrators (Important)

+ Characteristics:

- Less frequent users (daily or occasional)
- Typically have moderate to high technical expertise
- Focused on managing bus operations and data
- Responsible for system configuration and user management

+ Important Requirements:

- Route management tools (adding/editing stops, schedules)
- User management (adding/editing drivers and passengers)
- Real-time bus location monitoring and fleet management
- Performance analytics (on-time rates, ridership data)
- Secure access with role-based permissions

- Drivers (Important)

+ Characteristics:

- Frequent users (during work shifts)
- May have varying technical expertise
- Need a clear and functional interface while driving
- Responsible for passenger safety and adhering to schedules

+ Important Requirements:

- Clear route navigation with turn-by-turn instructions
- Passenger check-in/check-out functionality (optional)
- Two-way communication with administrators for reporting issues
- Ability to view service alerts and updates

1.5.4 Operating Environment

Mobile devices must utilize operating systems starting from iOS 15.0 or later. They must be granted permission to utilize features such as accessing the camera and video, obtaining precise location data even when the app is running in the background, accessing approximate location data only while the app is active on the screen, reading phone identification, accessing content from memory, controlling vibration, overlaying on other apps, having unrestricted network access, viewing network connections, preventing the device from sleeping, utilizing the install

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referrer API, viewing Wi-Fi connections, recognizing physical activity, displaying notifications, and receiving data from the internet.

1.5.5 Design and Implementation Constraints

- Regulatory Policies:

+ Compliance with local transportation regulations and data privacy laws, such as General Data Protection Regulation (GDPR), California Consumer Privacy Act (CCPA), or equivalent regional regulations.

+ Adherence to accessibility standards to ensure the system is usable by individuals with disabilitie, including those with visual, auditory, motor, or cognitive impairments. This includes compliance with accessibility guidelines such as the Web Content Accessibility Guidelines (WCAG) established by the World Wide Web Consortium (W3C).

- Hardware Limitations:

+ Compatibility with mobile devices running iOS 15.0 or later.

+ Consideration of hardware capabilities such as GPS for accurate location tracking and internet connectivity for real-time updates.

- Interfaces to Other Applications:

+ Integration with existing transportation management systems and databases for data exchange and synchronization.

+ APIs for payment gateways for online ticket booking functionality (e.g., Momo & ZaloPay)

- Specific Technologies, Tools, and Databases:

+ Use of specific programming languages (e.g., Swift and Objective-C for iOS development).

+ Utilization of framework and technologies (e.g., HTML for structural layout of user interfaces, ReactJS for front-end development and Flutter for cross-platform app development).

+ Utilization of Oracle SQL databases for storing user information, bus routes data, and transaction records.

- Communications Protocols:

+ The system ensures data security between the mobile app, GPS devices and server by using HTTPS.

+ If using a separate database, MQTT will be used to secure communication between the server and database, ideal for large-scale data management.

+ The system needs to support real-time data transfer from the buses to the central server, as the app users require up-to-date information on bus locations and arrival times.

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+ The data transferred should be compact and efficient to minimize bandwidth usage and ensure responsive performance.
+ The communication protocol should be able to handle disconnections and reconnections seamlessly, ensuring that data is not lost and the app users can still access the bus tracking information.
+ The communication devices installed on the buses should have low power consumption, as they need to operate continuously while the buses are in service.
+ WebSocket will be used as a system communication protocol.

- Design conventions or programming standards:

+ Ensure the app is responsive and optimized for various iOS device sizes and screen resolutions.
+ Follow the Human Interface Guidelines (HIG) established by Apple for iOS apps.
+ Utilize the Model-View-Controller (MVC) architectural pattern, which is the recommended approach for iOS app development.
+ Separate the app's concerns into distinct layers (Model, View, Controller) to promote maintainability and testability.
+ Apply the Dependency Injection principle to decouple components and enable easier testing and modification
+ Follow the Swift Language Guide and the Apple Developer documentation for coding best practices.
+ Use consistent naming conventions for classes, functions, variables, and other entities (e.g., camelCase for methods and properties, PascalCase for types).
+ Write clear and concise comments to document the purpose and behavior of the codebase.

- Parallel operations:

+ Use Git as the version control system and establish a branching strategy like Git Flow or Trunk-Based Development.
+ Integrate code coverage tools to monitor the percentage of the codebase covered by tests and ensure high test coverage.
+ Perform regular manual testing on various iOS devices and iOS versions to catch any platform-specific issues.
+ Ensure the app's code is well-structured and documented, making it easier for the team to understand and extend the functionality in the future.
+ The bus tracking system needs to be able to track and report the real-time location of multiple buses at once to provide a seamless user experience.

- Audit function:

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+ The system needs to log any significant events, such as bus arrivals/departures, route changes, or any anomalies or incidents that occur during operation. This data can be used for analysis and to investigate any issues that arise.

- Safety and security considerations

+ The app and associated systems must be designed with robust cybersecurity measures to protect against hacking, data breaches, and other malicious attacks that could compromise the safety and security of user data and other privacy data. This includes implementing strong access controls, encryption, and network security protocols to safeguard the sensitive data and critical infrastructure of the bus tracking system.

+ The application and system need to implement data anonymization, access controls, and other privacy-preserving measures.

- The criticality of bus applications tracking:

+ The bus tracking app provides real-time information on bus locations, arrival times, and schedule updates, which is essential for passengers to plan their journeys effectively and reduce waiting times.

The data collected by the bus tracking app can be used for comprehensive analysis and reporting, enabling transportation authorities to make data-driven decisions to improve service, optimize routes, and allocate resources more effectively.

1.6 Specific Requirements

1.6.1 External Interface Requirements

1.6.1.1 User Interfaces

- Home screen:

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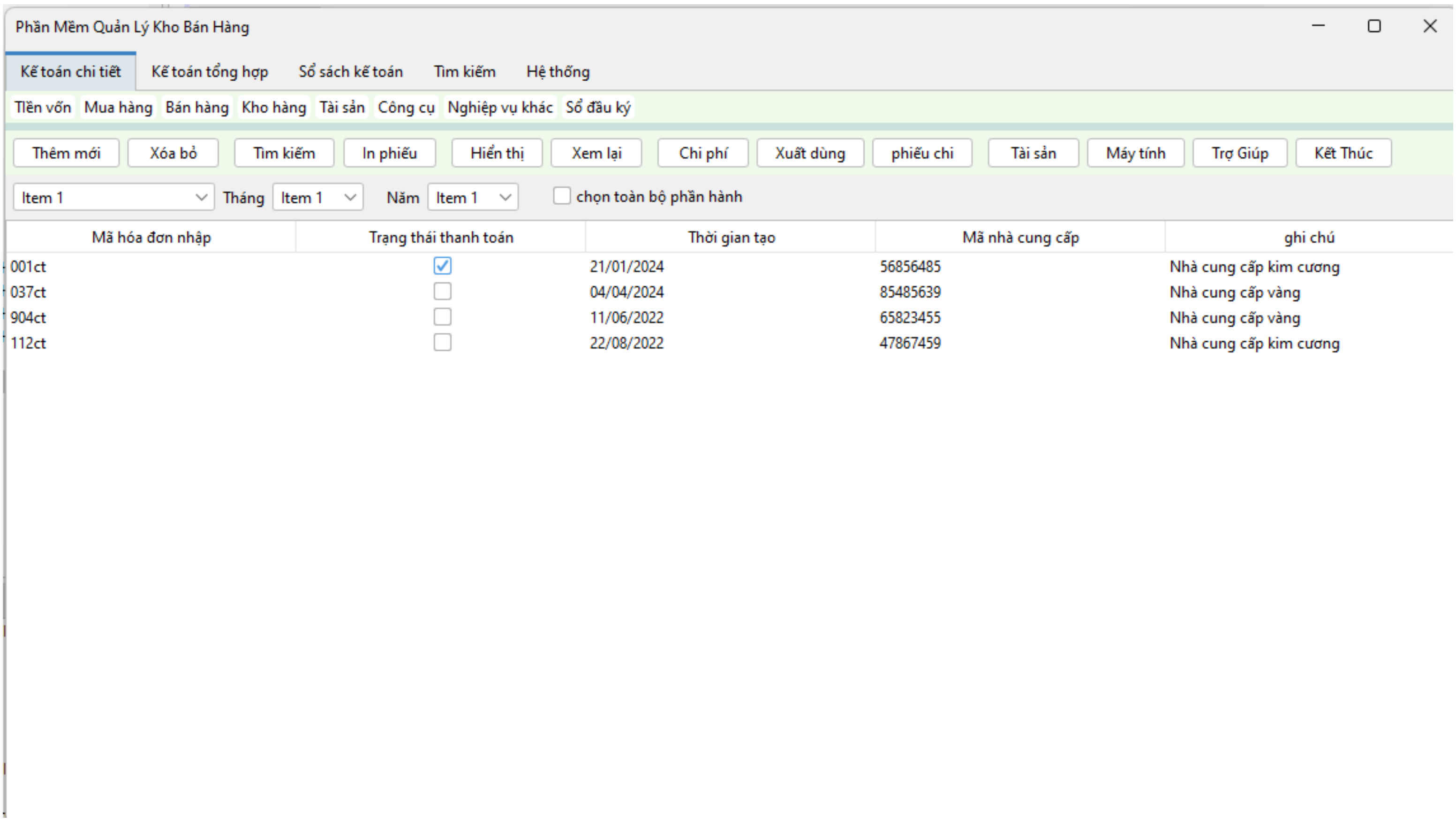

1.6.2 Hardware Interfaces

The bus tracking app will require the following hardware interfaces:

1. Mobile Application: This application is tailored for iOS smartphones. Users must download the app onto their smartphones and sign in to access its features.
2. GPS Integration: Utilizing GPS technology, the app tracks bus locations in real-time. Integrated within the smartphone, the app can access location data seamlessly.
3. Internet Connection: To access real-time location data and other information, a stable internet connection is required for effective app usage.
4. Bus Tracking Devices: Bus drivers are equipped with tracking devices installed in their buses. These devices transmit location data to the app, facilitating real-time updates.

1.6.3 Software Interaces:

The software interface for the bus tracking application includes the following components:

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+ User Interface: The application's user interface should be designed to be simple and user-friendly. It should include features such as login using Gmail or Facebook accounts to create user profiles, bus selection, and real-time tracking of bus locations on the map.
+ Map Interface: The application should integrate with a mapping API like Google Maps to provide an interactive map interface for users. The map should display accurate bus routes, bus stops, and real-time bus locations via GPS.
+ Database Interface: The application should interact with a database system such as Oracle Database to store user account information and retrieve data related to users, bus routes, bus locations, schedules, and ticket statuses. The database should be reliable and scalable to handle a large number of users and buses.
+ API Interface: The application should provide an API interface to allow third-party developers to integrate their applications with the bus tracking app. The API should be well-documented and secure to prevent unauthorized access.
+ Payment Gateway Interface: The application should also integrate with a payment gateway like Zalopay or Momo to enable users to pay bus fares online. The payment gateway should be secure and reliable to protect user data and transactions.
+ Notification Interface: The application should provide a notification interface to send push notifications to users about bus delays, route changes, or other important updates.

1.6.3.1Communications Interfaces

- The app should be able to send email notifications to users regarding their bus arrival and departure times, service updates, and other relevant information.
- The email format should follow a consistent template, including the app's logo, branding, and relevant details about the user's bus trip.
- Email delivery should be reliable and secure, utilizing standard email protocols by using IMAP (Internet Message Access Protocol).
- Users should have the option to customize their email notification preferences, such as the frequency and types of updates they receive.

*Integrate with backend system:

The mobile application will integrate directly with the system's backend server through APIs.

The APIs will be built on REST or GraphQL standards, allowing mobile applications to retrieve and exchange data about schedules, bus locations, passenger information, and more.

Ensure security and privacy through appropriate authentication and authorization mechanisms.

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Integration with other systems:

- The application can integrate with other systems such as GPS systems, driver management systems, etc. to collect and share information.
- This integration will be done through APIs, ensuring the openness and scalability of the application.

*Electronic Forms:

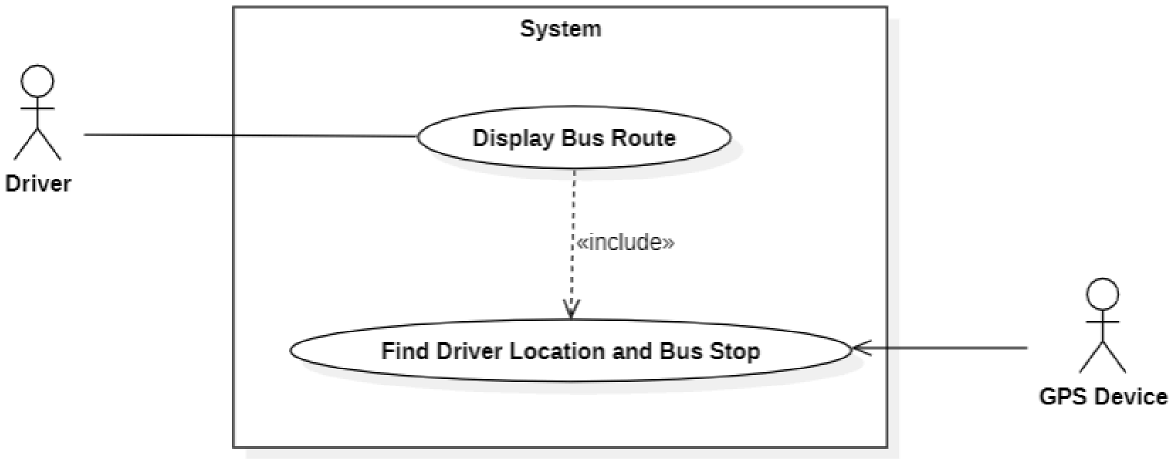
- The app should provide electronic forms for users to submit feedback, report issues, or request additional features.
- The form data should be securely transmitted to the app's backend server, using WebSockets method.
- The form design should be user-friendly, with clear instructions and validation to ensure the submission of accurate and complete information

*Data Transfer Rates and Synchronization:

- The app should be designed to handle fluctuating data transfer rates, especially in areas with poor network connectivity or high network traffic.
- The app should implement efficient data compression and optimization techniques to minimize the amount of data transferred while maintaining the necessary level of detail and responsiveness.
- The app should also have robust synchronization mechanisms to ensure that the user's device and the backend server are always in sync, even in the event of intermittent network connectivity.

1.6.4 Functional Requirement

1.6.4.1 Function 1: Quản lý khách hàng



Use Case: Display Bus Route	**ID:** UC001

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Main actor: Driver | **Priority:** Essential

Brief description: *A valid driver logged on to the system. She/he wants to see the information about the bus route.*

Trigger: Driver **Type:** external

Relationship:
+Association: Driver – Display Bus Route
+Include: Find Driver Location and Bus Stop
+Extend: None **+Generalization:** None

Normal flow:

1. The Driver triggers the request to view the bus route information.
2. The system requests turning on GPS device
3. Include "Find Driver Location and Bus Stop".
4. The system collects the data of bus stop location and driver location from the GPS device.
5. Checking the driver's ID and fetching relevant information like the driver's details, route trip, and bus stop information from the database.
6. Displaying the Home page with all the driver's basic information.
7. Navigating to the Journeys Page where upcoming route trips are displayed or allowing the driver to view past route trip details
   7.1. The system uses the driver's ID to determine the relevant bus route information.
   7.2. Starting the journey, which initiates a Google map view showing the starting point, ending destination, trip information, and the bus's real-time location.
   7.3. Updating the time, driver's location, and next bus stop information during the journey.
   7.4 Detecting if the driver has reached the final destination and displaying a "Finished trip" notification.
   7.5 The system will update route trip data and go to Step 8.
8. The system will check if the validity of driver's work time
   8.1 Checking if the driver's work time has exceeded a certain threshold (20 minutes after the last finished trip). If it is true, the trip will start the new updated trip or else the system will prompt an "Out of work message" and return to the Home Page.
   8.2 Checking if the driver's work time has exceeded a certain threshold of route operation time . If it is false, the driver is allowed to continue the next trip or else the system will prompt an "Out of work message" and return to the Home Page.

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Exceptional flows:
At any time, the Driver may choose to cancel the request. At which point, the processing is discontinued, no changes are made to the display or route information, and the Driver is notified that the request has been canceled.

Use Case: Find Driver Location and Bus Stop	**ID:** UC002
Main actor: GPS Device	**Priority:** Essential
Brief description: *A GPS Device addresses the driver location on the map.*	
Trigger: GPS Device	**Type:** Internal
Relationship: **+Association:** GPS Device – Find Driver Location **+Include:** None **+Extend:** None	**+Generalization:** None
Normal flow: 1. The GPS Device finds the bus and bus stop location. 2. The GPS Device collects the bus and bus stop data. 3. The GPS Device sends the data information to the system.	

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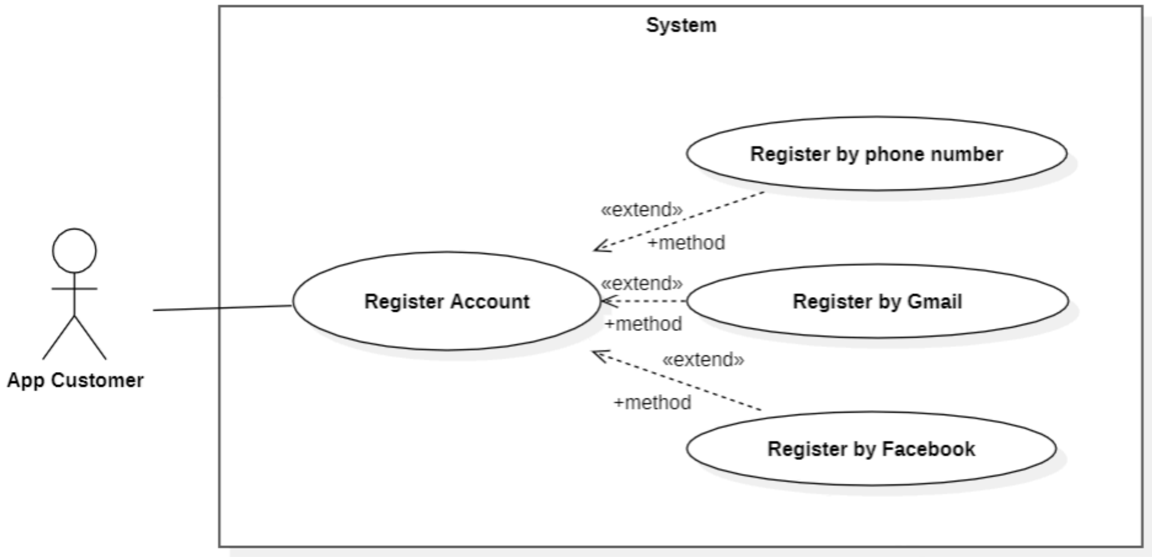
1.6.4.2 Function 2: Quản lý nhà cung cấp



Use Case: Register Account	**ID:** UC003
Main actor: App Customer	**Priority:** Essential
Brief description: *The App Customer registers an account (Gmail, Facebook or directly).*	
Trigger: App Customer	**Type:** external
Relationship: **+Association:** App Customer – Register Account **+Include:** None **+Extend:** Register by Gmail, Register by Facebook, Register directly **+Generalization:** None	
Preconditions: The user's device connected to the Internet when the user register a new account.	
Post Conditions: The user's details are stored in the database.	
Normal flow: 1. The use case starts when the user clicks the "Create Account" from profile before login page. 2. The system displays the sign up page that allows users to register (Gmail, Facebook or phone number). 3. <<method>>	

	Name	Code	Data Type	Length	Precision	P
1	AccountID	ACCOUNTID	char(7)	7		☑
2	AName	ANAME	varchar(20)	20		☐
3	AGender	AGENDER	varchar(3)	3		☐
4	ADoB	ADOB	date			☐
5	APhoneNumber	APHONENUMB	char(10)	10		☐
6	AEmail	AEMAIL	varchar(20)	20		☐

4. The system displays the user's profile page.

Use Case: Register by phone number	**ID:** UC004
Main actor: App Customer	**Priority:** Essential
Brief description: *The App Customer registers an account by using their phone number.*	
Trigger: App Customer	**Type:** external
Relationship: **+Association:** App Customer – Register Account **+Include:** None **+Extend:** Register Account **+Generalization:** None	
Normal flow: 1. System asks the user for his/her mobile number. 2. The user enters his/her mobile number and clicks on the "Get OTP" button. 3. System asks the user for verification code sent by SMS to the mobile number. 4. The user enters a verification code. 5. System verifies the verification code. It is ok. 6. The user enters his/her information (includes username, gender, age) or skip. 7. The system updates the information.	
Exceptional flow: **1. Invalid mobile number:** In step 3, when the user enters his/her mobile number and clicks on the "Get OTP" button, if the mobile number is invalid, an error message is displayed. The user then edits the mobile number. **2. Enter wrong verification code:** a. In step 6, if the user enters the wrong verification code, a message is displayed to inform the wrong code and resend another code after 20 seconds. b. The system sends OTP code to Zalo or via phone call. c. Users re-enter a verification code.	

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	Name	Code	Data Type	Length	Precision	P
1	AccountID	ACCOUNTID	char(7)	7		☑
2	AName	ANAME	varchar(20)	20		☐
3	AGender	AGENDER	varchar(3)	3		☐
4	ADoB	ADOB	date			☐
5	APhoneNumber	APHONENUMB	char(10)	10		☐
6	AEmail	AEMAIL	varchar(20)	20		☐

Use Case: Register by Gmail	**ID:** UC005
Main actor: App Customer	**Priority:** Essential

Brief description: *The App Customer registers an account by using their Gmail account.*

Trigger: App Customer **Type:** External

Relationship:

+Association: App Customer – Register Account

+Include: None

+Extend: Register Account **+Generalization:** None

Preconditions: User has already had a Google account.

Postconditions: User account is created successfully.

Normal flow:

1. A pop-up notification asks users to move to the Gmail app.
2. Users enter the "OPEN" button.
3. System asks the user for his/her Gmail account
4. The user enters his/her account and clicks on the "Continue" button.
5. System verifies the verification account. It is ok.
6. System requires the user to check the box "Agree to allow collection of user information"and clicks on the "Continue" button.
7. The system automatically retrieves information from the user's account.

Exceptional flow:

1. **Invalid Gmail account:** In step 1, when the user enters his/her Gmail account and clicks on the "Continue" button, if the Gmail account is invalid, an error message is displayed. The user then edits the Gmail account. If the user entered invalid more than 3 times, the system returns to the sign up page.
2. **Do not check the agree box:** In step 4, if the user do not allow to collect information from Gmail account, the screen will come back to the sign up page.

Use Case: Register by Facebook	**ID:** UC006
Main actor: App Customer	**Priority:** Essential

	Name	Code	Data Type	Length	Precision	P
1	AccountID	ACCOUNTID	char(7)	7		✓
2	AName	ANAME	varchar(20)	20		
3	AGender	AGENDER	varchar(3)	3		
4	ADoB	ADOB	date			
5	APhoneNumber	APHONENUMB	char(10)	10		
6	AEmail	AEMAIL	varchar(20)	20		

Brief description: *The App Customer registers an account by using their Facebook account.*

Trigger: App Customer **Type:** External

Relationship:
+Association: App Customer – Register Account
+Include: None
+Extend: Register Account **+Generalization:** None

Preconditions: User has already had a Facebook account.

Postconditions: User account is created successfully.

Normal flow:

1. A pop-up notification asks users to move to the Facebook app.
2. Users enter the "OPEN" button.
3. System asks the user for his/her Gmail account
4. The user enters his/her account and clicks on the "Continue" button.
5. System verifies the verification account. It is ok.
6. System requires the user to check the box "Agree to allow collection of user information"and clicks on the "Continue" button.
7. The system automatically retrieves information from the user's account.

Exceptional flow:

1. **Invalid Facebook account:** In step 1, when the user enters his/her Facebook account and clicks on the "Continue" button, if the Facebook account is invalid, an error message is displayed. The user then edits the Facebook account. If the user entered invalid more than 3 times, the system returns to the sign up page.
2. **Do not check the agree box:** In step 4, if the user do not allow to collect information from Gmail account, the screen will come back to the sign up page.

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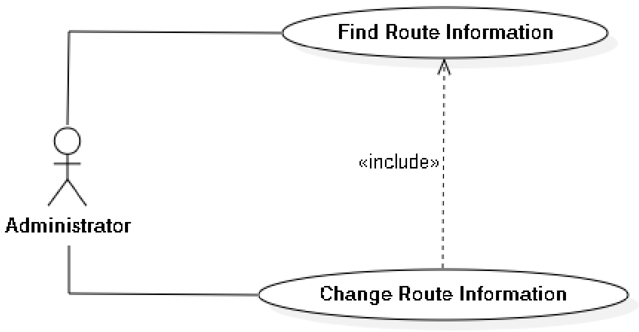
	Name	Code	Data Type	Length	Precision	P
1	AccountID	ACCOUNTID	char(7)	7		☑
2	AName	ANAME	varchar(20)	20		☐
3	AGender	AGENDER	varchar(3)	3		☐
4	ADoB	ADOB	date			☐
5	APhoneNumber	APHONENUMB	char(10)	10		☐
6	AEmail	AEMAIL	varchar(20)	20		☐

1.6.4.3 Function 3: Quản lý hàng hóa



Use Case: Change Route Information	**ID:** UC007
Main actor: Manager	**Priority:** Essential
Brief description: *A valid manager logged on to the system. She/he wants to change some information about the bus route.*	
Trigger: Manager	**Type:** External
Relationship: **+Association:** Manager – Change Route Information **+Include:** Find Route Information **+Extend:** None **+Generalization:** None	
Normal flow: 1. Include "Find Route Information". 2. The Manager selects the part of route information to change. 3. The Manager updates information of selected paths and requests that the system saves the entered values. 4. The system validates the entered route information. 4.1. The system describes which entered data was invalid and presents the Manager with suggestions of entering valid data. 4.2. The system prompts the Manager to re-enter the invalid information. 4.3. The Manager re-enters the information and the system re-validates it	

	Name	Code	Data Type	Length	Precision	P
1	AccountID	ACCOUNTID	char(7)	7		☑
2	AName	ANAME	varchar(20)	20		☐
3	AGender	AGENDER	varchar(3)	3		☐
4	ADoB	ADOB	date			☐
5	APhoneNumber	APHONENUMB	char(10)	10		☐
6	AEmail	AEMAIL	varchar(20)	20		☐

4.4. If valid information is entered, go to step 5; else go to step 4.1 or cancel the change route information request. If the Manager entered invalid data or chose to cancel the change request, there is no change to the Route's record.

5. The updated route information is stored in the Route's record. The system notifies the Manager that the route information has been updated.

Exceptional flows:

1. At any time, the Manager may choose to cancel the information update. The system ensures that any partially entered and modified data is discarded, and the Route's record remains unchanged.
2. At which point, the process is discontinued, the Route's record unchanged, and the Manager is notified that the request has been cancelled.

Use Case: Find Route Information	**ID:** UC008
Main actor: Manager	**Priority:** Essential

Brief description: *A valid manager logged on to the system. She/he wants to find the information of the bus route.*

Trigger: Manager **Type:** External

Relationship:
+Association: Manager – Find Route Information
+Include: Find Route Information
+Extend: None **+Generalization:** None

Normal flow:

1. The Manager entered the ID number of the bus route.
2. The system finds the information of the route.
3. The system displays information on a form, including details such as ID, name, operation time, ticket price and any other relevant information.

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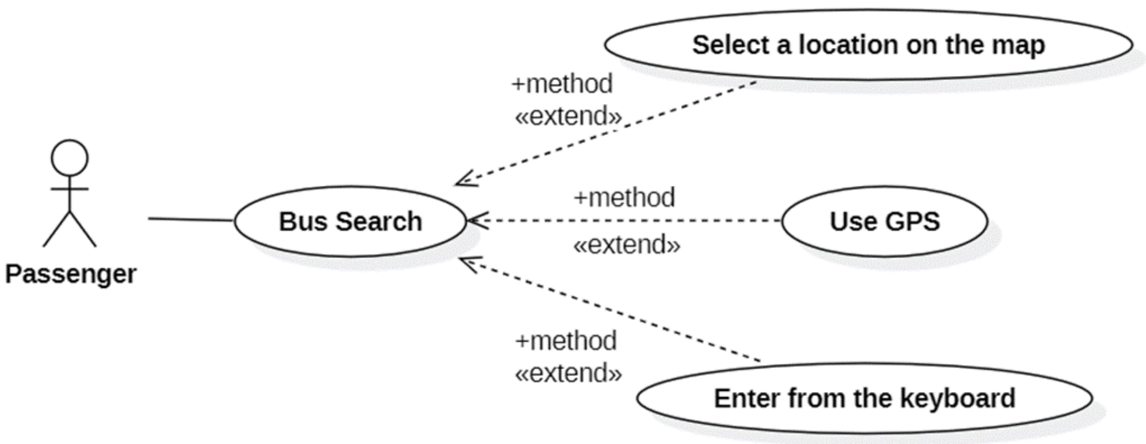
	Name	Code	Data Type	Length	Precision	P
1	AccountID	ACCOUNTID	char(7)	7		☑
2	AName	ANAME	varchar(20)	20		☐
3	AGender	AGENDER	varchar(3)	3		☐
4	ADoB	ADOB	date			☐
5	APhoneNumber	APHONENUMB	char(10)	10		☐
6	AEmail	AEMAIL	varchar(20)	20		☐

Exceptional flows:

1. If the Manager enters an ID that does not match any route in the system, the system provides an error message and prompts the Manager to re-enter a valid route ID.
2. If the customer enters "Cancel", the system confirms the cancellation and returns to the initial state without displaying route information.

1.6.4.4 Function 4: Quản lý hàng hóa



Use Case: Bus Route Search	**ID:** UC009
Main actor: Passenger	**Priority:** Essential
Brief description: *The Passenger inputs the starting point and destination to find the nearest suitable bus routes.*	
Trigger: Customer	**Type:** External
Relationship: + **Association:** Passenger- Search for suitable bus routes + **Include:** Enter the starting location and destination. + **Extend:** Use GPS, Enter from the keyboard, select a location on the map +**Generalization:** None	
Normal Flow: 1. Check the operating hours of the bus. 2. If still within operating hours, allow passengers to enter search criteria. 3. Customers input their departure, and destination, or select other search	

	Name	Code	Data Type	Length	Precision	P
1	AccountID	ACCOUNTID	char(7)	7		✓
2	AName	ANAME	varchar(20)	20		
3	AGender	AGENDER	varchar(3)	3		
4	ADoB	ADOB	date			
5	APhoneNumber	APHONENUMB	char(10)	10		
6	AEmail	AEMAIL	varchar(20)	20		

methods.

Sub 1:<Use GPS> if the passenger selects to search by GPS

Sub 2:<Enter from the keyboard> if the passenger selects to search by Enter from the keyboard.

Sub 3:<select a location on the map> if the passenger selects to search by selecting a location on the map.

4. If the departure is correct (has a name on the map), display relevant suggestions.
5. If the destination is correct (has a name on the map), display relevant suggestions.
6. Display the selected location on the map.
7. Find suitable bus routes passing through that route.
8. If suitable bus routes exist, display station locations along the routes and nearby stations on those routes starting from the chosen location.
9. Display a list of bus routes, and the total travel time the starting station to the ending station.
10. Passengers select the bus route.
11. Display detailed information about each movement step and its distance.

Subflows:

Sub 1: <Use GPS>

1. The system displays a notification requesting passengers to turn on their GPS on the screen.
2. Passengers turn on their GPS.
3. The system accesses the GPS system to retrieve the passenger's location and then displays it on the map.

Sub 2: <Enter from the keyboard>

1. When passengers enter the initial phrases, the system generates and displays a list of locations that match the entered keywords.
2. If the passenger selects a location, it is displayed on the map.
3. If the location entered by the passenger is not found in the database, a message indicating that the location is not suitable is displayed.

Sub 3: <Select a location on the map

1. The system displays a map.
2. Passengers search for and select the exact location on the map.

Exceptional Flow: If the departure and destination are not found, notify them and requires passenger to search again

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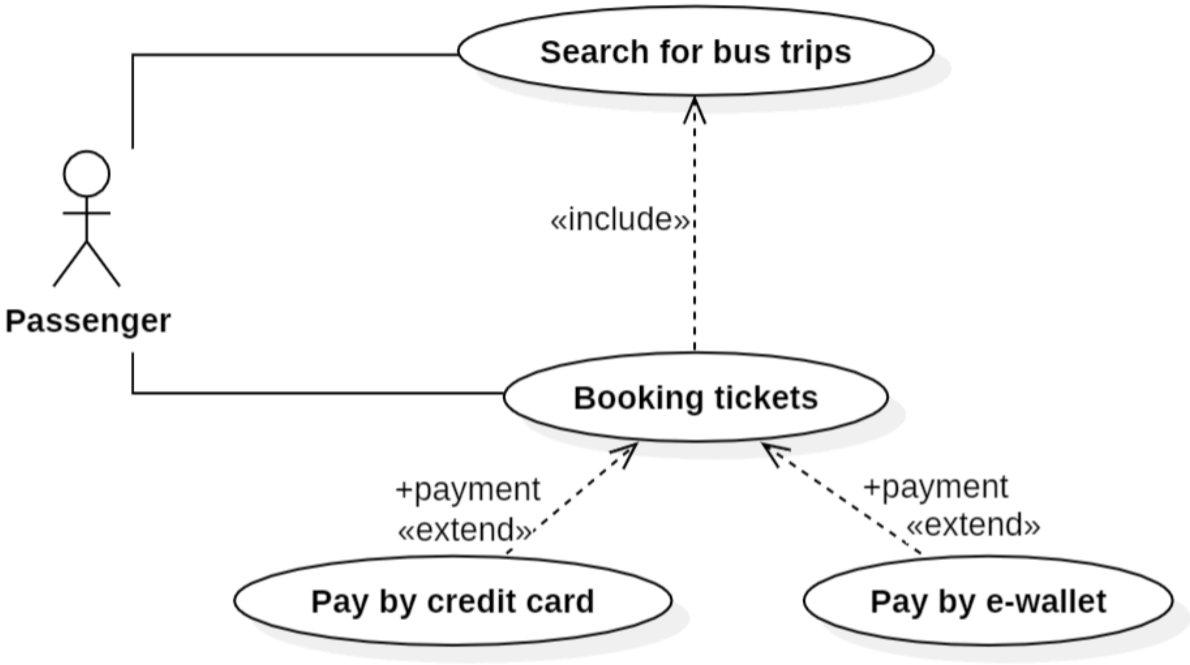
	Name	Code	Data Type	Length	Precision	P
1	AccountID	ACCOUNTID	char(7)	7		☑
2	AName	ANAME	varchar(20)	20		☐
3	AGender	AGENDER	varchar(3)	3		☐
4	ADoB	ADOB	date			☐
5	APhoneNumber	APHONENUMB	char(10)	10		☐
6	AEmail	AEMAIL	varchar(20)	20		☐

1.6.4.5 Function 5: Quản lý hóa đơn



Use Case: Online Booking	**ID:** UC010
Main actor: Passenger	**Priority:** Essential
Brief description: *A valid passenger logged on to the app. She/he wants to book a bus ticket.*	
Trigger: Passenger	**Type:** external
Relationship: **+Association:** Passenger – Book tickets **+Include:** Bus Routes Searching **+Extend:** Pay by credit card, Pay by e-wallet	**+Generalization:** None
Normal flow: 1. Include “Bus Routes Searching”. 2. Passenger selects the bus trip that they want to book. 3. Passenger selects the bus seat. 4. Passenger confirms ticket booking by entering their information (name, phone number) 5. System prompts for payment by credit card, debit card, e-wallet app. 6. <<payment>> 7. System displays a confirmation message and provides ticket details.	

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	Name	Code	Data Type	Length	Precision	P
1	AccountID	ACCOUNTID	char(7)	7		☑
2	AName	ANAME	varchar(20)	20		☐
3	AGender	AGENDER	varchar(3)	3		☐
4	ADoB	ADOB	date			☐
5	APhoneNumber	APHONENUMB	char(10)	10		☐
6	AEmail	AEMAIL	varchar(20)	20		☐

Excemptional flow:

1. **No available bus trips:** If there aren't any bus trips available for the passenger's chosen route or date, the system will prompt a message letting them know and ask them to go back and amend their search criteria.
2. **Payment failure:** If the system fails to confirm the payment, it will show an error message on the screen. The passenger's money will be returned to their account.

Use Case: Bus Routes Searching	**ID:** UC011
Main actor: Passenger	**Priority:** Essential
Brief description: *Passenger initiates a search for bus routes.*	
Trigger: Passenger	**Type:** external
Relationship: **+Association:** Passenger – Book tickets **+Include:** None **+Extend:** None	**+Generalization:** None
Preconditions: The passenger's device connected to the Internet and the booking feature has been accessed.	
Normal flow: 1. Passenger inputs the desired origin and destination for the bus trip by entering, using GPS or selecting on the map. 2. System displays available bus routes matching the specified criteria.	
Exceptional flow: **1. If there are no routes matching the specified criteria:** + The system informs the Passenger that no matching routes were found. + Passenger can choose to revise the search criteria or use different searching methods.	

Use Case: Pay by credit card	**ID:** UC012
Main actor: Passenger	**Priority:** Essential
Brief description: *Passenger pays by credit card for ticket booked.*	
Trigger: Passenger	**Type:** external
Relationship: **+Association:** Passenger – Book tickets **+Include:** None **+Extend:** Booking tickets	**+Generalization:** None

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	Name	Code	Data Type	Length	Precision	P
1	AccountID	ACCOUNTID	char(7)	7		☑
2	AName	ANAME	varchar(20)	20		☐
3	AGender	AGENDER	varchar(3)	3		☐
4	ADoB	ADOB	date			☐
5	APhoneNumber	APHONENUMB	char(10)	10		☐
6	AEmail	AEMAIL	varchar(20)	20		☐

Normal flow:

1. Passenger selects payment by credit card
2. Passenger selects the bank to make payment
3. System enter information about credit card (cnumber, name, exp date, CVV)
4. Passenger confirms the information.
5. System receives confirmation from the e-wallet app and finalizes the payment

Exceptional flow:

1. **Incorrect credit card information:** If the passenger enters incorrect credit card information (e.g., card number, expiration date, CVV), the system will likely prompt the passenger to re-enter the information. After a certain number of failed attempts, the system may lock the passenger's account or require them to contact customer support.
2. **Insufficient funds:** If the passenger's credit card does not have sufficient funds to cover the cost of the ticket, the system will decline the transaction and notify the passenger. The passenger may then need to choose a different payment method or add funds to their credit card.
3. **Network timeout:** If there is a network timeout during the transaction, the system may display an error message and ask the passenger to try again.
4. **Payment declined by bank:** The passenger's bank may decline the transaction for a variety of reasons, such as suspected fraud or suspicious activity. The passenger will need to contact their bank to resolve the issue.

Use Case: Pay by e-wallet	**ID:** UC013
Main actor: Passenger	**Priority:** Essential
Brief description: *Passenger pays by e-wallet app for ticket booked.*	
Trigger: Passenger	**Type:** external
Relationship: **+Association:** Passenger – Book tickets **+Include:** None **+Extend:** Booking tickets **+Generalization:** None	
Normal flow: 1. Passenger selects payment by e-wallet app (Momo, ZaloPay).. 2. System requires opening e-wallet app that the passenger chose. 3. Passenger confirms the requirement. 4. Passenger authorizes the payment within the e-wallet app	

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	Name	Code	Data Type	Length	Precision	P
1	AccountID	ACCOUNTID	char(7)	7		✓
2	AName	ANAME	varchar(20)	20		
3	AGender	AGENDER	varchar(3)	3		
4	ADoB	ADOB	date			
5	APhoneNumber	APHONENUMB	char(10)	10		
6	AEmail	AEMAIL	varchar(20)	20		

5. System receives confirmation from the e-wallet app and finalizes the payment.

Exceptional flow:

1. **Passenger cancels payment:** If the passenger cancels the payment within the e-wallet app, the payment won't be completed and the passenger will be directed back to the mobile ticketing system.
2. **Insufficient funds in e-wallet:** If there are insufficient funds in the passenger's e-wallet to cover the cost of the ticket, the e-wallet app will notify the passenger and the payment won't be completed. The passenger can then choose to top up their e-wallet or use a different payment method.

1.6.5 Nonfunctional Requirements

1.6.5.1Performance

- Timing Requirements:

+ Response Time: The maximum allowable time for the system to respond to user interactions, such as loading bus route information, login/signup request response or processing ticket bookings, is not exceed 3 seconds on average.

+ Processing Time: The system processes user requests, such as updating bus routes or retrieving real-time location data, within 5 seconds on average to ensure a responsive user experience.

- Maximum Capacity:

+ Concurrent Users: The system supports a minimum of 1000 concurrent users accessing the service simultaneously during peak usage periods.

+ Bandwidth: The system is able to handle a minimum bandwidth of 1 Mbps to accommodate data transfer requirements for real-time updates and multimedia content.

+ Transaction Volume: The system is able to execute a minimum of 100 transactions per second to ensure timely processing of user requests, such as ticket bookings and route updates.

1.6.5.2Reliability

- Availability:

+ The system's availability is maintained at 99.9% uptime on an annual basis, excluding planned maintenance windows. This ensures continuous accessibility for users without significant downtime.

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	Name	Code	Data Type	Length	Precision	P
1	AccountID	ACCOUNTID	char(7)	7		✓
2	AName	ANAME	varchar(20)	20		
3	AGender	AGENDER	varchar(3)	3		
4	ADoB	ADOB	date			
5	APhoneNumber	APHONENUMB	char(10)	10		
6	AEmail	AEMAIL	varchar(20)	20		

+ The system's downtime, excluding planned maintenance, should not exceed 1 hour per month to minimize service disruptions and ensure reliability.

- **Resilience:** The system has built-in redundancy and failover mechanisms to recover from hardware failures or unexpected events within 1 hour, ensuring minimal disruption to service continuity and data integrity.

1.6.5.3 Safety and Security

- Security:

+ User Authentication: The system enforces strong authentication mechanisms, such as username and password authentication with encryption, to prevent unauthorized access to user accounts and sensitive data. This requirement also aligns with various data protection laws, including the European Union's General Data Protection Regulation (GDPR), which mandates the protection of personal data through appropriate security measures.

+ Data Access Control: Access to data is restricted based on user roles and permissions, ensuring that only authorized users can view or modify specific information. This requirement is in compliance with privacy laws such as the California Consumer Privacy Act (CCPA), which requires organizations to implement measures to control access to personal information and prevent unauthorized disclosure.

- Integrity:

+ Data Encryption: The system encrypts sensitive data, such as user credentials and payment information, during transmission over the network to prevent unauthorized interception or tampering. This practice is essential for compliance with regulations like the Health Insurance Portability and Accountability Act (HIPAA), which mandates the protection of electronic protected health information (ePHI) through encryption and other security measures.

+ Audit Trail: The system maintains an audit trail of user activities and system events to track changes and ensure data integrity and accountability. This requirement is mandated by financial regulations such as the Sarbanes-Oxley Act (SOX), which requires organizations to maintain accurate and complete records of financial transactions and auditing trails to detect and prevent fraud. Additionally, audit trail functionalities support compliance with data protection laws by providing transparency and accountability in data handling processes.

1.6.5.4Adaptability and Portability

- **Portability:** This mobile application is currently designed and optimized for exclusive use on Apple devices running iOS version 15.0 or later versions. This includes iPhones and iPads. While the application prioritizes this platform for

	Name	Code	Data Type	Length	Precision	P
1	AccountID	ACCOUNTID	char(7)	7		✓
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3	AGender	AGENDER	varchar(3)	3		
4	ADoB	ADOB	date			
5	APhoneNumber	APHONENUMB	char(10)	10		
6	AEmail	AEMAIL	varchar(20)	20		

optimal performance and user experience, the development team acknowledges the potential for future portability considerations. Continued user demand and market trends will be evaluated to determine the feasibility of expanding the application to other mobile operating systems.

- Adaptability:

1. Device Compatibility:

+ The app utilizes responsive design principles to ensure a smooth user experience on all iPhone screen sizes, currently supporting iPhones with displays ranging from 4.7 inches (iPhone SE) to 6.12 inches (iPhone 15 Pro Max).

2. Scalable Data Management:

+ The underlying database structure is designed to accommodate future growth in data volume. This includes the ability to integrate information for:

- Up to 1,000 additional bus routes.
- Real-time updates for bus locations and arrival times.
- Historical trip data for up to one year.

3. Transaction Volume Handling:

+ The app is architected to handle increasing user traffic. This includes the ability to scale to accommodate:

- A projected growth of 10,000 new users per month
- A peak transaction volume of 5,000 user requests per minute during rush hour

4. Flexible Report Formatting:

+ The app can generate reports in various formats to cater to different user needs. This includes:

- User-friendly trip itineraries displayed on the map.
- Detailed bus schedule tables with customizable filters (e.g., by route, stop, or time).
- Downloadable reports in CSV format for further analysis.

1.6.5.5Business Rules

1. User Roles and Permissions:

The application will have three primary user roles:

- Passenger: This is the default role for all users. Passengers can view bus routes, schedules, real-time bus locations (if available), and trip itineraries. They may also be able to access downloadable reports (if applicable) and submit feedback on the app's functionality.
- Administrator: This role is restricted to authorized personnel (e.g., bus company employees). Administrators can manage all aspects of the app,

	Name	Code	Data Type	Length	Precision	P
1	AccountID	ACCOUNTID	char(7)	7		☑
2	AName	ANAME	varchar(20)	20		☐
3	AGender	AGENDER	varchar(3)	3		☐
4	ADoB	ADOB	date			☐
5	APhoneNumber	APHONENUMB	char(10)	10		☐
6	AEmail	AEMAIL	varchar(20)	20		☐

including adding and editing bus route information, schedules, real-time location updates, and user data access permissions. They can also access anonymized user data for analytics purposes (with appropriate privacy safeguards).

- Driver: This role is for authorized bus drivers. Drivers can update the app with their real-time location and potential delays (if applicable). They cannot modify route information or access user data.

2. Data Access and Management:

- All user data, including login credentials, trip history (if applicable), and search preferences, will be stored securely following relevant data privacy regulations.
- Only authorized administrators will have access to manage and modify bus route information, schedules, real-time location data, and user data access permissions.
- Driver access will be limited to updating their own bus's real-time location and potential delays.

3. User Interactions and Transactions:

- Passengers can search for bus routes and schedules by specifying origin, destination, and desired travel time.
- The app will display trip itineraries with estimated travel times and potential transfer points.
- Passengers may be able to set up alerts for real-time bus arrival notifications (if applicable).
- The app will not currently support functionalities like booking tickets or fare payments. These functionalities may be considered for future development based on user needs and business requirements.

4. Driver Responsibilities and Updates:

- Drivers are responsible for updating their bus's real-time location within the app (if applicable).
- Drivers can also report any potential delays affecting their route (e.g., traffic congestion, accidents).
- Driver updates will be reflected in real-time for passengers to adjust their travel plans if necessary.

5. System Availability and Maintenance:

- The application will strive for high availability with minimal downtime for scheduled maintenance or unforeseen technical issues.
- In case of maintenance or outages, users will be notified through appropriate channels (e.g., in-app messages).

6. User Feedback and Dispute Resolution:

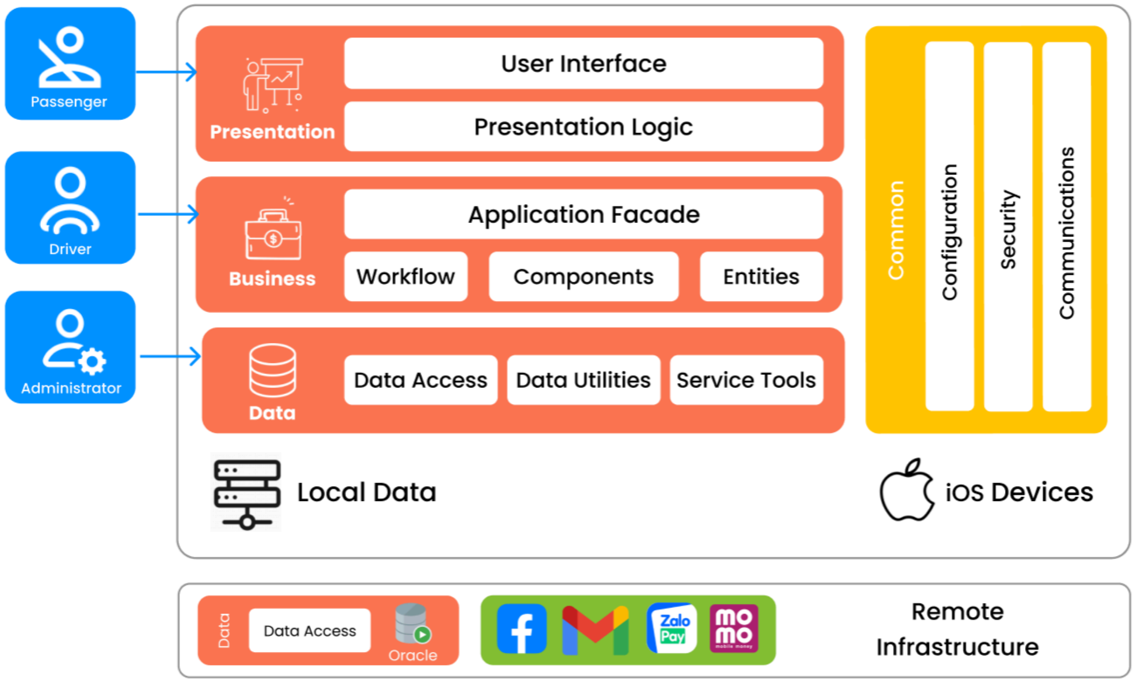
	Name	Code	Data Type	Length	Precision	P
1	AccountID	ACCOUNTID	char(7)	7		☑
2	AName	ANAME	varchar(20)	20		☐
3	AGender	AGENDER	varchar(3)	3		☐
4	ADoB	ADOB	date			☐
5	APhoneNumber	APHONENUMB	char(10)	10		☐
6	AEmail	AEMAIL	varchar(20)	20		☐

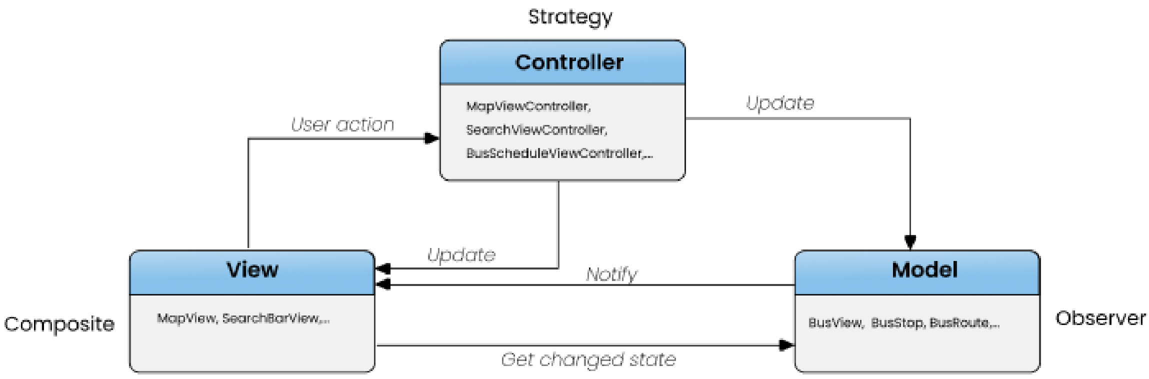
- The app will provide a mechanism for all users (passengers, administrators, and drivers) to submit feedback and report any issues they encounter.
- Administrators will review user feedback and address reported issues in a timely manner.
- A clear dispute resolution process will be established to handle any user concerns related to data accuracy, functionality, or driver updates.

2. Software Design

2.1 Application Architecture

2.1.1 Application Architecture Diagram





2.1.2 Description

	Name	Code	Data Type	Length	Precision	P
1	AccountID	ACCOUNTID	char(7)	7		☑
2	AName	ANAME	varchar(20)	20		☐
3	AGender	AGENDER	varchar(3)	3		☐
4	ADoB	ADOB	date			☐
5	APhoneNumber	APHONENUMB	char(10)	10		☐
6	AEmail	AEMAIL	varchar(20)	20		☐

- The application architecture diagram above consists of 2 model, including 3-layers model and MVC model:

+ 3-layers model: Illustrates for the Client - Server architecture.

- This model divides the application into 3 layers: presentation layer responsible for user interaction and interface design., business layer manages core application logic and business rules, and data layer handles data storage, retrieval, and manipulation.
- For our bus application, we will utilize the iOS platform for seamless operation on Apple devices. The backend database will be Oracle SQL, ensuring robust data management and storage capabilities. Our client-server model will be hosted on Amazon Web Services (AWS), providing scalable and reliable infrastructure for hosting and managing the application.
- In terms of authentication, users will have the option to login/signup using their Facebook or Gmail accounts, leveraging their existing credentials for ease of access. Additionally, for payment processing, we will integrate payment gateways such as Zalpay and Momo to facilitate secure and convenient transactions within the app. Furthermore, we will incorporate Google Maps API to provide mapping services, enabling users to view bus routes, nearby stops, and navigate their journeys effectively.

+ MVC model: Illustrates for iOS app development architecture.

- Model: Represents the application's data structure and business logic. It's responsible for managing data, rules for modifying or processing data, and interactions with the database, including BusView, BusStop, and BusRoute
- View: Responsible for rendering UI elements and displaying data to the user, listen to changes in the Model and updates the UI accordingly. For eaxmple, MapView for showing bus routes and SearchBarView for user input.
- Controller: Acts as an interface between the Model and View components. Processes user actions, manipulates data using the Model, and returns output for display in the View, as MapViewController, SearchViewController, BusScheduleViewController.

2.2 Data Design

2.2.1 Data Description

- Information Domain to Data Structures:

+ Information Domain: The Bus Application's information domain encompasses data related to stops, drivers, passengers, accounts, routes, bookings, and tickets. It includes details about the locations where buses halt, information on the drivers

■

	Name	Code	Data Type	Length	Precision	P
1	AccountID	ACCOUNTID	char(7)	7		☑
2	AName	ANAME	varchar(20)	20		☐
3	AGender	AGENDER	varchar(3)	3		☐
4	ADoB	ADOB	date			☐
5	APhoneNumber	APHONENUMB	char(10)	10		☐
6	AEmail	AEMAIL	varchar(20)	20		☐

operating the buses, descriptions of bus routes, records of passenger reservations, and specifics of issued tickets.

+ Data Structures: These structured formats efficiently store and manage data within the Bus Application, facilitating access, retrieval, and manipulation of information.

- Conversion Process:

+ Mapping Function: The information domain of the Bus Application is mapped to Oracle data structures through a defined mapping function. This function establishes how elements in the application's domain correspond to elements within Oracle's relational database model.

+ Data Morphology: Oracle's relational database model captures the morphology or shape of information. Tables within the Oracle database represent entities such as stops, drivers, routes, bookings, and tickets, with rows representing individual records and columns representing attributes.

+ Semantic Interpretation: Within Oracle's relational database model, relationships between entities are defined through primary and foreign keys, providing semantic interpretation to the data. For example, a schema in Oracle defines relationships between tables, such as the association between routes and stops.

+ Irregular Structure: Oracle's relational database model can accommodate irregular or custom data representations through the creation of specialized tables or views to address specific requirements within the Bus Application.

- Major Data Entities and Storage:

+ Stops: Locations where buses halt, stored within Oracle's relational database as a table with attributes such as stop ID, name, and address.

+ Drivers: Information about the drivers operating buses, stored in an Oracle table with attributes including driver ID, name, and contact details.

+ Passengers: Information about the passengers reserving bus tickets, stored in an Oracle table with attributes including passenger's name, phone number and email.

+ Accounts: Records of user app accounts, including attributes such as account ID, account name,...

+ Routes: Descriptions of paths followed by buses, stored in an Oracle table with attributes defining route details such as route ID, total seats, distance, and operational times.

+ Bookings: Records of passenger reservations, stored in an Oracle table with attributes such as booking ID, booking date, and method,....

+ Tickets: Details of issued tickets, stored in an Oracle table with attributes defining ticket information such as ticket ID, seat number, type, and price.

- These entities are organized and processed within Oracle's relational database using tables, primary and foreign keys, indexes, and other database structures to ensure efficient data management and retrieval.

- Oracle's relational database management system offers robust capabilities for storing, querying, and managing data within the Bus Application, providing

■

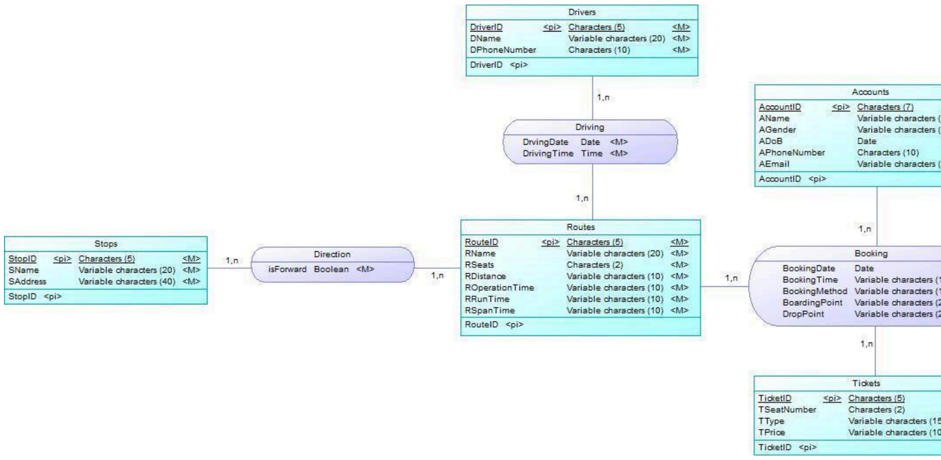
	Name	Code	Data Type	Length	Precision	P
1	AccountID	ACCOUNTID	char(7)	7		☑
2	AName	ANAME	varchar(20)	20		☐
3	AGender	AGENDER	varchar(3)	3		☐
4	ADoB	ADOB	date			☐
5	APhoneNumber	APHONENUMB	char(10)	10		☐
6	AEmail	AEMAIL	varchar(20)	20		☐

scalability, reliability, and performance for handling the application's data requirements.

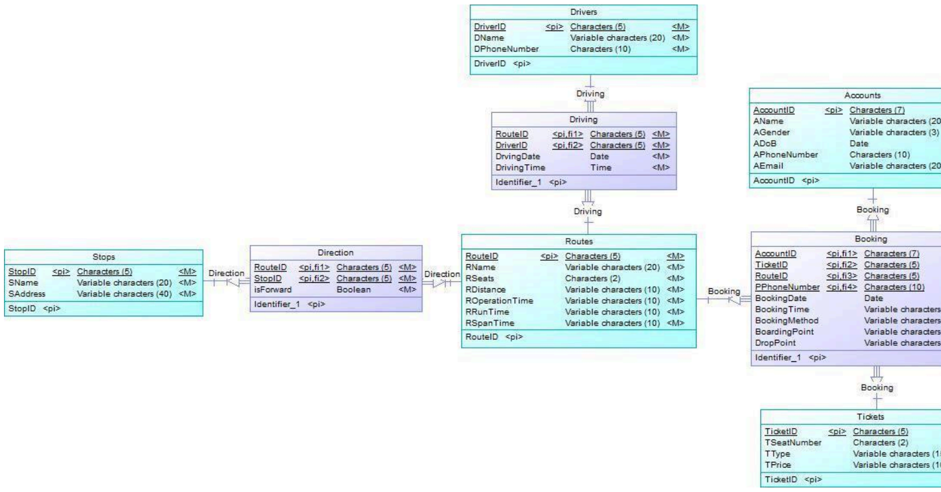
■

	Name	Code	Data Type	Length	Precision	P
1	AccountID	ACCOUNTID	char(7)	7		☑
2	AName	ANAME	varchar(20)	20		☐
3	AGender	AGENDER	varchar(3)	3		☐
4	ADoB	ADOB	date			☐
5	APhoneNumber	APHONENUMB	char(10)	10		☐
6	AEmail	AEMAIL	varchar(20)	20		☐

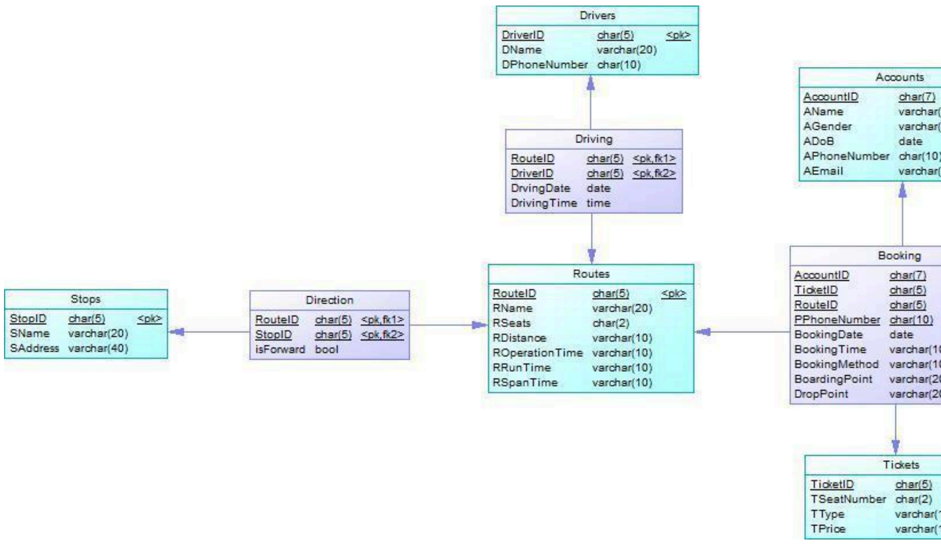
2.2.1.1 Conceptual Data Model



2.2.1.2 Logical Data Model



2.2.1.3 Physical Data Model



2.2.2 Data Dictionary

	Name	Code	Data Type	Length	Precision	P
1	AccountID	ACCOUNTID	char(7)	7		☑
2	AName	ANAME	varchar(20)	20		☐
3	AGender	AGENDER	varchar(3)	3		☐
4	ADoB	ADOB	date			☐
5	APhoneNumber	APHONENUMB	char(10)	10		☐
6	AEmail	AEMAIL	varchar(20)	20		☐

- Account:

■

- Booking:

	Name	Code	Data Type	Length	Precision	P	F	M
1	PassengerID	PASSENGERID	char(7)	7		✓	✓	✓
2	TicketID	TICKETID	char(5)	5		✓	✓	✓
3	RouteID	ROUTEID	char(5)	5		✓	✓	✓
4	BookingDate	BOOKINGDATE	date					✓
5	BookingTime	BOOKINGTIME	varchar(10)	10				✓
6	BookingMethod	BOOKINGMET	varchar(10)	10				✓

- Direction:

	Name	Code	Data Type	Length	Precision	P	F	M
1	RouteID	ROUTEID	char(5)	5		✓	✓	✓
2	StopID	STOPID	char(5)	5		✓	✓	✓
3	isForward	ISFORWARD	bool					✓

- Driver:

	Name	Code	Data Type	Length	Precision	P	F	M
1	DriverID	DRIVERID	char(5)	5		✓		✓
2	DName	DNAME	varchar(20)	20				✓
3	DPhoneNumber	DPHONENUM	char(10)	10				✓

- Driving:

	Name	Code	Data Type	Length	Precision	P	F	M
1	RouteID	ROUTEID	char(5)	5		✓	✓	✓
2	DriverID	DRIVERID	char(5)	5		✓	✓	✓
3	DrvingDate	ATTRIBUTE_35	date					✓
4	DrivingTime	ATTRIBUTE_36	time					✓

- Passengers:

	Name	Code	Data Type	Length	Precision	P	F	M
1	PName	PNAME	varchar(30)	30				✓
2	PPhoneNumber	PPHONENUMB	char(10)	10		✓		✓
3	PEmail	PEMAIL	varchar(20)	20				✓

- Routes:

■

	Name	Code	Data Type	Length	Precision	P
1	AccountID	ACCOUNTID	char(7)	7		✓
2	AName	ANAME	varchar(20)	20		
3	AGender	AGENDER	varchar(3)	3		
4	ADoB	ADOB	date			
5	APhoneNumber	APHONENUMB	char(10)	10		
6	AEmail	AEMAIL	varchar(20)	20		

	Name	Code	Data Type	Length	Precision	P	F	M
1	RouteID	ROUTEID	char(5)	5		✓		✓
2	RName	RNAME	varchar(20)	20				✓
3	RSeats	RSEATS	char(2)	2				✓
4	RDistance	RDISTANCE	varchar(10)	10				✓
5	ROperationTime	ROPERATIONT	varchar(10)	10				✓
6	RRunTime	RRUNTIME	varchar(10)	10				✓
7	RSpanTime	RSPANTIME	varchar(10)	10				✓

- Stops:

	Name	Code	Data Type	Length	Precision	P	F	M
1	StopID	STOPID	char(5)	5		✓		✓
2	SName	SNAME	varchar(20)	20				✓
3	SAddress	SADDRESS	varchar(40)	40				✓

- Tickets:

	Name	Code	Data Type	Length	Precision	P	F	M
1	TicketID	TICKETID	char(5)	5		✓		✓
2	TSeatNumber	TSEATNUMBE	char(2)	2				✓
3	TType	TTYPE	varchar(15)	15				✓
4	TPrice	TPRICE	varchar(10)	10				✓

2.3 Detailed Design

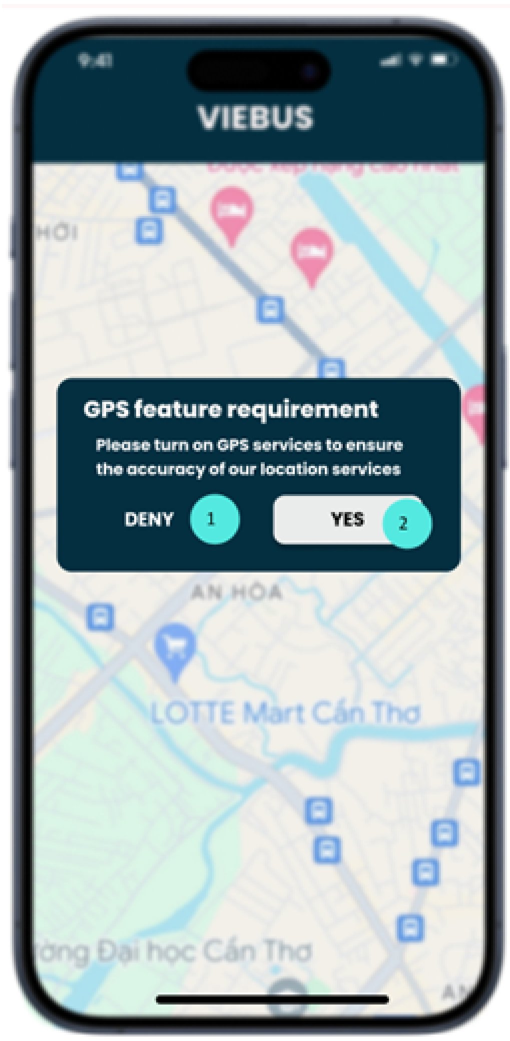
2.3.1 Function 1 : Quản lý khách hàng

- *Purpose*: The platform allow the driver to see the information of the upcoming or previous route trips. Drivers can also track the route they drive on and their location.

- *Interface:*

  + *Interface* 1: Activating GPS requirements

	Name	Code	Data Type	Length	Precision	P
1	AccountID	ACCOUNTID	char(7)	7		✓
2	AName	ANAME	varchar(20)	20		
3	AGender	AGENDER	varchar(3)	3		
4	ADoB	ADOB	date			
5	APhoneNumber	APHONENUMB	char(10)	10		
6	AEmail	AEMAIL	varchar(20)	20		



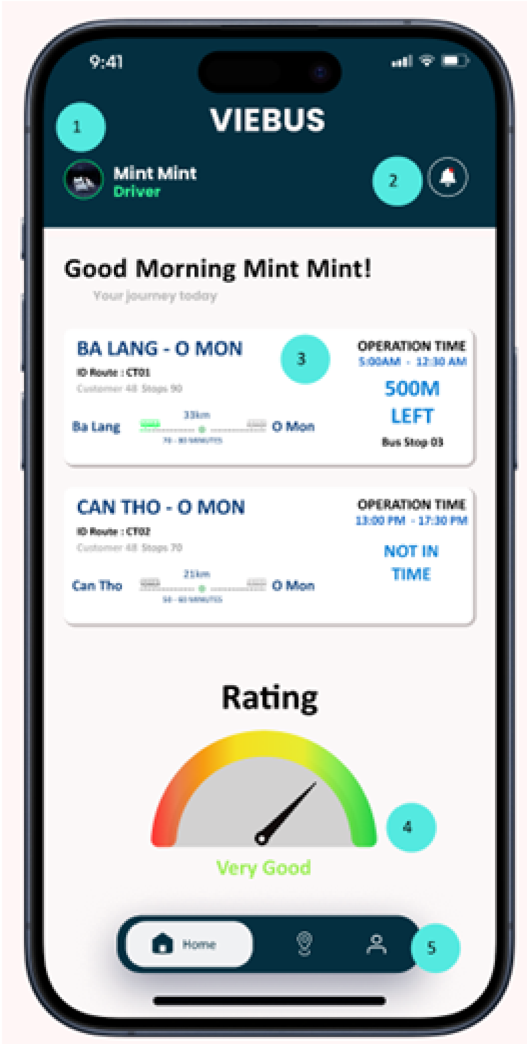
- *The components of interface 1*:

No.	Controller type	Note
1	*Button*	*Turn on the GPS of device*
2	*Button*	*Cancel requirement and navigate to not signed in interface*

■

	Name	Code	Data Type	Length	Precision	P
1	AccountID	ACCOUNTID	char(7)	7		☑
2	AName	ANAME	varchar(20)	20		☐
3	AGender	AGENDER	varchar(3)	3		☐
4	ADoB	ADOB	date			☐
5	APhoneNumber	APHONENUMB	char(10)	10		☐
6	AEmail	AEMAIL	varchar(20)	20		☐

\+ *Interface* 2: Driver Home Page



\- The components of interface 2:

No.	Controller type	Note
1	*Image*	*Navigating to "Driver profile" interface*
2	*Button*	*Show the new changes or message*
3	*Table*	*Display driver route trip information*

■

	Name	Code	Data Type	Length	Precision	P
1	AccountID	ACCOUNTID	char(7)	7		☑
2	AName	ANAME	varchar(20)	20		☐
3	AGender	AGENDER	varchar(3)	3		☐
4	ADoB	ADOB	date			☐
5	APhoneNumber	APHONENUMB	char(10)	10		☐
6	AEmail	AEMAIL	varchar(20)	20		☐

4	*Gauges*	*Display driver rating*
5	*Button*	*Navigate to "Home Page" interface (3)*

- *Data to be used*:

No.	**Table name / Data structure**	**Method**			
		Add	**Modify**	**Delete**	**Query**
1	*Drivers*				X
2	*Driving*				X
3	*Routes*				X

+ Interface 3.1 and 3.2: Journeys Page (Upcoming or previous route trip)

■

	Name	Code	Data Type	Length	Precision	P
1	AccountID	ACCOUNTID	char(7)	7		✓
2	AName	ANAME	varchar(20)	20		
3	AGender	AGENDER	varchar(3)	3		
4	ADoB	ADOB	date			
5	APhoneNumber	APHONENUMB	char(10)	10		
6	AEmail	AEMAIL	varchar(20)	20		

- The components of interface 3.1 and 3.2:

No.	Controller type	Note
1	*Type header*	*Navigate to "Home Page" interface (3)*
2	*Button*	*Display past route trip in last week*
3	*Button*	*Display upcoming route trip*

	Name	Code	Data Type	Length	Precision	P
1	AccountID	ACCOUNTID	char(7)	7		✓
2	AName	ANAME	varchar(20)	20		
3	AGender	AGENDER	varchar(3)	3		
4	ADoB	ADOB	date			
5	APhoneNumber	APHONENUMB	char(10)	10		
6	AEmail	AEMAIL	varchar(20)	20		

4	*Button*	*Display all detail of chosen route trip information*
5	*Button*	*Navigate to "Journeys" interface (4.1)*

- Data to be used:

No.	Table name / Data structure	Method			
		Add	Modify	Delete	Query
1	*Driving*				X
2	*Routes*				X

+ Interface 4.1: Display detailed trip information (Not starting trip)

■

	Name	Code	Data Type	Length	Precision	P
1	AccountID	ACCOUNTID	char(7)	7		☑
2	AName	ANAME	varchar(20)	20		☐
3	AGender	AGENDER	varchar(3)	3		☐
4	ADoB	ADOB	date			☐
5	APhoneNumber	APHONENUMB	char(10)	10		☐
6	AEmail	AEMAIL	varchar(20)	20		☐

- The components of interface 4.1:

No.	Controller type	Note
1	*Type header*	*Return to "Journeys" interface (4.1)*
2	*Button*	*Display route trip information on map of Google Map*
3	*Circle shape*	*Show the mark of location (green for currently or already visited, red for ending location, gray for unvisit)*

■

	Name	Code	Data Type	Length	Precision	P
1	AccountID	ACCOUNTID	char(7)	7		☑
2	AName	ANAME	varchar(20)	20		☐
3	AGender	AGENDER	varchar(3)	3		☐
4	ADoB	ADOB	date			☐
5	APhoneNumber	APHONENUMB	char(10)	10		☐
6	AEmail	AEMAIL	varchar(20)	20		☐

4	*Button*	*Start working, tracking the location and mapping bus route for driver*
5	*Button*	*Navigate to "Journeys" interface (4.1)*

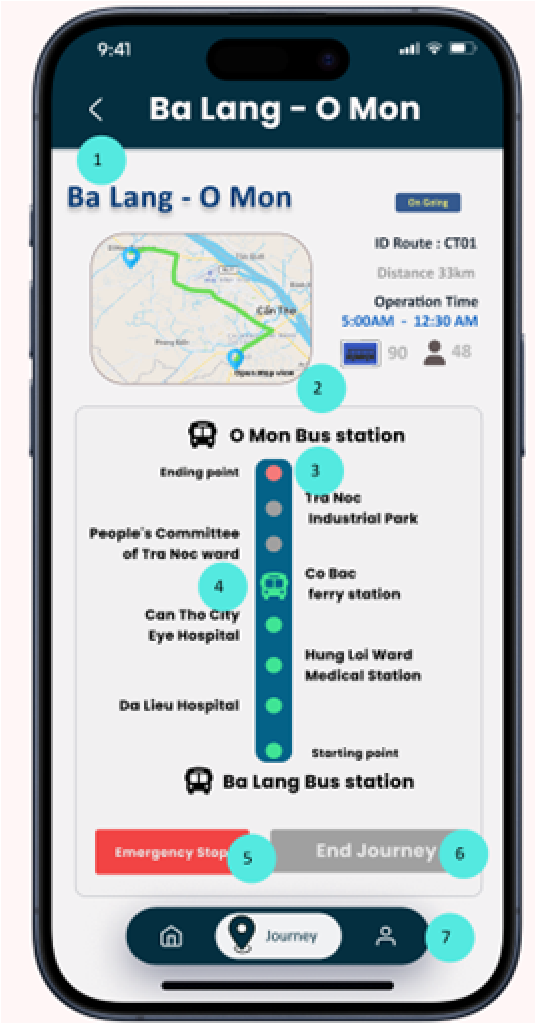
- Data to be used:

No.	**Table name / Data structure**	**Method**			
		Add	**Modify**	**Delete**	**Query**
1	*Driving*				X
2	*Routes*				X
3	*Direction*				X

+ Interface 4.2: Display detailed trip information (Starting trip)

■

	Name	Code	Data Type	Length	Precision	P
1	AccountID	ACCOUNTID	char(7)	7		☑
2	AName	ANAME	varchar(20)	20		☐
3	AGender	AGENDER	varchar(3)	3		☐
4	ADoB	ADOB	date			☐
5	APhoneNumber	APHONENUMB	char(10)	10		☐
6	AEmail	AEMAIL	varchar(20)	20		☐



- The components of interface 4.2:

No.	Controller type	Note
1	*Type header*	*Return to "Journeys" interface (4.1)*
2	*Button*	*Display route trip information on map of Google Map*
3	*Circle shape*	*Show the mark of location (green for currently or already visited, red for ending location, gray for unvisit)*
4	*Icon*	*Showing which bus stops is the bus at (This icon is mark as green)*

▪

	Name	Code	Data Type	Length	Precision	P
1	AccountID	ACCOUNTID	char(7)	7		✓
2	AName	ANAME	varchar(20)	20		
3	AGender	AGENDER	varchar(3)	3		
4	ADoB	ADOB	date			
5	APhoneNumber	APHONENUMB	char(10)	10		
6	AEmail	AEMAIL	varchar(20)	20		

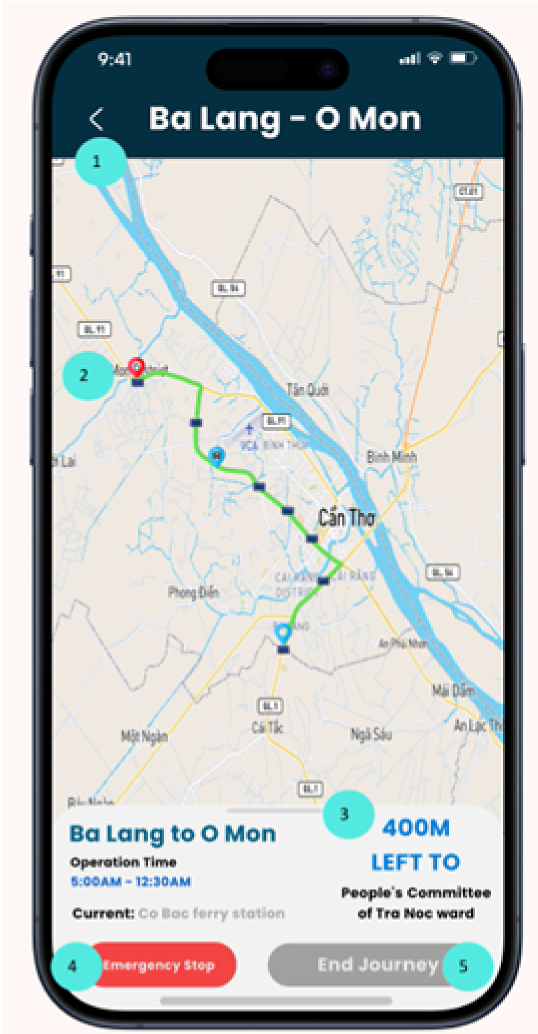
5	*Button*	*To make an emergency stop for the serious problem*
6	*Button*	*Ending the trip if the bus is arrived last destination*
7	*Button*	*Navigate to "Journeys" interface (4.1)*

- Data to be used:

No.	Table name / Data structure	Method			
		Add	**Modify**	**Delete**	**Query**
1	*Driving*				X
2	*Routes*				X
3	*Direction*				X

+ Interface 5: Displaying the trip in map format

	Name	Code	Data Type	Length	Precision	P
1	AccountID	ACCOUNTID	char(7)	7		✓
2	AName	ANAME	varchar(20)	20		
3	AGender	AGENDER	varchar(3)	3		
4	ADoB	ADOB	date			
5	APhoneNumber	APHONENUMB	char(10)	10		
6	AEmail	AEMAIL	varchar(20)	20		



- The components of interface 5:

No.	Controller type	Note
1	*Type header*	*Return to "Detailed Journeys" interface (5.2)*
2	*Icon*	*Show the last destination as red, first starting location as blue or bus current location with the bus icon on the top*
3	*Pull-down bar*	*Return to "Detailed Journeys" interface (5.2)*
4	*Button*	*To make an emergency stop for the serious problem*

	Name	Code	Data Type	Length	Precision	P
1	AccountID	ACCOUNTID	char(7)	7		✓
2	AName	ANAME	varchar(20)	20		
3	AGender	AGENDER	varchar(3)	3		
4	ADoB	ADOB	date			
5	APhoneNumber	APHONENUMB	char(10)	10		
6	AEmail	AEMAIL	varchar(20)	20		

5	*Button*	*Ending the trip if the bus is arrived last destination*

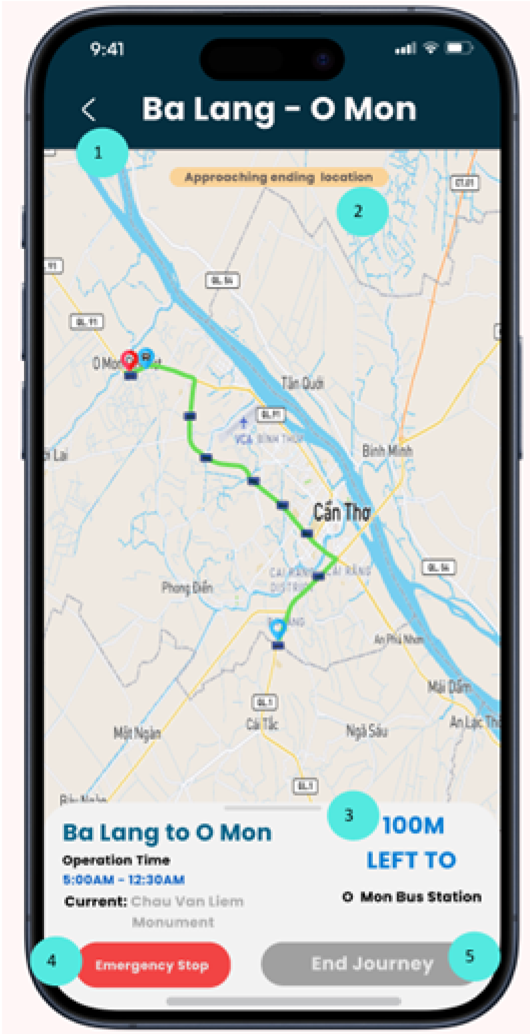
- Data to be used:

No.	**Table name / Data structure**	**Method**			
		Add	**Modify**	**Delete**	**Query**
1	*Driving*				X
2	*Routes*				X
3	*Direction*				X

+ Interface 6.1: Pop-up notification for drivers to know if they are near the destination

■

	Name	Code	Data Type	Length	Precision	P
1	AccountID	ACCOUNTID	char(7)	7		✓
2	AName	ANAME	varchar(20)	20		
3	AGender	AGENDER	varchar(3)	3		
4	ADoB	ADOB	date			
5	APhoneNumber	APHONENUMB	char(10)	10		
6	AEmail	AEMAIL	varchar(20)	20		



- The components of interface 6.1:

No.	Controller type	Note
1	*Type header*	*Return to "Detailed Journeys" interface (5.2)*
2	*Text*	*A notification tells driver that he/she is near to the last location*
2	*Icon*	*Show the last destination as red, first starting location as blue or bus current location with the bus icon on the top*
3	*Pull-down bar*	*Return to "Detailed Journeys" interface (5.2)*

■

	Name	Code	Data Type	Length	Precision	P
1	AccountID	ACCOUNTID	char(7)	7		☑
2	AName	ANAME	varchar(20)	20		☐
3	AGender	AGENDER	varchar(3)	3		☐
4	ADoB	ADOB	date			☐
5	APhoneNumber	APHONENUMB	char(10)	10		☐
6	AEmail	AEMAIL	varchar(20)	20		☐

4	*Button*	*To make an emergency stop for the serious problem*
5	*Button*	*Ending the trip if the bus is arrived last destination*

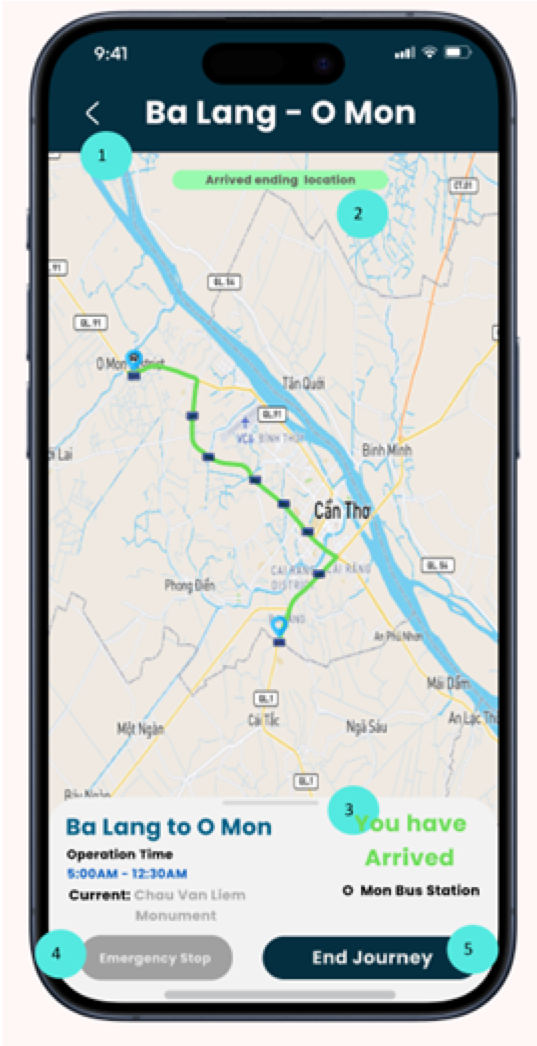
- Data to be used:

No.	Table name / Data structure	Method			
		Add	Modify	Delete	Query
1	*Driving*				X
2	*Routes*				X
3	*Direction*				X

+ Interface 6.2 : Pop-up notification for drivers to know if they are arriving the destination

■

	Name	Code	Data Type	Length	Precision	P
1	AccountID	ACCOUNTID	char(7)	7		☑
2	AName	ANAME	varchar(20)	20		☐
3	AGender	AGENDER	varchar(3)	3		☐
4	ADoB	ADOB	date			☐
5	APhoneNumber	APHONENUMB	char(10)	10		☐
6	AEmail	AEMAIL	varchar(20)	20		☐



- The components of interface 6.2:

No.	Controller type	Note
1	*Type header*	*Return to "Detailed Journeys" interface (5.2)*
2	*Text*	*A notification tells driver that he/she has reached the last location*
2	*Icon*	*Show the last destination as red, first starting location as blue or bus current location with the bus icon on the top*
3	*Pull-down bar*	*Return to "Detailed Journeys" interface (5.2)*

	Name	Code	Data Type	Length	Precision	P
1	AccountID	ACCOUNTID	char(7)	7		☑
2	AName	ANAME	varchar(20)	20		☐
3	AGender	AGENDER	varchar(3)	3		☐
4	ADoB	ADOB	date			☐
5	APhoneNumber	APHONENUMB	char(10)	10		☐
6	AEmail	AEMAIL	varchar(20)	20		☐

4	*Button*	*To make an emergency stop for the serious problem*
5	*Button*	*Ending the trip if the bus is arrived last destination*

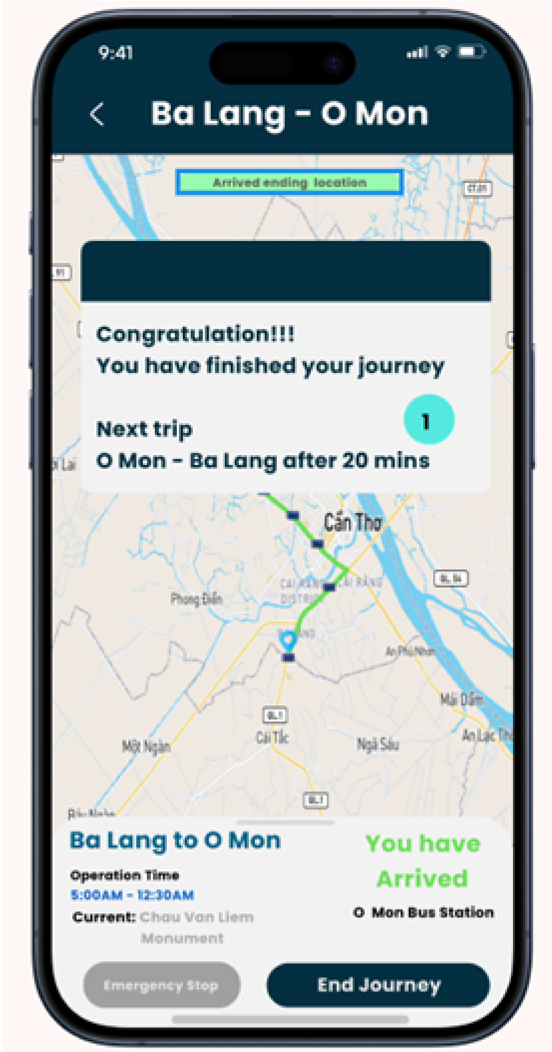
- Data to be used:

No.	**Table name / Data structure**	**Method**			
		Add	**Modify**	**Delete**	**Query**
1	*Driving*				X
2	*Routes*				X
3	*Direction*				X

+ Interface 7: Message for driver when they finished the trip

■

	Name	Code	Data Type	Length	Precision	P
1	AccountID	ACCOUNTID	char(7)	7		✓
2	AName	ANAME	varchar(20)	20		
3	AGender	AGENDER	varchar(3)	3		
4	ADoB	ADOB	date			
5	APhoneNumber	APHONENUMB	char(10)	10		
6	AEmail	AEMAIL	varchar(20)	20		



- The components of interface 7:

No.	Controller type	Default value	Note
1	*Text*	*"Congratulation!!! You have finished your journey Next trip" + reverse route trip + "after 20 mins"*	*A message for driver when he/she finish the trip*

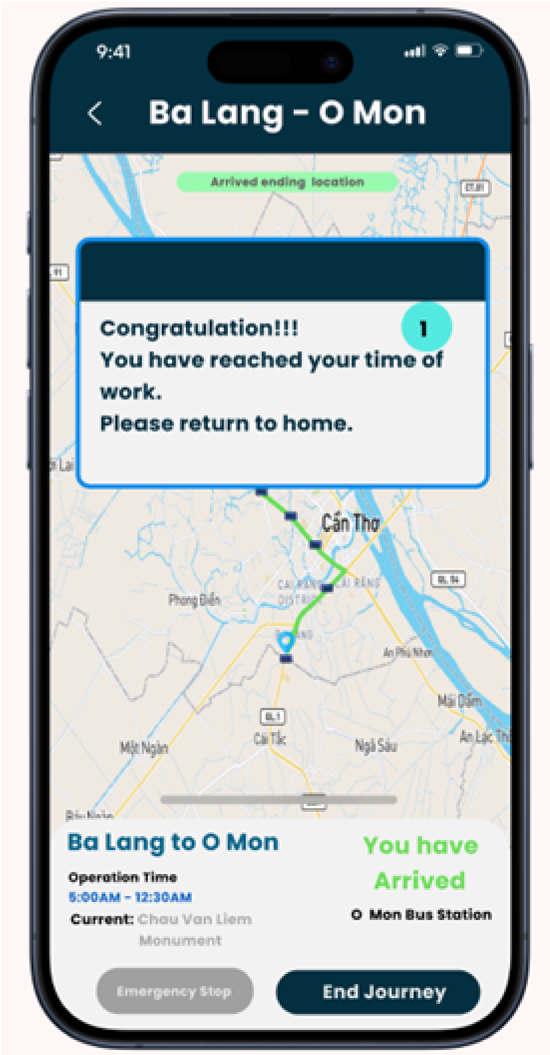
■

	Name	Code	Data Type	Length	Precision	P
1	AccountID	ACCOUNTID	char(7)	7		☑
2	AName	ANAME	varchar(20)	20		☐
3	AGender	AGENDER	varchar(3)	3		☐
4	ADoB	ADOB	date			☐
5	APhoneNumber	APHONENUMB	char(10)	10		☐
6	AEmail	AEMAIL	varchar(20)	20		☐

- Data to be used:

No.	Table name / Data structure	Method			
		Add	Modify	Delete	Query
1	*Driving*				X
2	*Routes*				X
3	*Direction*				X

+ Interface 8:



	Name	Code	Data Type	Length	Precision	P
1	AccountID	ACCOUNTID	char(7)	7		✓
2	AName	ANAME	varchar(20)	20		
3	AGender	AGENDER	varchar(3)	3		
4	ADoB	ADOB	date			
5	APhoneNumber	APHONENUMB	char(10)	10		
6	AEmail	AEMAIL	varchar(20)	20		

- The components of interface 8:

No.	Controller type	Default value	Note
1	Text	"Congratulation!!! You have reached your time of work. Please return to home."	A message for driver when he/she is out of working time

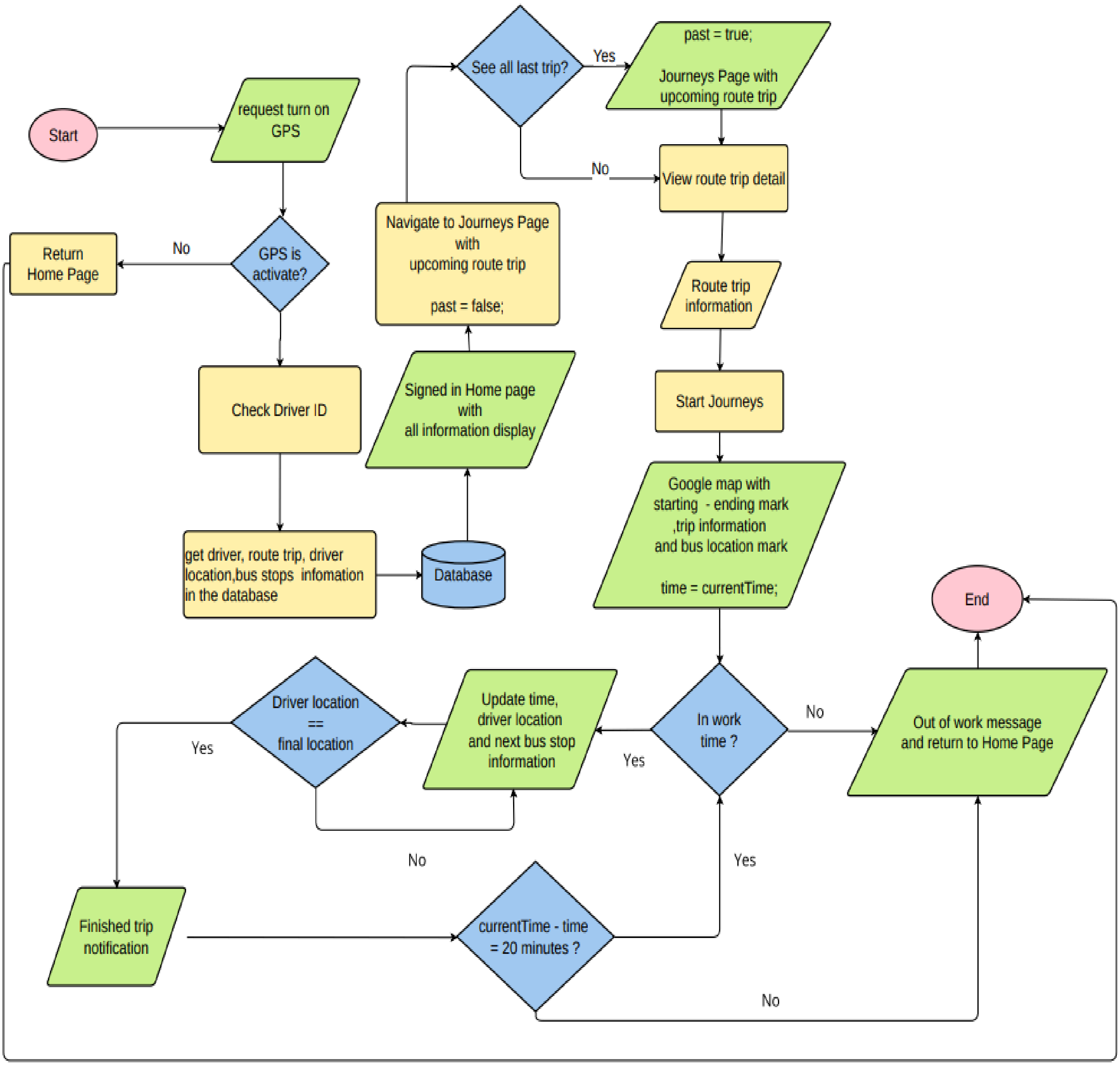
- Data to be used:

No.	Table name / Data structure	Method			
		Add	Modify	Delete	Query
1	Driving				X
2	Routes				X

- *Process*:

■

	Name	Code	Data Type	Length	Precision	P
1	AccountID	ACCOUNTID	char(7)	7		☑
2	AName	ANAME	varchar(20)	20		☐
3	AGender	AGENDER	varchar(3)	3		☐
4	ADoB	ADOB	date			☐
5	APhoneNumber	APHONENUMB	char(10)	10		☐
6	AEmail	AEMAIL	varchar(20)	20		☐

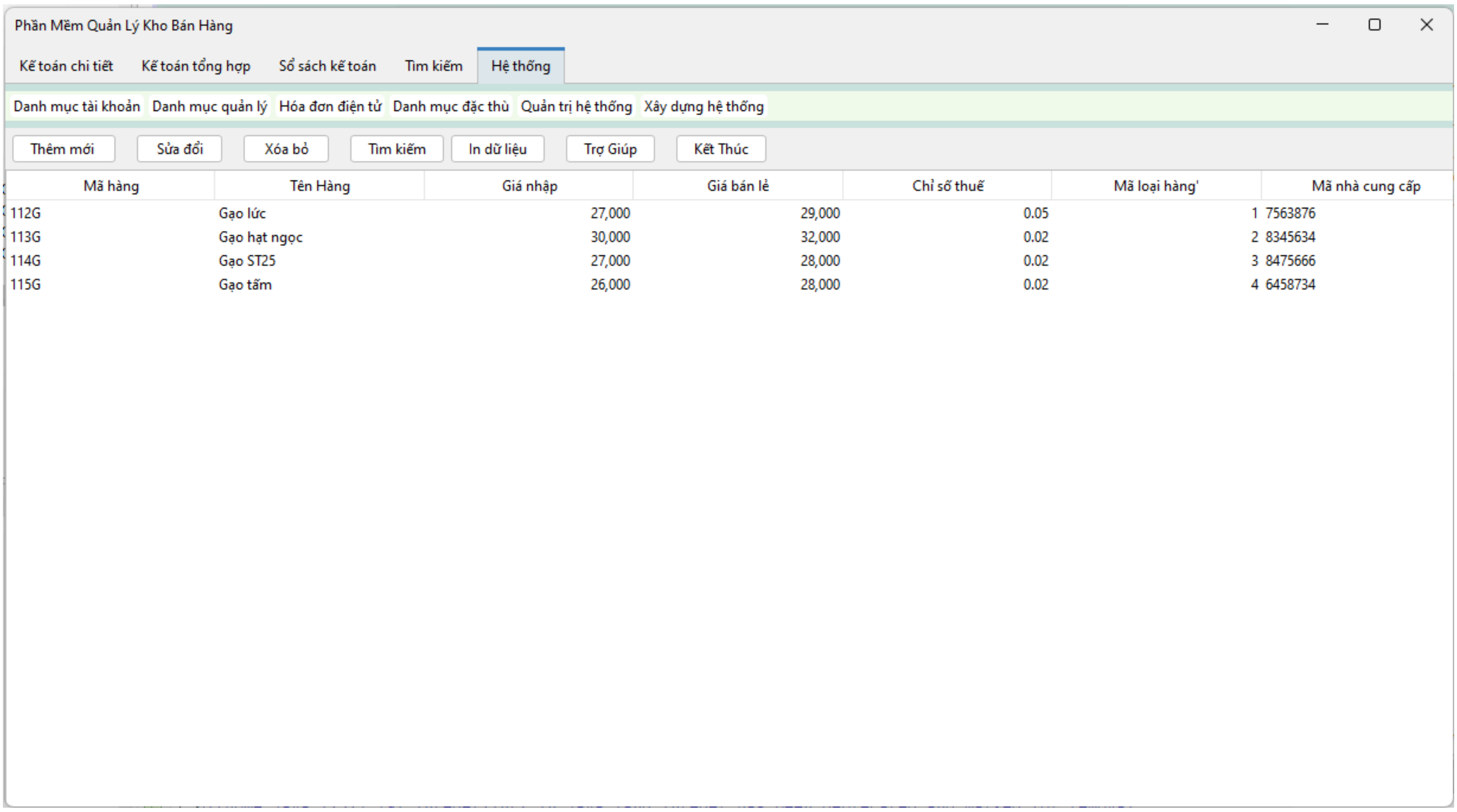


- *Constrains*: Functional requirements - Function 1 (Real Time Bus Routes Tracking)

	Name	Code	Data Type	Length	Precision	P
1	AccountID	ACCOUNTID	char(7)	7		✓
2	AName	ANAME	varchar(20)	20		
3	AGender	AGENDER	varchar(3)	3		
4	ADoB	ADOB	date			
5	APhoneNumber	APHONENUMB	char(10)	10		
6	AEmail	AEMAIL	varchar(20)	20		

2.3.2 Function 2 : Quản lý hàng hóa

- *Purpose*: The platform allows users to choose to register or log in and the methods for users to choose from are phone number, facebook or gmail.
- *Interface*:



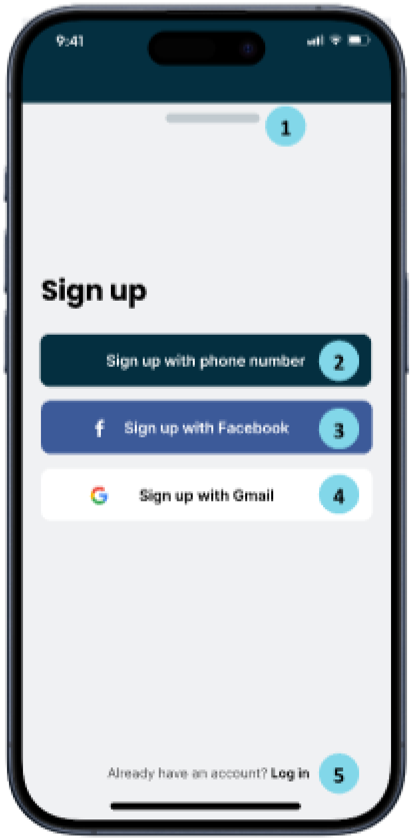
+ Interface 1: Profile page before login

- *The components of interface 1*:

No.	Controller type	Note
1	*Button*	*Go to sign in interface*
2	*Button*	*Go to "Sign up" interface (2)*
3	*Button*	*Go to profile page*

	Name	Code	Data Type	Length	Precision	P
1	AccountID	ACCOUNTID	char(7)	7		✓
2	AName	ANAME	varchar(20)	20		
3	AGender	AGENDER	varchar(3)	3		
4	ADoB	ADOB	date			
5	APhoneNumber	APHONENUMB	char(10)	10		
6	AEmail	AEMAIL	varchar(20)	20		

+ Interface 2: Sign-up Page



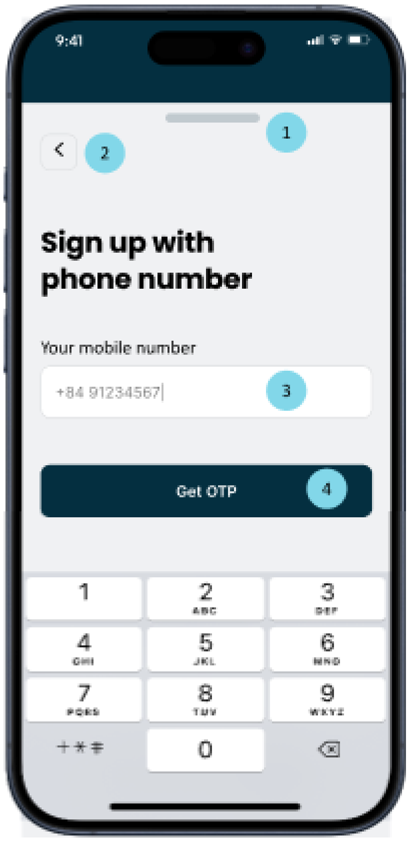
- *The components of interface 2:.*

No.	Controller type	Note
1	*Pull-down bar*	*Return to profile before login interface (1)*
2	*Button*	*Navigating to sign up with phone interface (3.1)*
3	*Button*	*Navigating to "Sign up with Facebook" interface (4.1)*
4	*Button*	*Navigating to "Sign up with Gmail" interface (4.2)*
5	*Button*	*Navigating to login interface*

■

	Name	Code	Data Type	Length	Precision	P
1	AccountID	ACCOUNTID	char(7)	7		✓
2	AName	ANAME	varchar(20)	20		
3	AGender	AGENDER	varchar(3)	3		
4	ADoB	ADOB	date			
5	APhoneNumber	APHONENUMB	char(10)	10		
6	AEmail	AEMAIL	varchar(20)	20		

+ Interface 3.1: Sign up with phone number (Entering phone number and getting OTP)



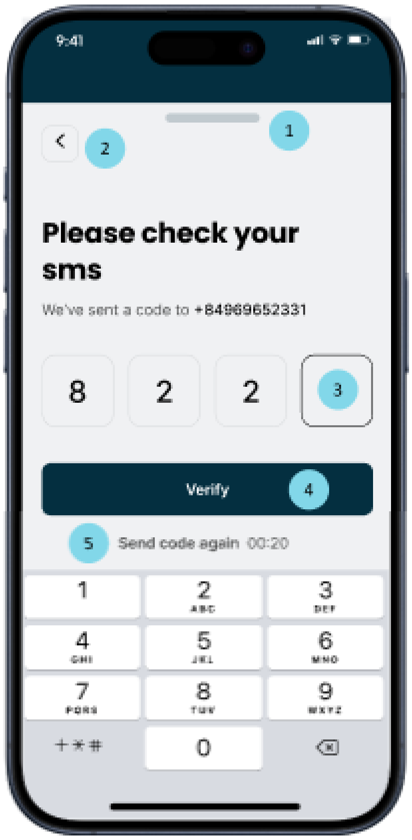
- *The components of interface 3.1*:

No.	Controller type	Note
1	*Pull-down bar*	*Return to profile before login interface (1)*
2	*Type header*	*Navigate to "Sign up" interface (2)*
3	*Textbox*	*Enter user phone number*
4	*Button*	*Receiving OTP code from the server via SMS text*

+ Interface 3.2: Sign up with phone number (Validating phone number)

■

	Name	Code	Data Type	Length	Precision	P
1	AccountID	ACCOUNTID	char(7)	7		☑
2	AName	ANAME	varchar(20)	20		☐
3	AGender	AGENDER	varchar(3)	3		☐
4	ADoB	ADOB	date			☐
5	APhoneNumber	APHONENUMB	char(10)	10		☐
6	AEmail	AEMAIL	varchar(20)	20		☐



- *The components of interface 3.2*:

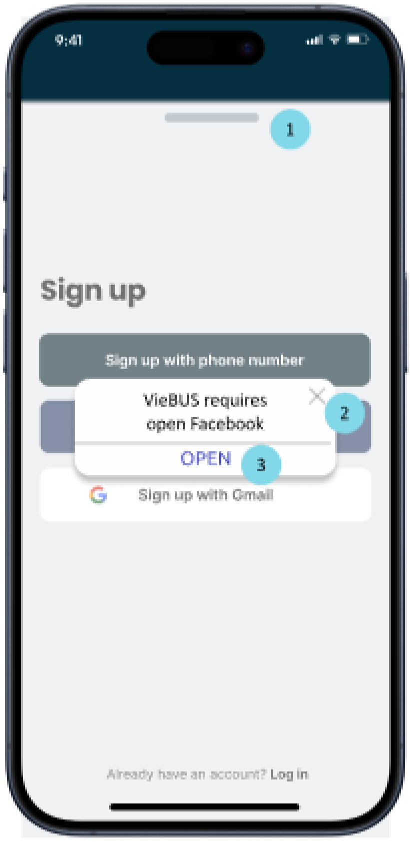
No.	Controller type	Note
1	*Pull-down bar*	*Return to profile before login interface (1)*
2	*Type header*	*Navigate to "Sign up with phone number" interface (3.1)*
3	*Pin input*	*Enter the code*
4	*Button*	*Verifying and creating an account by phone number*
5	*Button*	*Resending the SMS text code to the phone*

- *Data to be used*:

	Name	Code	Data Type	Length	Precision	P
1	AccountID	ACCOUNTID	char(7)	7		✓
2	AName	ANAME	varchar(20)	20		
3	AGender	AGENDER	varchar(3)	3		
4	ADoB	ADOB	date			
5	APhoneNumber	APHONENUMB	char(10)	10		
6	AEmail	AEMAIL	varchar(20)	20		

No.	Table name / Data structure	Method			
		Add	**Modify**	**Delete**	**Query**
1	*Accounts*	X			

+ Interface 4.1: Sign up with Facebook



- *The components of interface 4.1*:

No.	Controller type	Note
1	*Pull-down bar*	*Return to profile before login interface (1)*
2	*Type header*	*Turn off the notification*
3	*Button*	*Connecting with facebook app or navigate to*

■

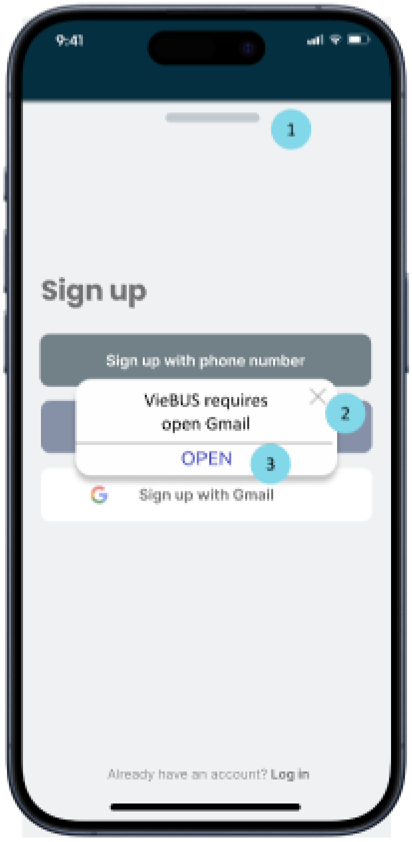
	Name	Code	Data Type	Length	Precision	P
1	AccountID	ACCOUNTID	char(7)	7		✓
2	AName	ANAME	varchar(20)	20		
3	AGender	AGENDER	varchar(3)	3		
4	ADoB	ADOB	date			
5	APhoneNumber	APHONENUMB	char(10)	10		
6	AEmail	AEMAIL	varchar(20)	20		

		facebook website to create account

- *Data to be used*:

No.	Table name / Data structure	Method			
		Add	**Modify**	**Delete**	**Query**
1	Accounts	X			

+ Interface 4.2: Sign up with Gmail



- *The components of interface 4.2*:

No.	Controller type	Note
1	*Pull-down bar*	*Return to profile before login interface (1)*

	Name	Code	Data Type	Length	Precision	P
1	AccountID	ACCOUNTID	char(7)	7		☑
2	AName	ANAME	varchar(20)	20		☐
3	AGender	AGENDER	varchar(3)	3		☐
4	ADoB	ADOB	date			☐
5	APhoneNumber	APHONENUMB	char(10)	10		☐
6	AEmail	AEMAIL	varchar(20)	20		☐

2	*Typeheader*	*Turn off the pop-up notification*
3	*Button*	*Connecting with Gmail app or navigate to Gmail website to create account*

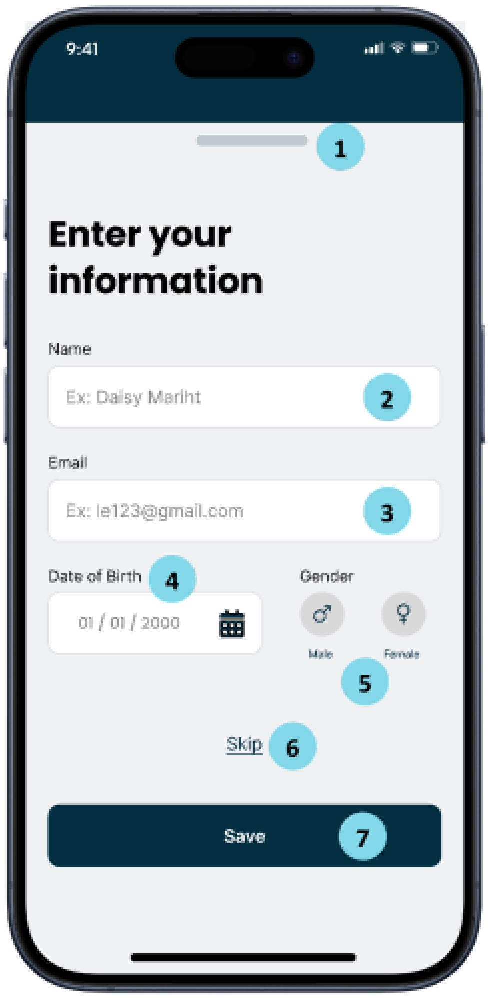
- *Data to be used*:

No.	**Table name / Data structure**	**Method**			
		Add	**Modify**	**Delete**	**Query**
1	*Accounts*	X			

+ Interface 5: Information form

■

	Name	Code	Data Type	Length	Precision	P
1	AccountID	ACCOUNTID	char(7)	7		✓
2	AName	ANAME	varchar(20)	20		
3	AGender	AGENDER	varchar(3)	3		
4	ADoB	ADOB	date			
5	APhoneNumber	APHONENUMB	char(10)	10		
6	AEmail	AEMAIL	varchar(20)	20		



- *The components of interface 5*:

No.	Controller type	Note
1	*Pull-down bar*	*Return to profile before login interface (1)*
2	*Textbox*	*Entering full-name*
3	*Textbox*	*Entering valid email*
4	*Textbox*	*Entering valid date or choosing date from the calendar via calendar icon*

	Name	Code	Data Type	Length	Precision	P
1	AccountID	ACCOUNTID	char(7)	7		✓
2	AName	ANAME	varchar(20)	20		
3	AGender	AGENDER	varchar(3)	3		
4	ADoB	ADOB	date			
5	APhoneNumber	APHONENUMB	char(10)	10		
6	AEmail	AEMAIL	varchar(20)	20		

5	*Button*	*Choosing the gender by clicking on male or female button*
6	*Hyperlink*	*Skipping the entering user information process and using the default value of each input field to create user account*
7	*Button*	*Saving, storing user information in the database and navigating to "User profile" interface (6)*

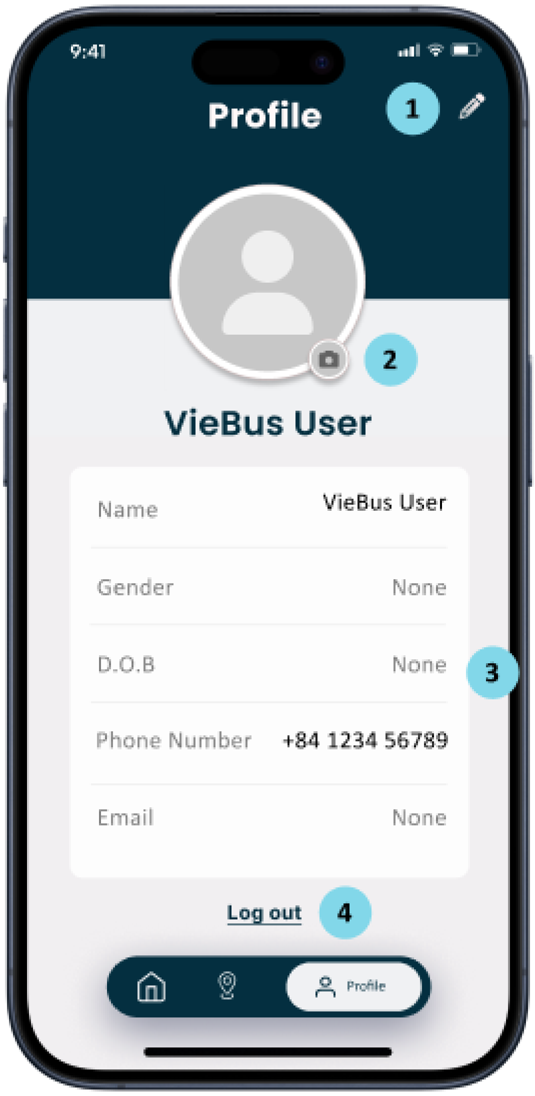
- *Data to be used*:

No.	**Table name / Data structure**	**Method**			
		Add	**Modify**	**Delete**	**Query**
1	*Accounts*	X			

\+ Interface 6: User profile

■

	Name	Code	Data Type	Length	Precision	P
1	AccountID	ACCOUNTID	char(7)	7		☑
2	AName	ANAME	varchar(20)	20		☐
3	AGender	AGENDER	varchar(3)	3		☐
4	ADoB	ADOB	date			☐
5	APhoneNumber	APHONENUMB	char(10)	10		☐
6	AEmail	AEMAIL	varchar(20)	20		☐



- *The components of interface 6*:

No.	Controller type	Default value	Note
1	*Button*		*Navigating to "Information form" interface (5) and update the user information in database*
2	*Button*		*Entering valid email*

	Name	Code	Data Type	Length	Precision	P
1	AccountID	ACCOUNTID	char(7)	7		☑
2	AName	ANAME	varchar(20)	20		☐
3	AGender	AGENDER	varchar(3)	3		☐
4	ADoB	ADOB	date			☐
5	APhoneNumber	APHONENUMB	char(10)	10		☐
6	AEmail	AEMAIL	varchar(20)	20		☐

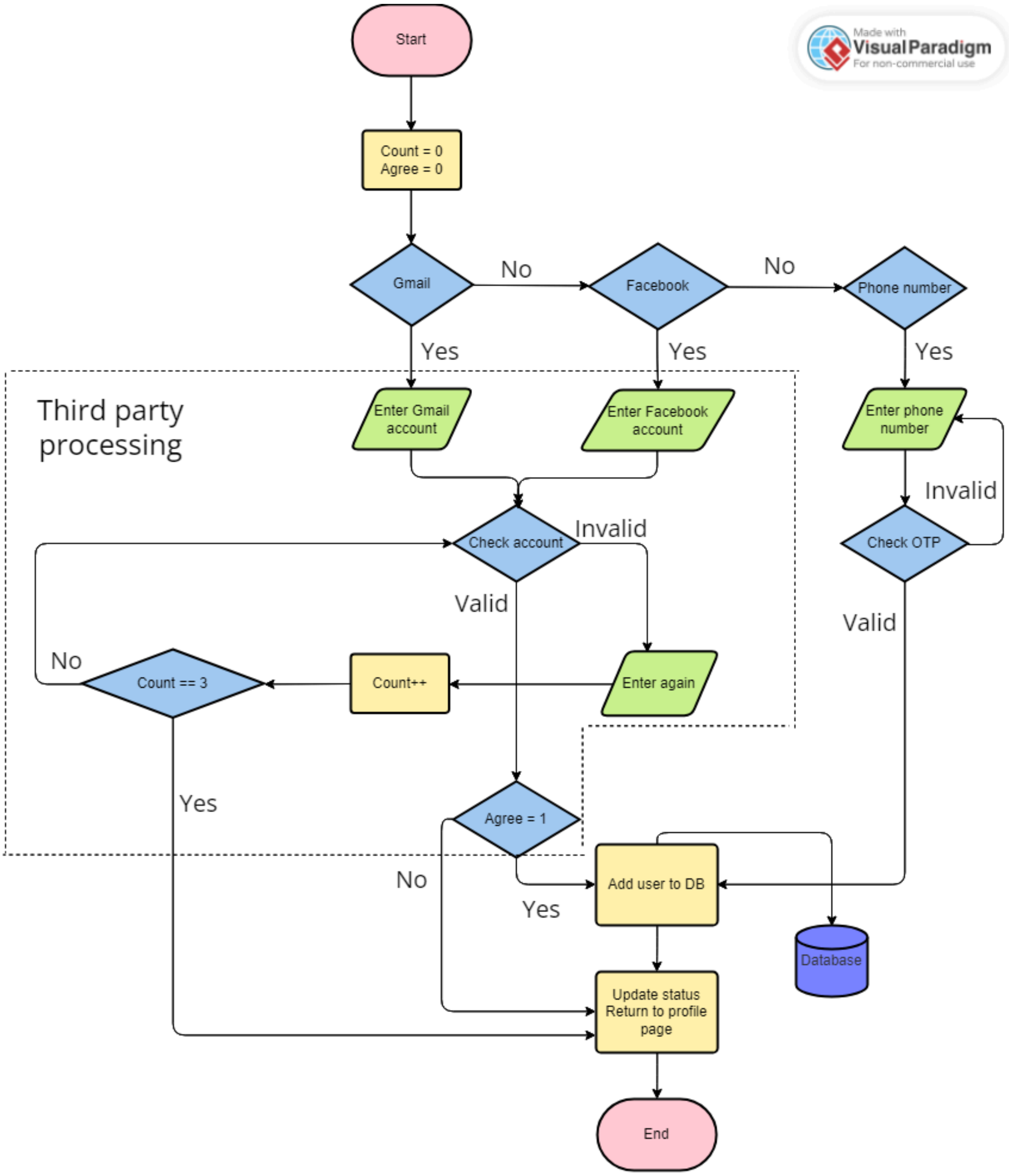
3	*Table*	*Name: VieBus User*	*Displaying a table of user information including full-name, gender, date of birth, phone number, email*
4	*Button*		*Logging out of account Navigating to "Profile page before login" interface (1).*

- *Data to be used*:

No.	**Table name / Data structure**	**Method**			
		Add	**Modify**	**Delete**	**Query**
1	*Accounts*		X		X

- *Process*:

	Name	Code	Data Type	Length	Precision	P
1	AccountID	ACCOUNTID	char(7)	7		✓
2	AName	ANAME	varchar(20)	20		
3	AGender	AGENDER	varchar(3)	3		
4	ADoB	ADOB	date			
5	APhoneNumber	APHONENUMB	char(10)	10		
6	AEmail	AEMAIL	varchar(20)	20		



- *Constrains*: Functional requirements - Function 2 (Account Registration).

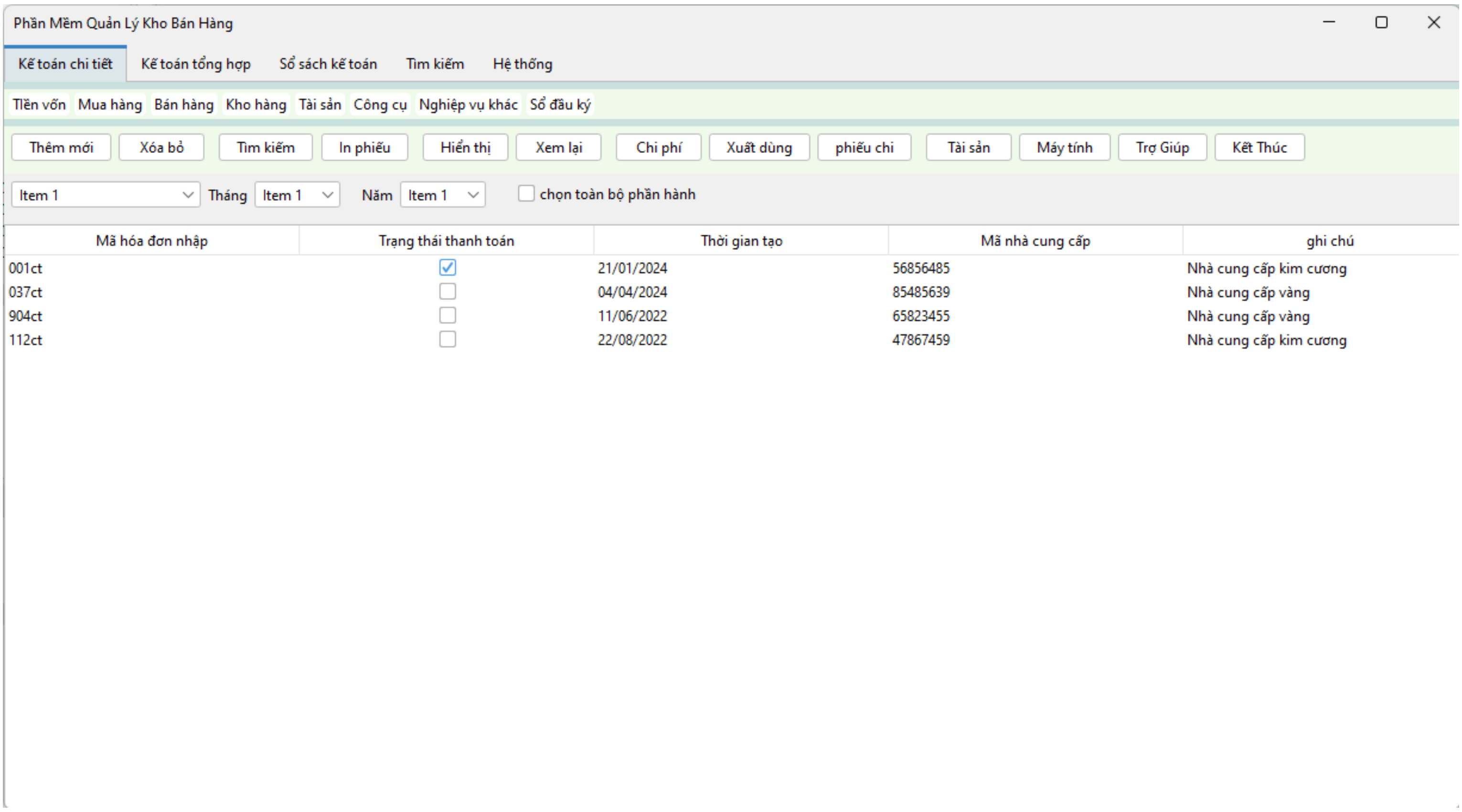
2.3.3 Function 3: Quản lý hóa đơn

- *Purpose:* The platform allows administrators to meticulously review and refine bus route information, facilitating precision in route planning and schedule adjustments. Through systematic editing, administrators can ensure that the

	Name	Code	Data Type	Length	Precision	P
1	AccountID	ACCOUNTID	char(7)	7		✓
2	AName	ANAME	varchar(20)	20		
3	AGender	AGENDER	varchar(3)	3		
4	ADoB	ADOB	date			
5	APhoneNumber	APHONENUMB	char(10)	10		
6	AEmail	AEMAIL	varchar(20)	20		

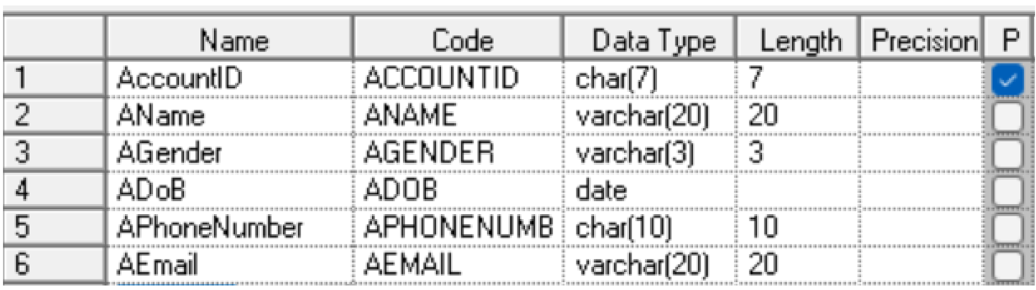
provided data aligns with current transit needs, thereby enhancing operational efficiency and passenger satisfaction.

- *Interface:*
    + Interface 1:



- *The component of interface 1:*

No.	Controller type	Default value	Note
1	*Button*		*Change either location*
2	*Text box*	*Current route number*	

▪ 

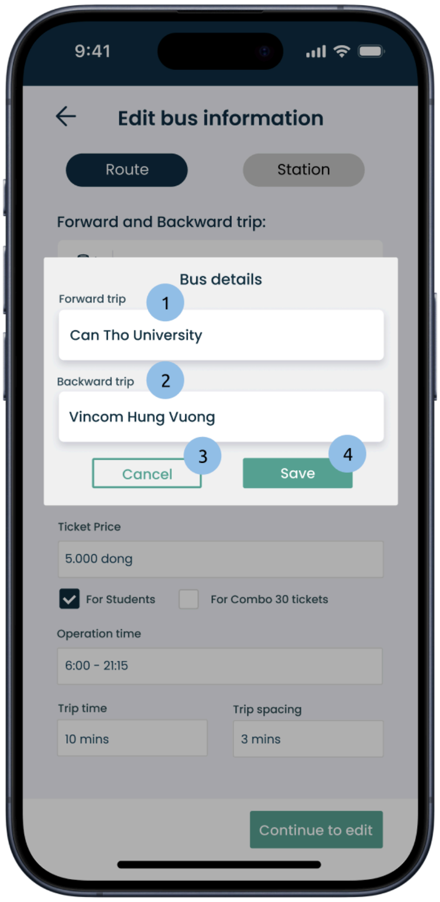
	Name	Code	Data Type	Length	Precision	P
1	AccountID	ACCOUNTID	char(7)	7		☑
2	AName	ANAME	varchar(20)	20		☐
3	AGender	AGENDER	varchar(3)	3		☐
4	ADoB	ADOB	date			☐
5	APhoneNumber	APHONENUMB	char(10)	10		☐
6	AEmail	AEMAIL	varchar(20)	20		☐

3	*Combo box*	*Current number of seats*	*Set: current number of seats. Values of "Number of seats" is 40,45,50,60*
4	*Text box*	*Current route name*	
5	*Text box*	*Current ticket price*	
6	*Check box*		*If the box is checked, the pricing for each category of ticket shall be as follows:* *Students = Ticket price*60%* *Combo = Ticket price*30* 75%*
7	*Text box*	*Current operation time*	
8	*Text box*	*Current trip time*	
9	*Text box*	*Current trip spacing*	
10	*Button*		*Navigate to interface 3*

- Data to be used:

No.	Table name / Data structure	Method			
		Add	Modify	Delete	Query
1	*Routes*		X		

	Name	Code	Data Type	Length	Precision	P
1	AccountID	ACCOUNTID	char(7)	7		✓
2	AName	ANAME	varchar(20)	20		
3	AGender	AGENDER	varchar(3)	3		
4	ADoB	ADOB	date			
5	APhoneNumber	APHONENUMB	char(10)	10		
6	AEmail	AEMAIL	varchar(20)	20		



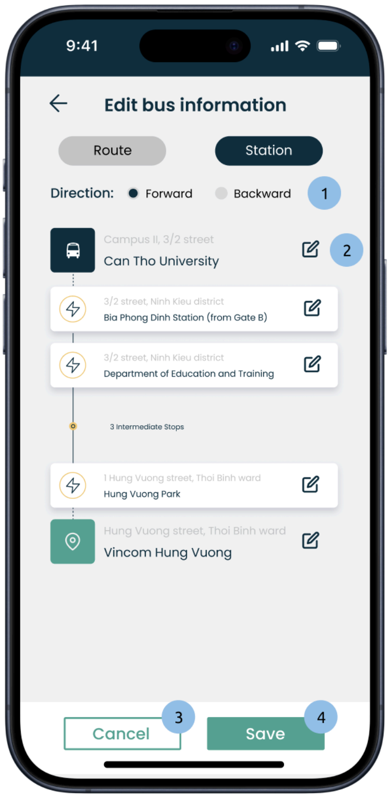
- The component of interface 2:

No.	Controller type	Default value	Note
1	*Text box*	*Current location*	*Set: current location*
2	*Text box*	*Current location*	*Set: current location*
3	*Button*		*Refresh all the text boxes to the default values*
4	*Button*		*Save and return to interface 1*

- Data to be used:

	Name	Code	Data Type	Length	Precision	P
1	AccountID	ACCOUNTID	char(7)	7		☑
2	AName	ANAME	varchar(20)	20		☐
3	AGender	AGENDER	varchar(3)	3		☐
4	ADoB	ADOB	date			☐
5	APhoneNumber	APHONENUMB	char(10)	10		☐
6	AEmail	AEMAIL	varchar(20)	20		☐

No.	Table name / Data structure	Method			
		Add	Modify	Delete	Query
1	*Routes*		X		



- The component of interface 3:

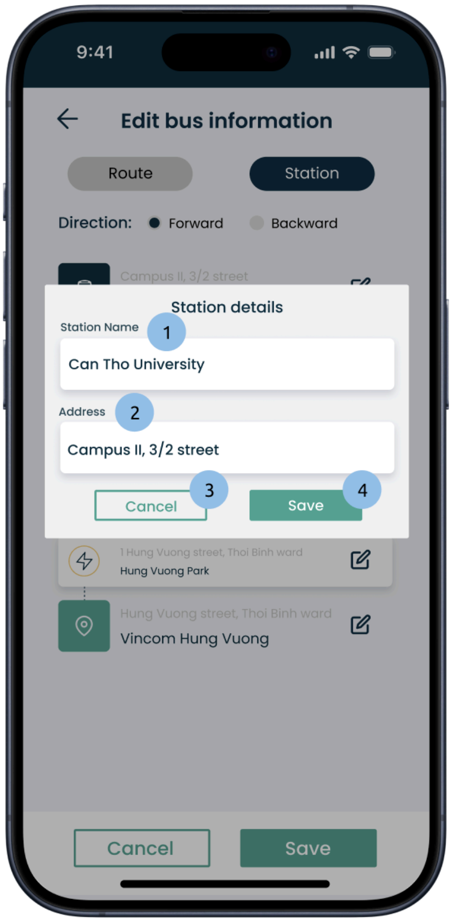
No.	Controller type	Default value	Note
1	*Check box*	*Check "forward" box*	
2	*Button*		*Allow editing details of each bus route*
3	*Button*		*Refresh all the changes*
4	*Button*		*Save and return to the main interface*

■

	Name	Code	Data Type	Length	Precision	P
1	AccountID	ACCOUNTID	char(7)	7		✓
2	AName	ANAME	varchar(20)	20		
3	AGender	AGENDER	varchar(3)	3		
4	ADoB	ADOB	date			
5	APhoneNumber	APHONENUMB	char(10)	10		
6	AEmail	AEMAIL	varchar(20)	20		

- Data to be used:

No.	Table name / Data structure	Method			
		Add	Modify	Delete	Query
1	*Stops*				X
2	*Direction*				X



- The component of interface 4:

No.	Controller type	Default value	Note
1	*Text box*	*Current location*	*Set: current location*
2	*Text box*	*Current address*	*Set: current address*

	Name	Code	Data Type	Length	Precision	P
1	AccountID	ACCOUNTID	char(7)	7		☑
2	AName	ANAME	varchar(20)	20		☐
3	AGender	AGENDER	varchar(3)	3		☐
4	ADoB	ADOB	date			☐
5	APhoneNumber	APHONENUMB	char(10)	10		☐
6	AEmail	AEMAIL	varchar(20)	20		☐

3	*Button*		*Refresh all the text boxes to the default values*
4	*Button*		*Save and return to interface 3*

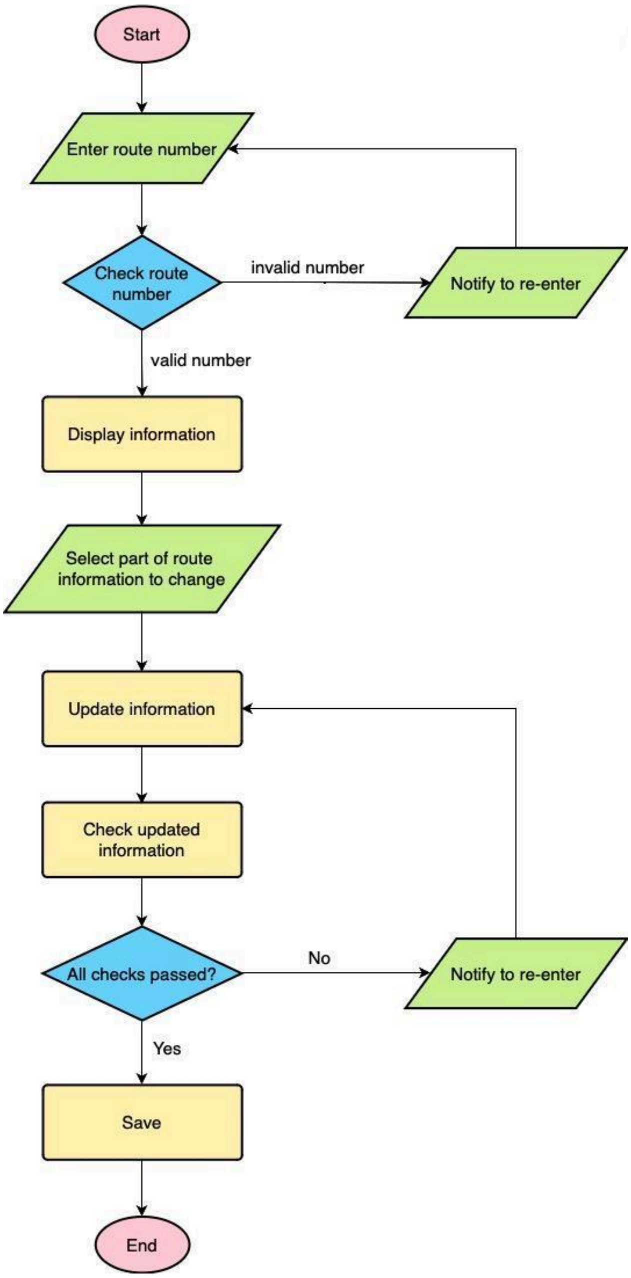
- Data to be used:

No.	**Table name / Data structure**	**Method**			
		Add	**Modify**	**Delete**	**Query**
1	*Stops*		X		

- Process:

■

	Name	Code	Data Type	Length	Precision	P
1	AccountID	ACCOUNTID	char(7)	7		☑
2	AName	ANAME	varchar(20)	20		☐
3	AGender	AGENDER	varchar(3)	3		☐
4	ADoB	ADOB	date			☐
5	APhoneNumber	APHONENUMB	char(10)	10		☐
6	AEmail	AEMAIL	varchar(20)	20		☐

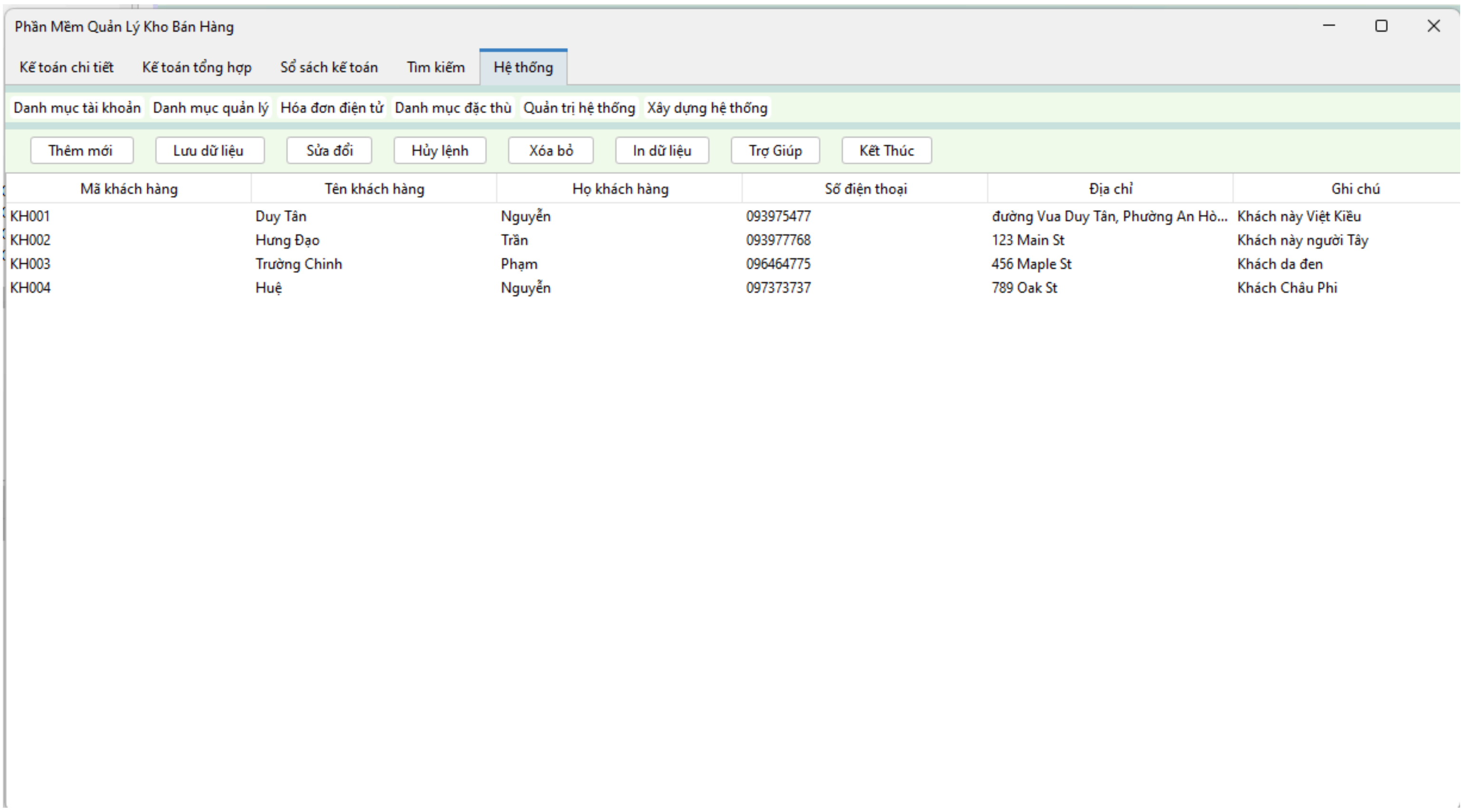


- *Constraints:* Functional Requirement - Function 3 (Bus Routes Editing)

2.3.4 Function 4: Quản lý khách hàng

- *Purpose*: The platform allows users to search for suitable routes between their current location and a destination using input methods such as keyboard input, GPS, or selecting a location on the map.
- *Interface*:
  + *Interface 1 (Bus Route Searching):*

	Name	Code	Data Type	Length	Precision	P
1	AccountID	ACCOUNTID	char(7)	7		✓
2	AName	ANAME	varchar(20)	20		
3	AGender	AGENDER	varchar(3)	3		
4	ADoB	ADOB	date			
5	APhoneNumber	APHONENUMB	char(10)	10		
6	AEmail	AEMAIL	varchar(20)	20		



- *The components of interface 1*:

No.	Controller type	Note
1	*Button*	*Departure points. Clicking will navigate to interface 2*
2	*Button*	*Destination points. Clicking will navigate to interface 2*
3	*Button*	*Search confirmation*
4	*Button*	*Choose from the available popular routes.*

■

	Name	Code	Data Type	Length	Precision	P
1	AccountID	ACCOUNTID	char(7)	7		☑
2	AName	ANAME	varchar(20)	20		☐
3	AGender	AGENDER	varchar(3)	3		☐
4	ADoB	ADOB	date			☐
5	APhoneNumber	APHONENUMB	char(10)	10		☐
6	AEmail	AEMAIL	varchar(20)	20		☐

5	*Button*	*The route has been found.*
6	*Button*	*Return to the homepage.*
7	*Button*	*Announcement*

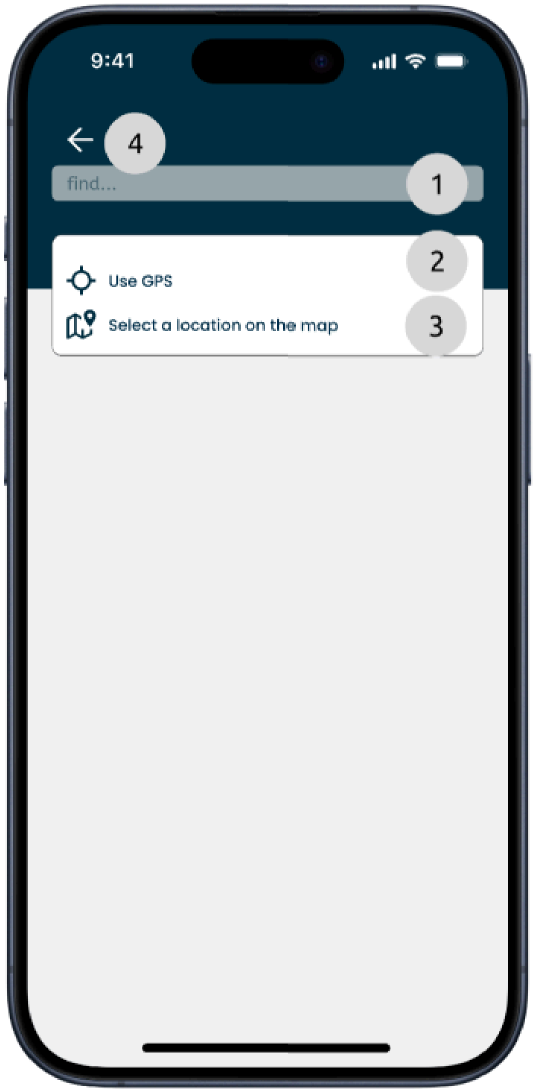
- Data to be used:

No.	Table name / Data structure	Method			
		Add	Modify	Delete	Query
1	*Routes*				x

\+ *Interface 2 (Search by typing on the keyboard or select other method):*

■

	Name	Code	Data Type	Length	Precision	P
1	AccountID	ACCOUNTID	char(7)	7		✓
2	AName	ANAME	varchar(20)	20		
3	AGender	AGENDER	varchar(3)	3		
4	ADoB	ADOB	date			
5	APhoneNumber	APHONENUMB	char(10)	10		
6	AEmail	AEMAIL	varchar(20)	20		



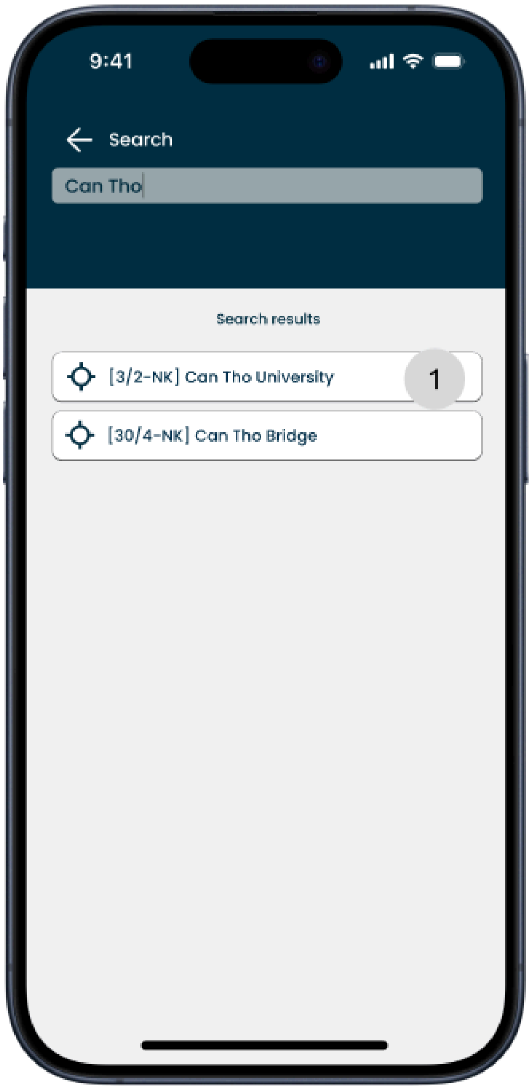
- *The components of interface 2*:

No.	Controller type	Note
1	*Box text*	*Enter locations.*
2	*Button*	*GPS search method*
3	*Button*	*Map selection method*
4	*Button*	*Return to interface 1*

\+ *Interface 3 (Search by typing on the keyboard):*

■

	Name	Code	Data Type	Length	Precision	P
1	AccountID	ACCOUNTID	char(7)	7		☑
2	AName	ANAME	varchar(20)	20		☐
3	AGender	AGENDER	varchar(3)	3		☐
4	ADoB	ADOB	date			☐
5	APhoneNumber	APHONENUMB	char(10)	10		☐
6	AEmail	AEMAIL	varchar(20)	20		☐



- *The components of interface 3*:

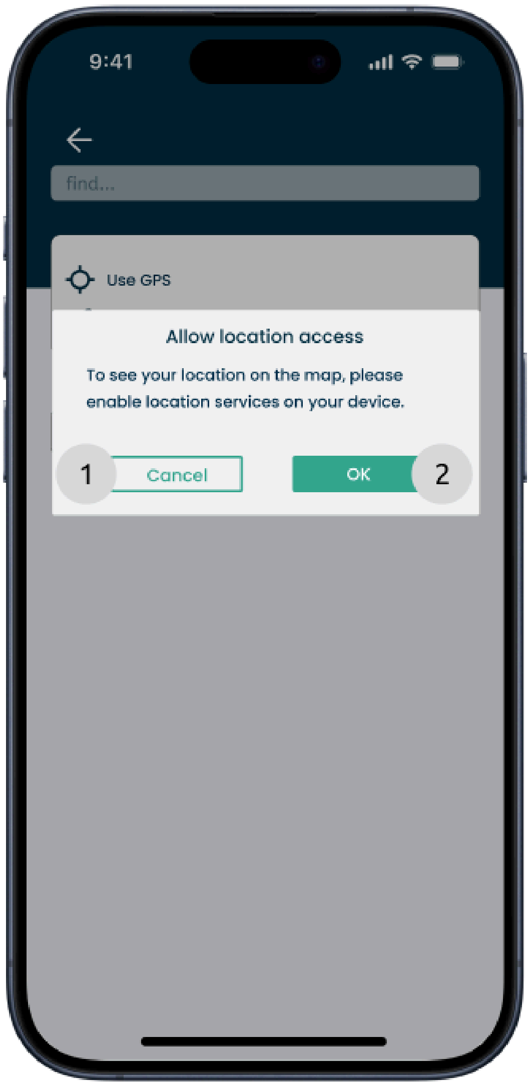
No.	Controller type	Note
1	*Button*	*Appropriate result*

- *Data to be used*:

No.	Table name / Data structure	Method			
		Add	**Modify**	**Delete**	**Query**
1	Stops				x

	Name	Code	Data Type	Length	Precision	P
1	AccountID	ACCOUNTID	char(7)	7		☑
2	AName	ANAME	varchar(20)	20		☐
3	AGender	AGENDER	varchar(3)	3		☐
4	ADoB	ADOB	date			☐
5	APhoneNumber	APHONENUMB	char(10)	10		☐
6	AEmail	AEMAIL	varchar(20)	20		☐

+ *Interface 4 (Searching using GPS):*



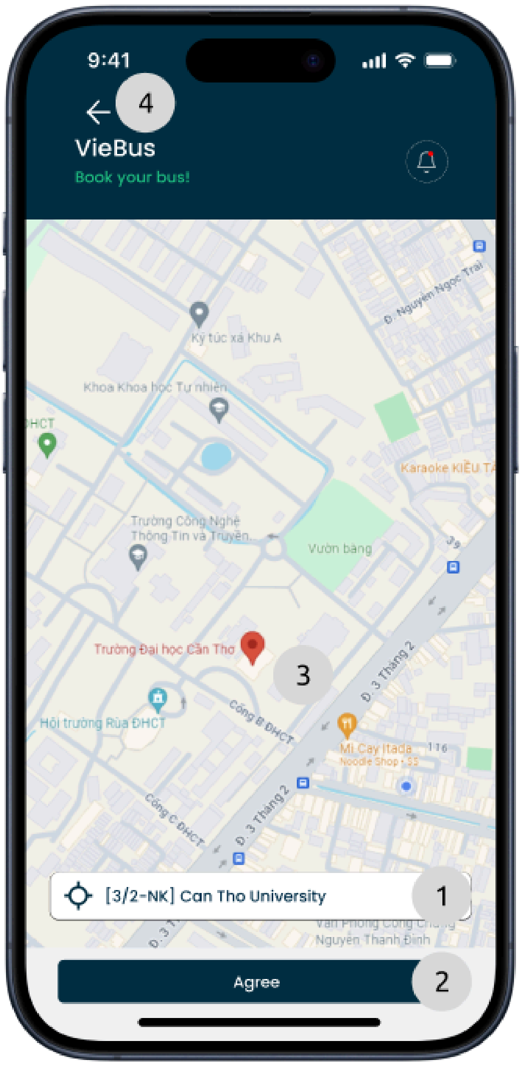
- *The components of interface 4*:

No.	Controller type	Note
1	*Button*	Access denied and return to the interface 2
2	*Button*	*Agree to access location.*

+ *Interface 5 (Searching by selecting on the map):*

■

	Name	Code	Data Type	Length	Precision	P
1	AccountID	ACCOUNTID	char(7)	7		✓
2	AName	ANAME	varchar(20)	20		
3	AGender	AGENDER	varchar(3)	3		
4	ADoB	ADOB	date			
5	APhoneNumber	APHONENUMB	char(10)	10		
6	AEmail	AEMAIL	varchar(20)	20		



- *The components of interface 5*:

No.	Controller type	Note
1	*Typeahead*	The location corresponds to the position of the red pin.
2	*Button*	*Agree on selecting that location.*
3	*Label*	Pin the red pin in the center of the screen.
4	*Button*	*Return to the interface 2*

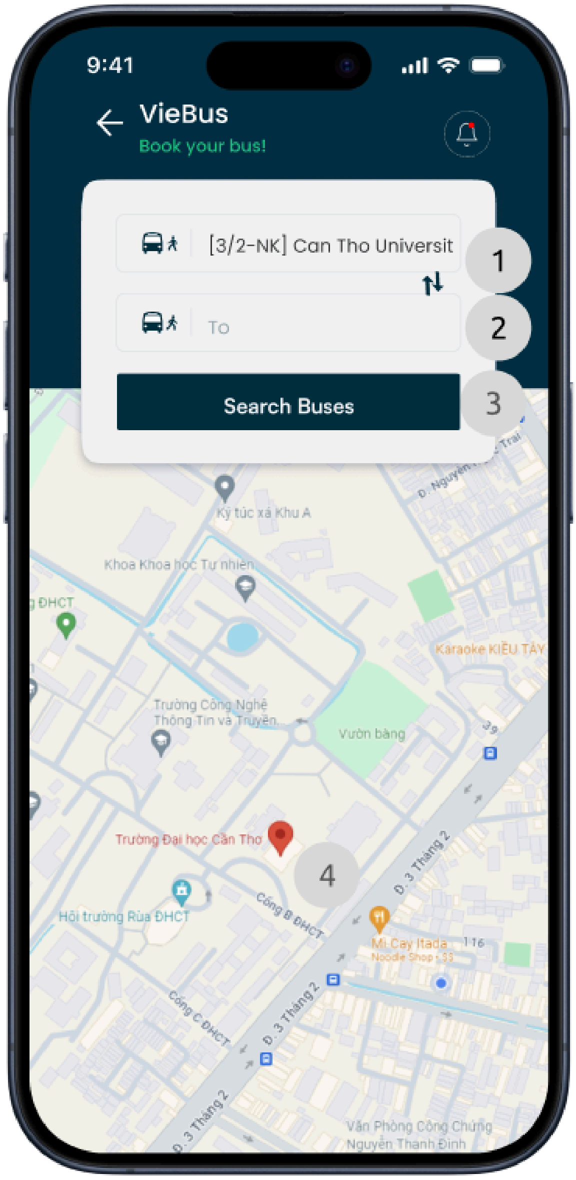
- *Data to be used*:

■

	Name	Code	Data Type	Length	Precision	P
1	AccountID	ACCOUNTID	char(7)	7		✓
2	AName	ANAME	varchar(20)	20		
3	AGender	AGENDER	varchar(3)	3		
4	ADoB	ADOB	date			
5	APhoneNumber	APHONENUMB	char(10)	10		
6	AEmail	AEMAIL	varchar(20)	20		

No.	Table name / Data structure	Method			
		Add	Modify	Delete	Query
1	*Stops*				x

+ *Interface 6 (Found the starting position):*



- *The components of interface 6*:

No.	Controller type	Note

■

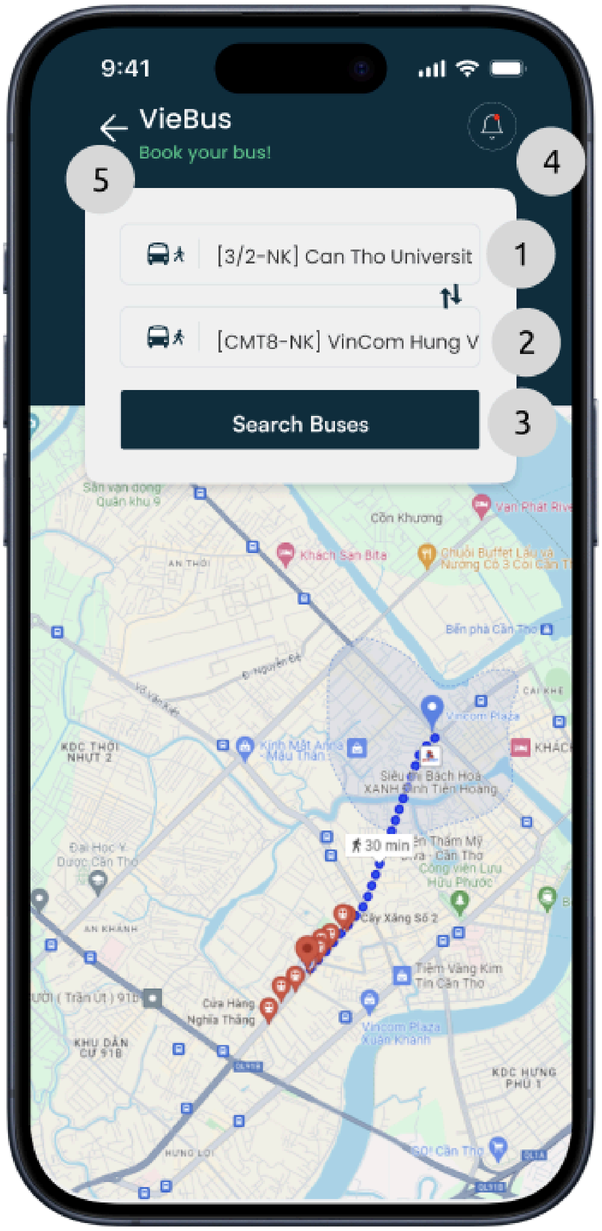
	Name	Code	Data Type	Length	Precision	P
1	AccountID	ACCOUNTID	char(7)	7		✓
2	AName	ANAME	varchar(20)	20		
3	AGender	AGENDER	varchar(3)	3		
4	ADoB	ADOB	date			
5	APhoneNumber	APHONENUMB	char(10)	10		
6	AEmail	AEMAIL	varchar(20)	20		

1	*Button*	*Departure points.Clicking will navigate to interface 2*
2	*Button*	*Destination points. Clicking will navigate to interface 2*
3	*Button*	*Search confirmation*
4	*Label*	*Location being searched*

\+ *Interface 7 (Appropriate route):*

■

	Name	Code	Data Type	Length	Precision	P
1	AccountID	ACCOUNTID	char(7)	7		☑
2	AName	ANAME	varchar(20)	20		☐
3	AGender	AGENDER	varchar(3)	3		☐
4	ADoB	ADOB	date			☐
5	APhoneNumber	APHONENUMB	char(10)	10		☐
6	AEmail	AEMAIL	varchar(20)	20		☐



- *The components of interface 6*:

No.	Controller type	Note
1	*Box text*	*Departure points.Clicking will navigate to interface 2*
2	*Box text*	*Destination points.Clicking will navigate to interface 2*
3	*Button*	*Search confirmation*

▪

	Name	Code	Data Type	Length	Precision	P
1	AccountID	ACCOUNTID	char(7)	7		☑
2	AName	ANAME	varchar(20)	20		☐
3	AGender	AGENDER	varchar(3)	3		☐
4	ADoB	ADOB	date			☐
5	APhoneNumber	APHONENUMB	char(10)	10		☐
6	AEmail	AEMAIL	varchar(20)	20		☐

4	*Button*	*Announcement*
5	*Button*	*Return to the homepage.*

- *Data to be used*:

No.	**Table name / Data structure**	**Method**			
		Add	**Modify**	**Delete**	**Query**
1	*Stop*				X
2	*Routes*				X

\+ *Interface 8 (List of suitable routes):*

■

	Name	Code	Data Type	Length	Precision	P
1	AccountID	ACCOUNTID	char(7)	7		☑
2	AName	ANAME	varchar(20)	20		☐
3	AGender	AGENDER	varchar(3)	3		☐
4	ADoB	ADOB	date			☐
5	APhoneNumber	APHONENUMB	char(10)	10		☐
6	AEmail	AEMAIL	varchar(20)	20		☐

- *The components of interface 7*:

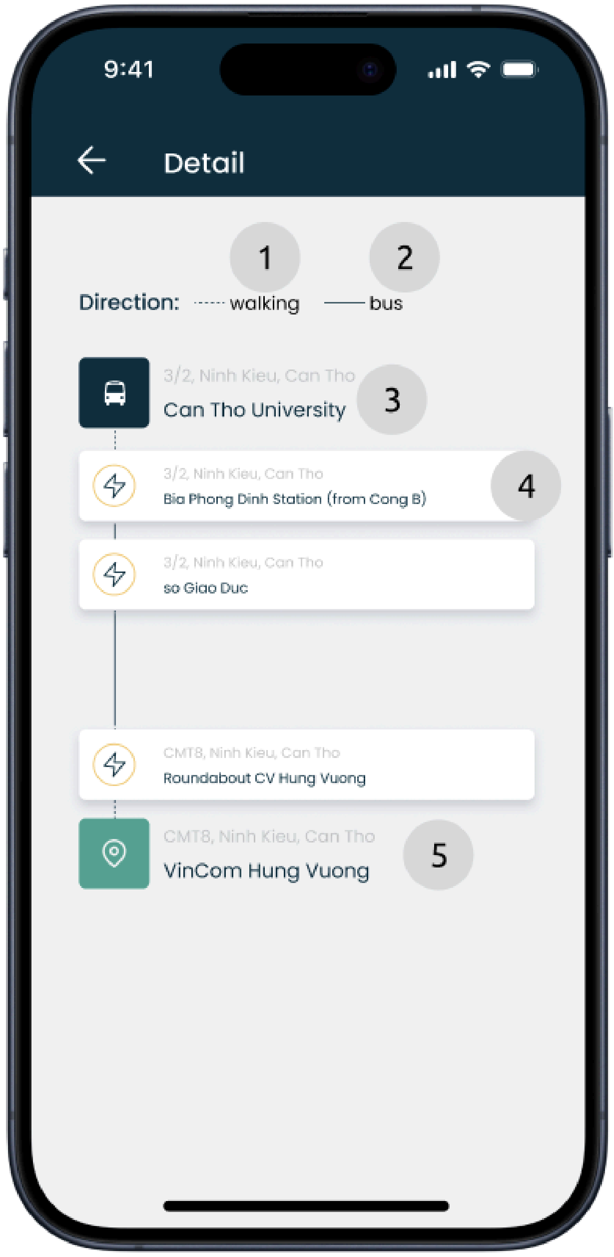
No.	Controller type	Note
1	*Button*	First route
2	*Button*	Second route
3	*Button*	*Refresh result*

	Name	Code	Data Type	Length	Precision	P
1	AccountID	ACCOUNTID	char(7)	7		✓
2	AName	ANAME	varchar(20)	20		
3	AGender	AGENDER	varchar(3)	3		
4	ADoB	ADOB	date			
5	APhoneNumber	APHONENUMB	char(10)	10		
6	AEmail	AEMAIL	varchar(20)	20		

- *Data to be used*:

No.	Table name / Data structure	Method			
		Add	Modify	Delete	Query
1	Routes				x

\+ *Interface 9 (Sequence of movement steps):*



- *The components of interface 9*:

	Name	Code	Data Type	Length	Precision	P
1	AccountID	ACCOUNTID	char(7)	7		✓
2	AName	ANAME	varchar(20)	20		
3	AGender	AGENDER	varchar(3)	3		
4	ADoB	ADOB	date			
5	APhoneNumber	APHONENUMB	char(10)	10		
6	AEmail	AEMAIL	varchar(20)	20		

No.	Controller type	Note
1	*Label*	Walking stage
2	*Label*	bus transportation stage
3	*Label*	*Starting point*
4	*Label*	*series of location landmarks*
5	*Label*	*destination*

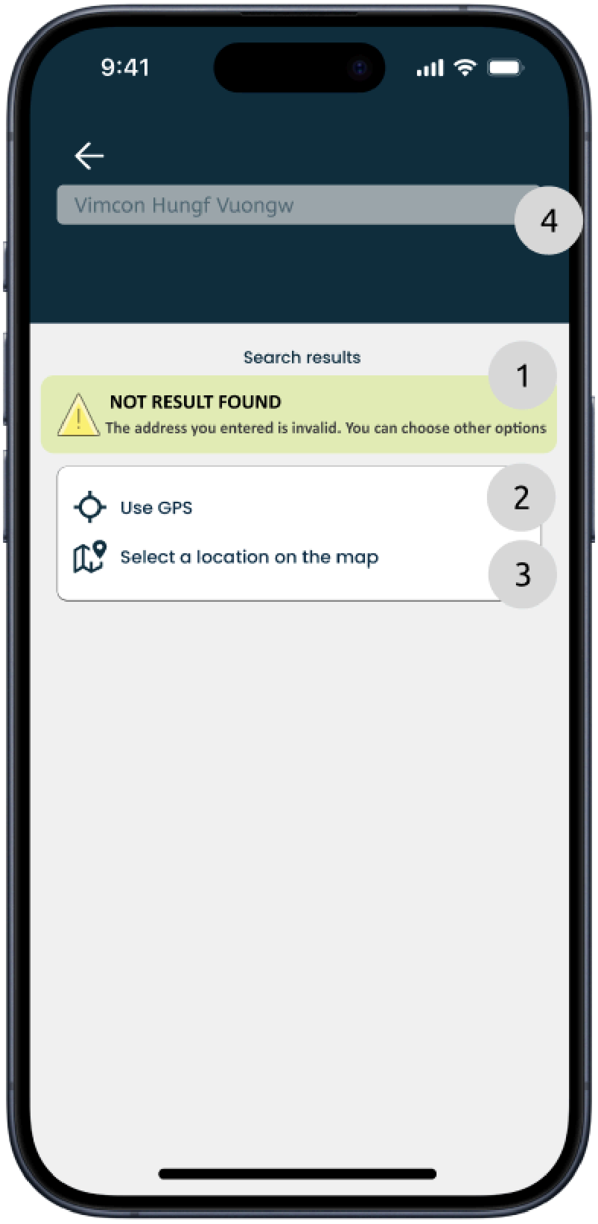
- *Data to be used*:

No.	Table name / Data structure	Method			
		Add	**Modify**	**Delete**	**Query**
1	*Stop*				X
2	*Routes*				X
3	*Direction*				X

+ *Interface 10 (Unable to find a suitable location):*

■

	Name	Code	Data Type	Length	Precision	P
1	AccountID	ACCOUNTID	char(7)	7		✓
2	AName	ANAME	varchar(20)	20		
3	AGender	AGENDER	varchar(3)	3		
4	ADoB	ADOB	date			
5	APhoneNumber	APHONENUMB	char(10)	10		
6	AEmail	AEMAIL	varchar(20)	20		



- *The components of interface 10*:

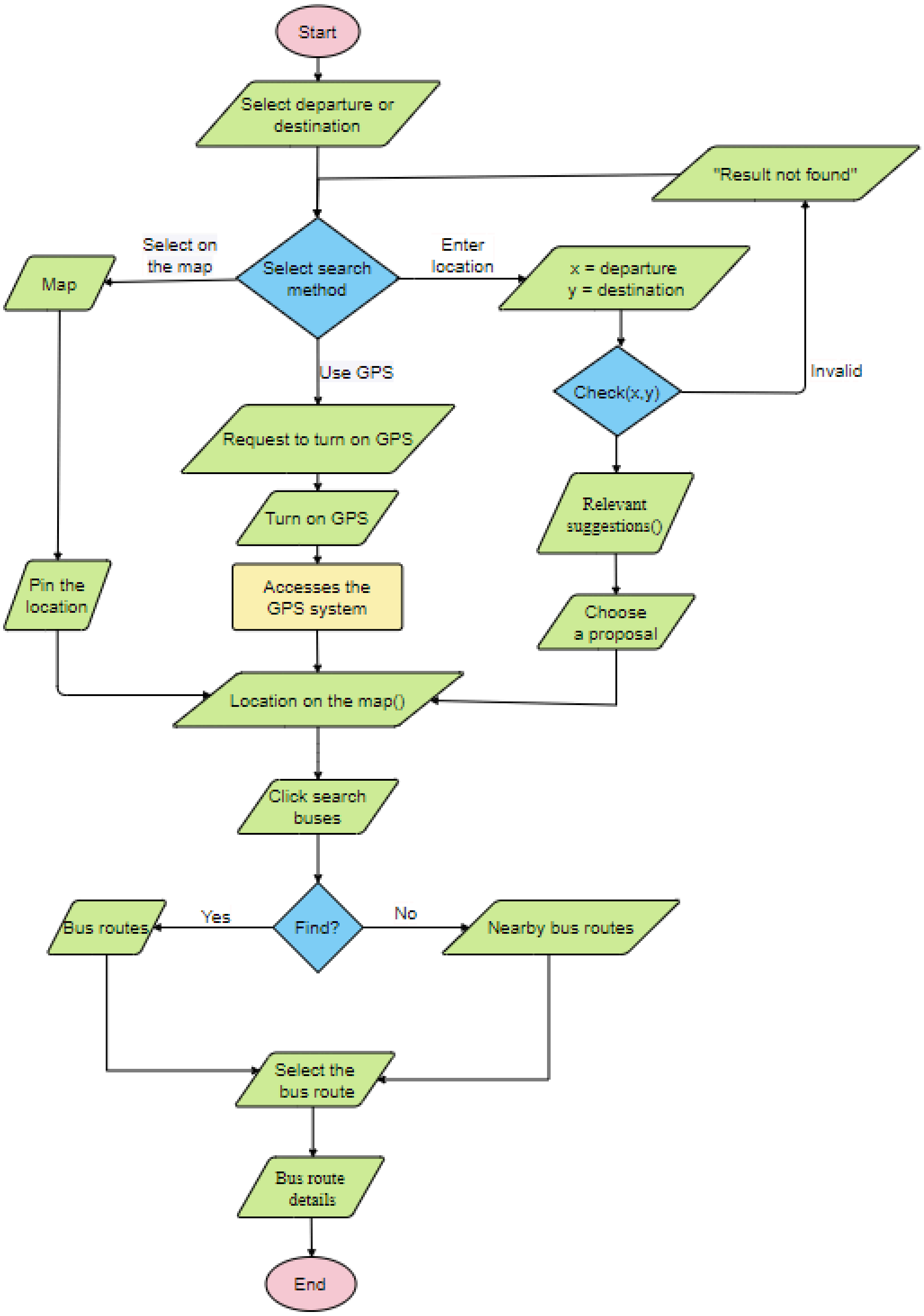
No.	Controller type	Note
1	*Label*	error message location not found
2	*Button*	*GPS search method*
3	*Button*	*Map selection method*

	Name	Code	Data Type	Length	Precision	P
1	AccountID	ACCOUNTID	char(7)	7		✓
2	AName	ANAME	varchar(20)	20		
3	AGender	AGENDER	varchar(3)	3		
4	ADoB	ADOB	date			
5	APhoneNumber	APHONENUMB	char(10)	10		
6	AEmail	AEMAIL	varchar(20)	20		

4	*Text Box*	*Enter location*

- *Process:*

	Name	Code	Data Type	Length	Precision	P
1	AccountID	ACCOUNTID	char(7)	7		☑
2	AName	ANAME	varchar(20)	20		☐
3	AGender	AGENDER	varchar(3)	3		☐
4	ADoB	ADOB	date			☐
5	APhoneNumber	APHONENUMB	char(10)	10		☐
6	AEmail	AEMAIL	varchar(20)	20		☐

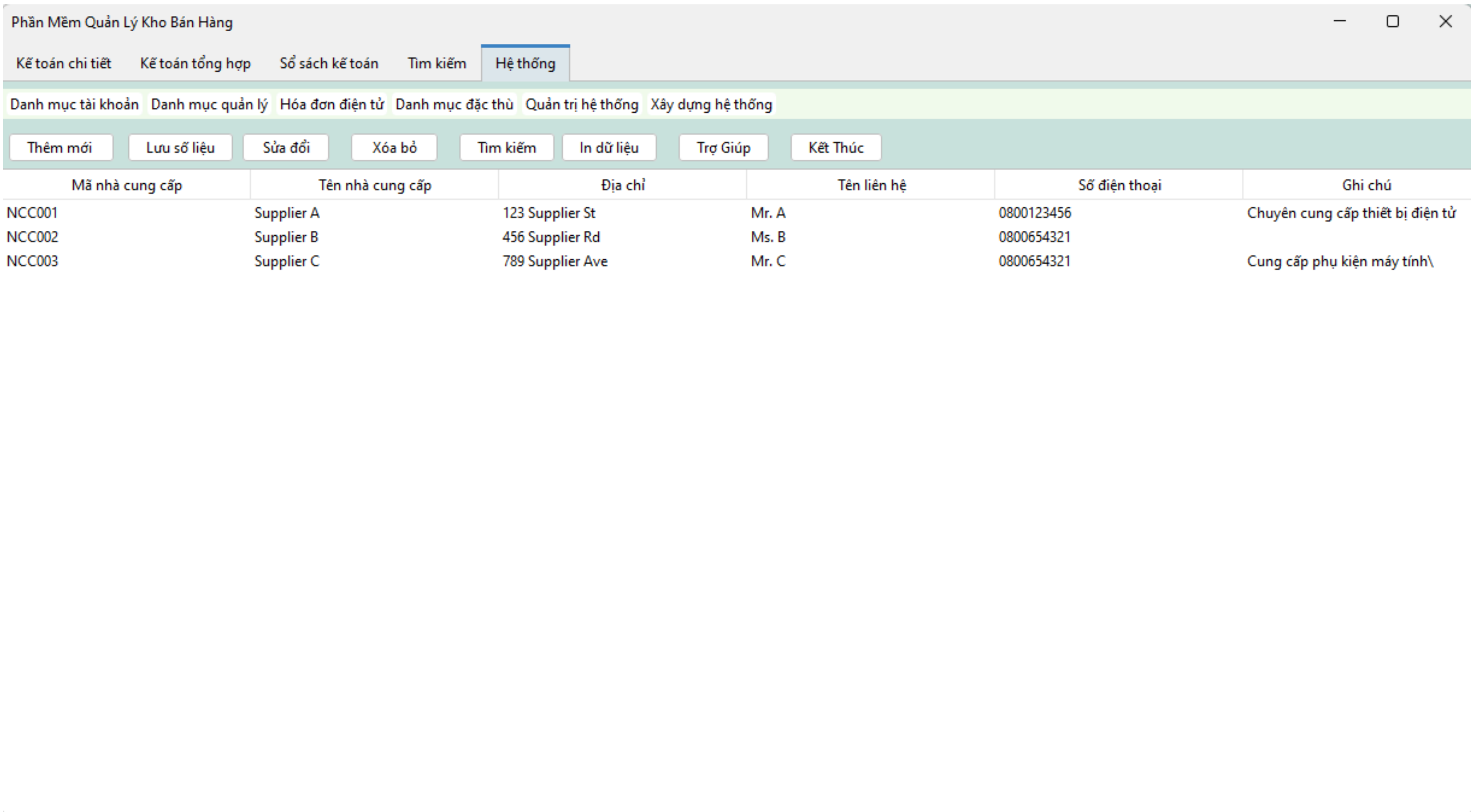


- *Constraints:* Functional Requirements - Function 4 (Bus Routes Searching)

2.3.5 Function 5 : Quản lý nhà cung cấp

	Name	Code	Data Type	Length	Precision	P
1	AccountID	ACCOUNTID	char(7)	7		☑
2	AName	ANAME	varchar(20)	20		☐
3	AGender	AGENDER	varchar(3)	3		☐
4	ADoB	ADOB	date			☐
5	APhoneNumber	APHONENUMB	char(10)	10		☐
6	AEmail	AEMAIL	varchar(20)	20		☐

- *Purpose:* The platform enables users to book bus tickets online, streamlining the ticketing process for commuters.
- Interface
    + *Interface 1 (Routes Searching):*



- The component of interface 1:

No.	Controller type	Note
1	*Combo box*	*List of departure points.*
2	*Combo box*	*List of destination points.*
3	*Combo box*	*Date to go*
4	*Button*	*Swap value of combo box 1 and 2*

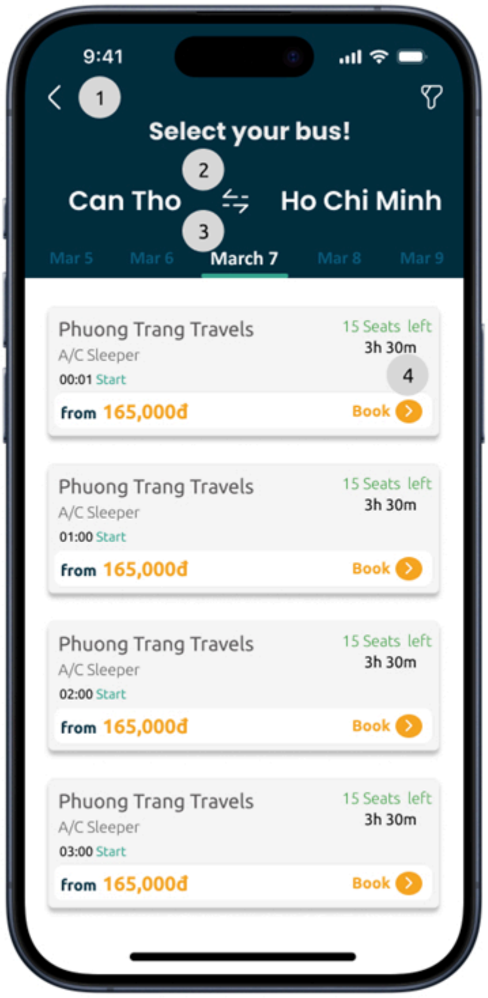
	Name	Code	Data Type	Length	Precision	P
1	AccountID	ACCOUNTID	char(7)	7		☑
2	AName	ANAME	varchar(20)	20		☐
3	AGender	AGENDER	varchar(3)	3		☐
4	ADoB	ADOB	date			☐
5	APhoneNumber	APHONENUMB	char(10)	10		☐
6	AEmail	AEMAIL	varchar(20)	20		☐

5	*Button*	*If founded go to interface 2a, otherwise go to interface 2b*

- Data to be used:

No.	Table name / Data structure	Method			
		Add	Modify	Delete	Query
1	*Routes*				X

\+ *Interface 2 (Select Routes):*



(a) (b)

- The components of interface 2:

	Name	Code	Data Type	Length	Precision	P
1	AccountID	ACCOUNTID	char(7)	7		☑
2	AName	ANAME	varchar(20)	20		☐
3	AGender	AGENDER	varchar(3)	3		☐
4	ADoB	ADOB	date			☐
5	APhoneNumber	APHONENUMB	char(10)	10		☐
6	AEmail	AEMAIL	varchar(20)	20		☐

No.	Controller type	Note
1	*Button*	*Return to interface 1*
2	*Button*	*Swap departure and destination.*
3	*Combo box*	*Drag left or right to change date*
4	*Button*	*Proceed to interface 3*
Default value (date) of No.3 is today.		

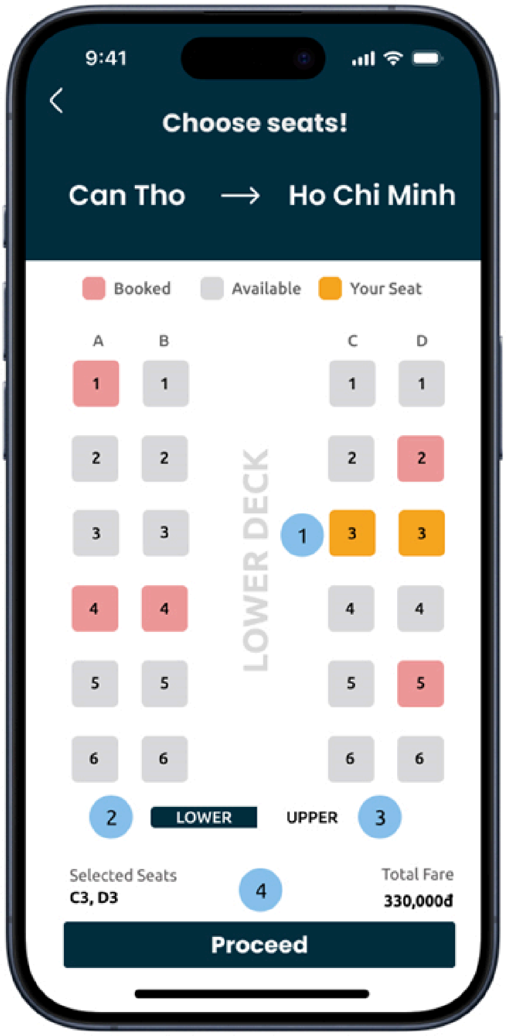
- Data to be used:

No.	Table name / Data structure	Method			
		Add	**Modify**	**Delete**	**Query**
1	*Routes*				X

\+ *Interface 3 (Select Seats):*

▪

	Name	Code	Data Type	Length	Precision	P
1	AccountID	ACCOUNTID	char(7)	7		☑
2	AName	ANAME	varchar(20)	20		☐
3	AGender	AGENDER	varchar(3)	3		☐
4	ADoB	ADOB	date			☐
5	APhoneNumber	APHONENUMB	char(10)	10		☐
6	AEmail	AEMAIL	varchar(20)	20		☐



- The components of interface 3:

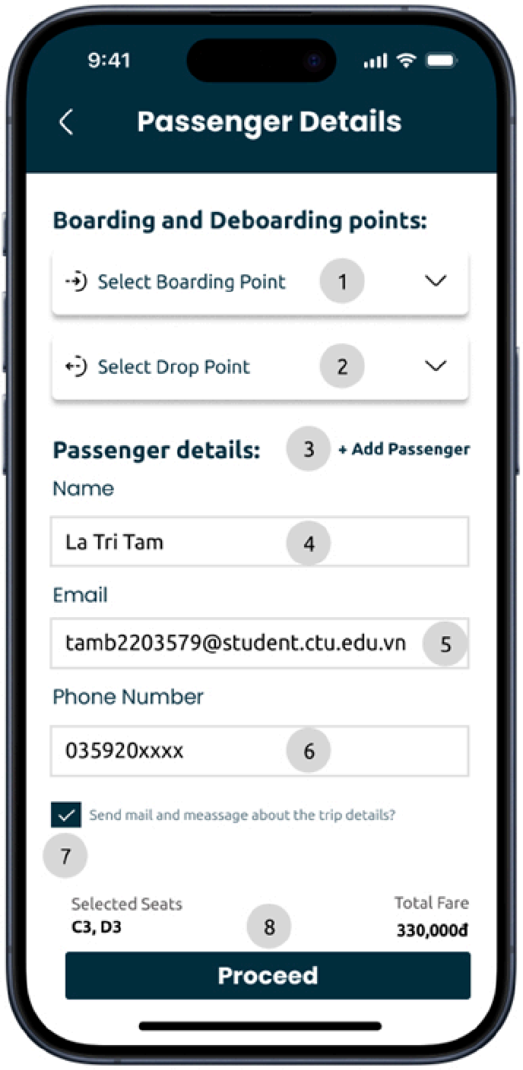
No.	Controller type	Default value	Note
1	*Button*	*Seat number*	*The box will turn to orange when we click*
2	*Button*	*Lower*	*Switch to lower stage*
3	*Button*	*Upper*	*Switch to upper stage*
4	*Button*	*Proceed*	*Proceed to interface 4*

- Data to be used:

	Name	Code	Data Type	Length	Precision	P
1	AccountID	ACCOUNTID	char(7)	7		☑
2	AName	ANAME	varchar(20)	20		☐
3	AGender	AGENDER	varchar(3)	3		☐
4	ADoB	ADOB	date			☐
5	APhoneNumber	APHONENUMB	char(10)	10		☐
6	AEmail	AEMAIL	varchar(20)	20		☐

No.	Table name / Data structure	Method			
		Add	Modify	Delete	Query
1	*Routes*				X

\+ *Interface 4 (Enter Details):*



- The components of interface 4:

No.	Controller type	Note

▪

	Name	Code	Data Type	Length	Precision	P
1	AccountID	ACCOUNTID	char(7)	7		☑
2	AName	ANAME	varchar(20)	20		☐
3	AGender	AGENDER	varchar(3)	3		☐
4	ADoB	ADOB	date			☐
5	APhoneNumber	APHONENUMB	char(10)	10		☐
6	AEmail	AEMAIL	varchar(20)	20		☐

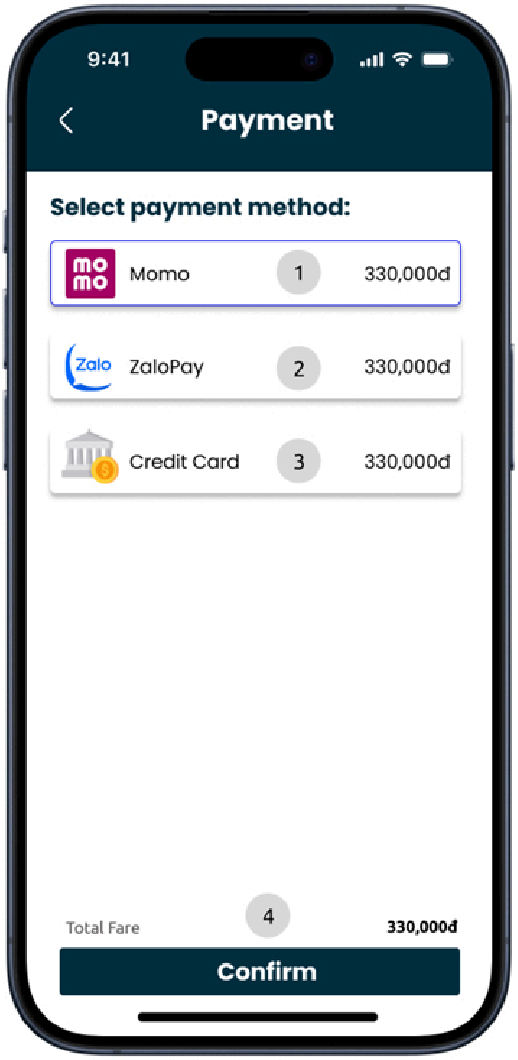
1	*Combo box*	Let the user choose detailed departure point from the combo box
2	*Combo box*	Let the user choose detailed destination point from the combo box
3	*Button*	*Add 1 more passenger details*
4	*Text box*	
5	*Text box*	
6	*Text box*	
7	*Check box*	*If checked the system will send a message about the trip details after booking completion.*
8	*Button*	*Proceed to interface 5*

- Data to be used:

No.	**Table name / Data structure**	**Method**			
		Add	**Modify**	**Delete**	**Query**
1	*Booking*	X			
2	*Passenger*	X			

\+ *Interface 5 (Select payment methods):*

	Name	Code	Data Type	Length	Precision	P
1	AccountID	ACCOUNTID	char(7)	7		☑
2	AName	ANAME	varchar(20)	20		☐
3	AGender	AGENDER	varchar(3)	3		☐
4	ADoB	ADOB	date			☐
5	APhoneNumber	APHONENUMB	char(10)	10		☐
6	AEmail	AEMAIL	varchar(20)	20		☐



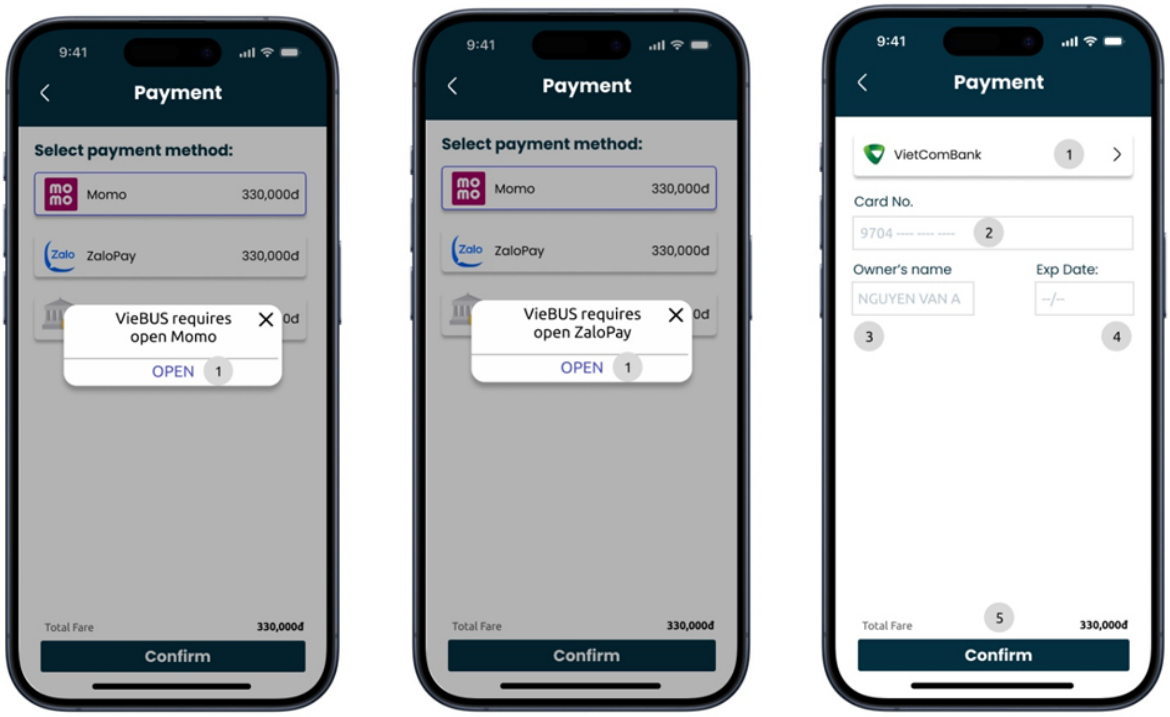
- *The components of interface 5:*

No.	Controller type	Note
1	*Button*	*Momo payment method*
2	*Button*	*ZaloPay payment method*
3	*Button*	*Credit card payment method*
4	*Button*	*Proceed to interface 6a, 6b, 6c according to the user choice.*

+ *Interface 6 (Payment confirm):*

■

	Name	Code	Data Type	Length	Precision	P
1	AccountID	ACCOUNTID	char(7)	7		☑
2	AName	ANAME	varchar(20)	20		☐
3	AGender	AGENDER	varchar(3)	3		☐
4	ADoB	ADOB	date			☐
5	APhoneNumber	APHONENUMB	char(10)	10		☐
6	AEmail	AEMAIL	varchar(20)	20		☐



(a) (b) (c)

- The components of interface 6:

No.	Controller type	Default value	Note
1a	*Button*		*Open Momo app*
1b	*Button*		*Open ZaloPay app*
1	*Combo box*		*Let the user choose from the bank list.*

-

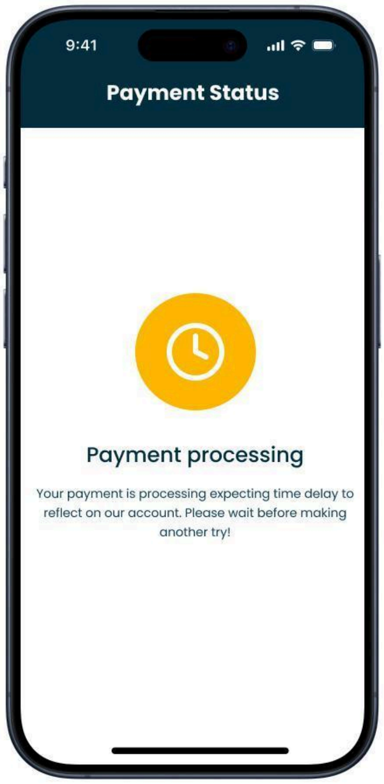
	Name	Code	Data Type	Length	Precision	P
1	AccountID	ACCOUNTID	char(7)	7		☑
2	AName	ANAME	varchar(20)	20		☐
3	AGender	AGENDER	varchar(3)	3		☐
4	ADoB	ADOB	date			☐
5	APhoneNumber	APHONENUMB	char(10)	10		☐
6	AEmail	AEMAIL	varchar(20)	20		☐

2	*Text box*	*9704 ---- ---- ----*	
3	*Text box*	*Nguyen Van A*	
4	*Text box*	*--/--*	
5	*Button*		*Proceed to interface 7*

+ *Interface 7:*

-

	Name	Code	Data Type	Length	Precision	P
1	AccountID	ACCOUNTID	char(7)	7		✓
2	AName	ANAME	varchar(20)	20		
3	AGender	AGENDER	varchar(3)	3		
4	ADoB	ADOB	date			
5	APhoneNumber	APHONENUMB	char(10)	10		
6	AEmail	AEMAIL	varchar(20)	20		



- The components of interface 7:

No.	Controller type	Default value	Note
1	*Button*	*View ticket*	*Proceed to interface 8*

- Data to be used:

No.	Table name / Data structure	Method			
		Add	Modify	Delete	Query
1	*Booking*	X			

+ *Interface 8:*

	Name	Code	Data Type	Length	Precision	P
1	AccountID	ACCOUNTID	char(7)	7		✓
2	AName	ANAME	varchar(20)	20		
3	AGender	AGENDER	varchar(3)	3		
4	ADoB	ADOB	date			
5	APhoneNumber	APHONENUMB	char(10)	10		
6	AEmail	AEMAIL	varchar(20)	20		

- The components of interface 8:

No.	Controller type	Note
1	*Button*	*Return to home*

- Data to be used:

No.	Table name / Data structure	Method			
		Add	Modify	Delete	Query
1	*Ticket*				X

■

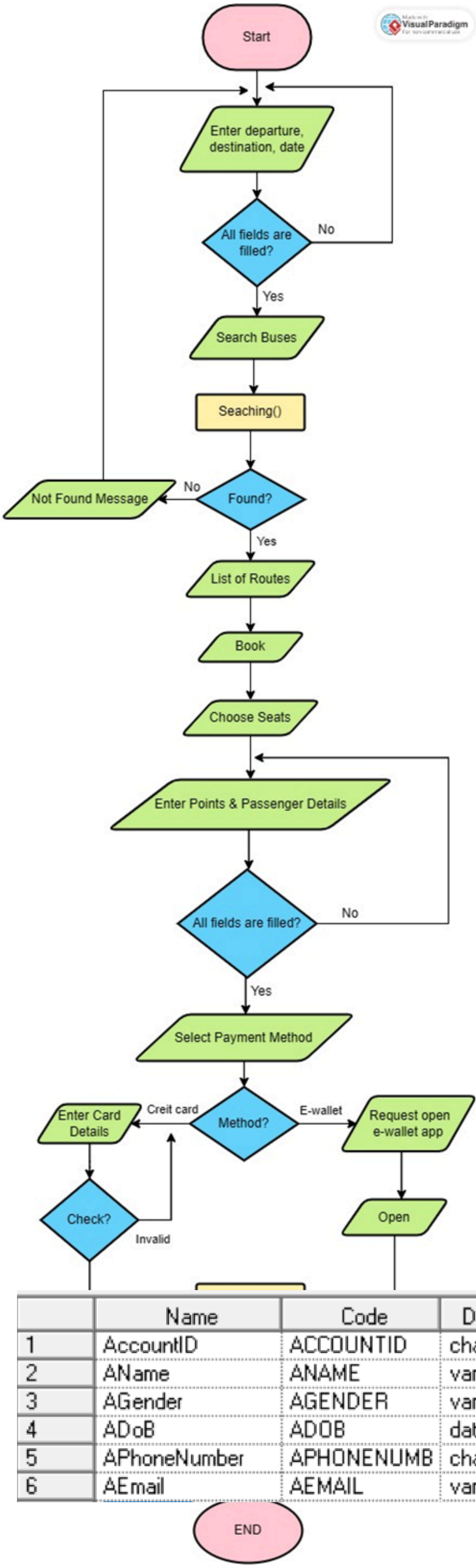
	Name	Code	Data Type	Length	Precision	P
1	AccountID	ACCOUNTID	char(7)	7		☑
2	AName	ANAME	varchar(20)	20		☐
3	AGender	AGENDER	varchar(3)	3		☐
4	ADoB	ADOB	date			☐
5	APhoneNumber	APHONENUMB	char(10)	10		☐
6	AEmail	AEMAIL	varchar(20)	20		☐

2	*Booking*				X

- *Process:*

■

	Name	Code	Data Type	Length	Precision	P
1	AccountID	ACCOUNTID	char(7)	7		✓
2	AName	ANAME	varchar(20)	20		
3	AGender	AGENDER	varchar(3)	3		
4	ADoB	ADOB	date			
5	APhoneNumber	APHONENUMB	char(10)	10		
6	AEmail	AEMAIL	varchar(20)	20		



-

	Name	Code	Data Type	Length	Precision	P
1	AccountID	ACCOUNTID	char(7)	7		✓
2	AName	ANAME	varchar(20)	20		
3	AGender	AGENDER	varchar(3)	3		
4	ADoB	ADOB	date			
5	APhoneNumber	APHONENUMB	char(10)	10		
6	AEmail	AEMAIL	varchar(20)	20		

- *Constraints:* Functional requirements - Function 5 (Online ticket booking)

2.4 References to Requirements

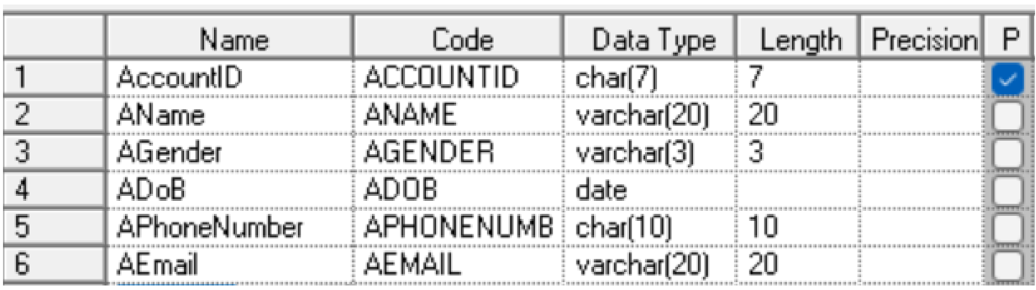
No.	Design	Functional Requirement
1	*Real-time Bus Routes Display*	Function 1
2	*Account Registration*	Function 2
3	*Bus Routes Editting*	Function 3
4	*Bus Routes Searching*	Function 4
5	*Online Ticket Booking*	Function 5

3. Unit Test

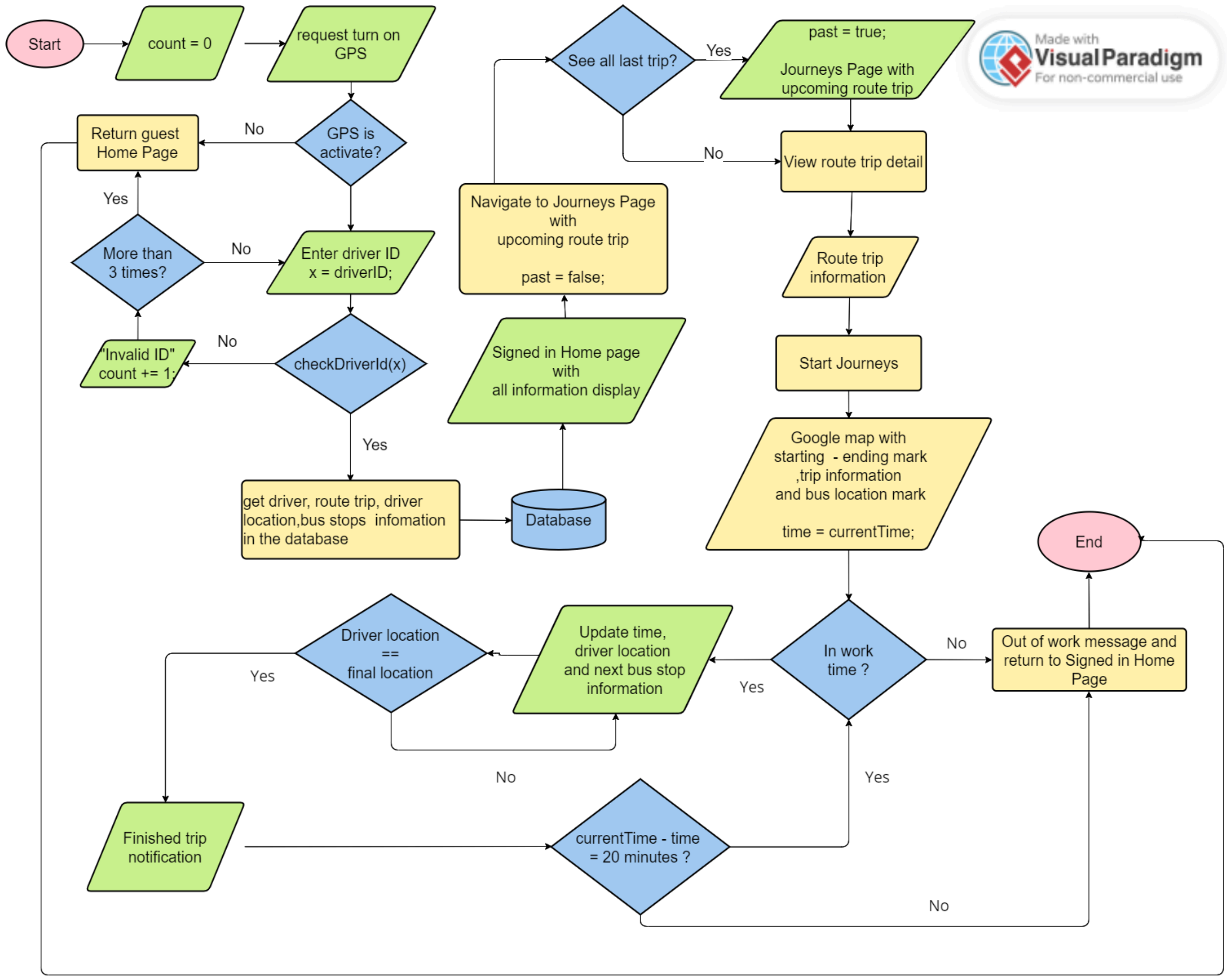
3.1 Test 1 - Real Time Bus Routes Tracking

Test ID:	TC001
Test description	Real Time Bus Routes Tracking Testing
Created by:	
Date of creation:	26/4/2024
Date of review:	30/4/2024
Test priority:	High
Pre-requisite:	
Post-requisite:	

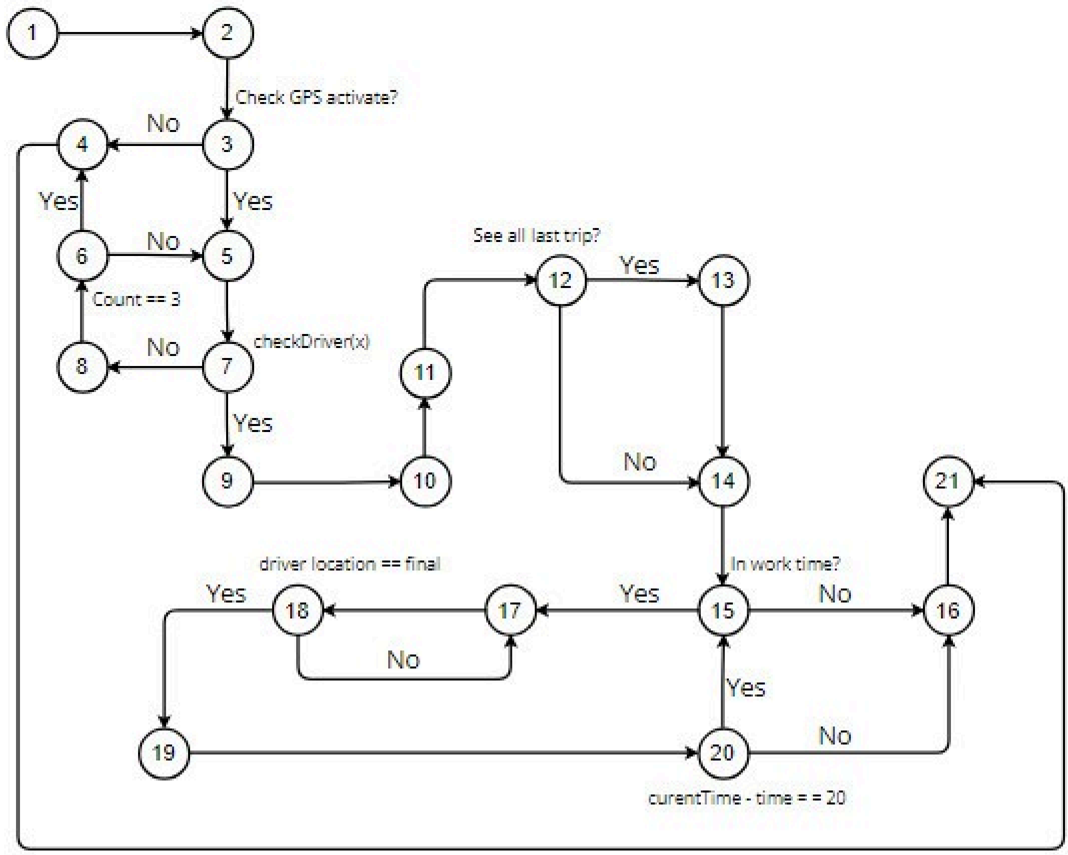
3.1.1 Drawing the flow graph

- 

	Name	Code	Data Type	Length	Precision	P
1	AccountID	ACCOUNTID	char(7)	7		☑
2	AName	ANAME	varchar(20)	20		☐
3	AGender	AGENDER	varchar(3)	3		☐
4	ADoB	ADOB	date			☐
5	APhoneNumber	APHONENUMB	char(10)	10		☐
6	AEmail	AEMAIL	varchar(20)	20		☐



	Na	Code	Data Type	Length	Precision	P
1	AccountID	ACCOUNTID	char(7)	7		☑
2	AName	ANAME	varchar(20)	20		☐
3	AGender	AGENDER	varchar(3)	3		☐
4	ADoB	ADOB	date			☐
5	APhoneNumber	APHONENUMB	char(10)	10		☐
6	AEmail	AEMAIL	varchar(20)	20		☐



3.1.2 Determining a basis set of independent paths

Path 1: 1 - 2 - 3 - 4 - 21 (Not activating GPS)

Path 2: 1 - 2 - 3 - 5 - 7 - 8 - 6 - 4 - 21 (Entering wrong driver ID for three 3 times)

Path 3: 1 - 2 - 3 - 5 - 7 - 9 - 10 - 11 - 12 - 13 - 14 - 15 - 16 - 21 (Start journeys unsuccessfully)

Path 4: 1 - 2 - 3 - 5 - 7 - 9 - 10 - 11 - 12 - 14 - 15 - 17 - 18 - 19 - 20 - 16 - 21 (Start journeys successfully, driver in break time and still in operation time)

Path 5: 1 - 2 - 3 - 5 - 7 - 9 - 10 - 11 - 12 - 14 - 15 - 17 - 18 - 17 - 18 - 19 - 20 - 16 - 21 (Continuingly updated driver location until he reach last destination and out of working time)

Path 6: 1 - 2 - 3 - 5 - 7 - 9 - 10 - 11 - 12 - 14 - 15 - 17 - 18 - 19 - 20 - 15 - 16 - 21 (the driver is not in the working time after a break)

■

	Name	Code	Data Type	Length	Precision	P
1	AccountID	ACCOUNTID	char(7)	7		✓
2	AName	ANAME	varchar(20)	20		
3	AGender	AGENDER	varchar(3)	3		
4	ADoB	ADOB	date			
5	APhoneNumber	APHONENUMB	char(10)	10		
6	AEmail	AEMAIL	varchar(20)	20		

Path 7: 1 - 2 - 3 - 5 - 7 - 8 - 6 - 5 - 7 - 9 - 10 - 11 - 12 - 13 - 14 - 15 - 16 - 21 (The driver login successfully after a try but he is not in his working time)
Path 8: :1 - 2 - 3 - 5 - 7 - 8 - 6 - 5 - 7 - 9 - 10 - 11 - 12 - 13 - 14 - 15 - 17 - 18 - 19 - 20 - 16 - 21 (Driver successfully sign in after a try and he is in break time)

3.1.3 Preparing test cases

▪

	Name	Code	Data Type	Length	Precision	P
1	AccountID	ACCOUNTID	char(7)	7		☑
2	AName	ANAME	varchar(20)	20		☐
3	AGender	AGENDER	varchar(3)	3		☐
4	ADoB	ADOB	date			☐
5	APhoneNumber	APHONENUMB	char(10)	10		☐
6	AEmail	AEMAIL	varchar(20)	20		☐

# Test Case	Input Data								Expected Result	Actual Result	Status (Pass/ Fail)
	GPS	Check DriverID (x)	Count	past	route trip detail	Start Journey	In work time	currentTime - time = 20 minutes			
1	No	Null	0	Null	Null	Null	Null	Null	Guest Home Page		
2	Yes	No	3	Null	Null	Null	Null	Null	"Invalid ID" and return Guest Home Page		
3	Yes	Yes	0	True	Yes	Yes	No	Null	Out of work message and return to Signed		

	Name	Code	Data Type	Length	Precision	P	F	M
1	AccountID	ACCOUNTID	char(7)	7		✓		✓
2	AName	ANAME	varchar(20)	20				✓
3	AGender	AGENDER	varchar(3)	3				✓
4	ADoB	ADOB	date					✓
5	APhoneNumber	APHONENUMB	char(10)	10				✓
6	AEmail	AEMAIL	varchar(20)	20				✓

									in Home Page		
4	Yes	Yes	0	False	Yes	Yes	Yes	No	Out of work message and return to Signed in Home Page		
5	Yes	Yes	0	False	Yes	Yes	No	Null	Out of work message and return to Signed in Home Page		
6	Yes	Yes	0	False	Yes	Yes	No	Yes	Out of work		

	Name	Code	Data Type	Length	Precision	P	F	M
1	AccountID	ACCOUNTID	char(7)	7		☑	☐	☑
2	AName	ANAME	varchar(20)	20		☐	☐	☑
3	AGender	AGENDER	varchar(3)	3		☐	☐	☑
4	ADoB	ADOB	date			☐	☐	☑
5	APhoneNumber	APHONENUMB	char(10)	10		☐	☐	☑
6	AEmail	AEMAIL	varchar(20)	20		☐	☐	☑

									message and return to Signed in Home Page		
7	Yes	Yes	0	False	Yes	Yes	No	Null	Out of work message and return to Signed in Home Page		
8	Yes	Yes	0	False	Yes	Yes	Yes	No	Out of work message and return to Signed in Home Page		

	Name	Code	Data Type	Length	Precision	P	F	M
1	AccountID	ACCOUNTID	char(7)	7		✓		✓
2	AName	ANAME	varchar(20)	20				✓
3	AGender	AGENDER	varchar(3)	3				✓
4	ADoB	ADOB	date					✓
5	APhoneNumber	APHONENUMB	char(10)	10				✓
6	AEmail	AEMAIL	varchar(20)	20				✓

-

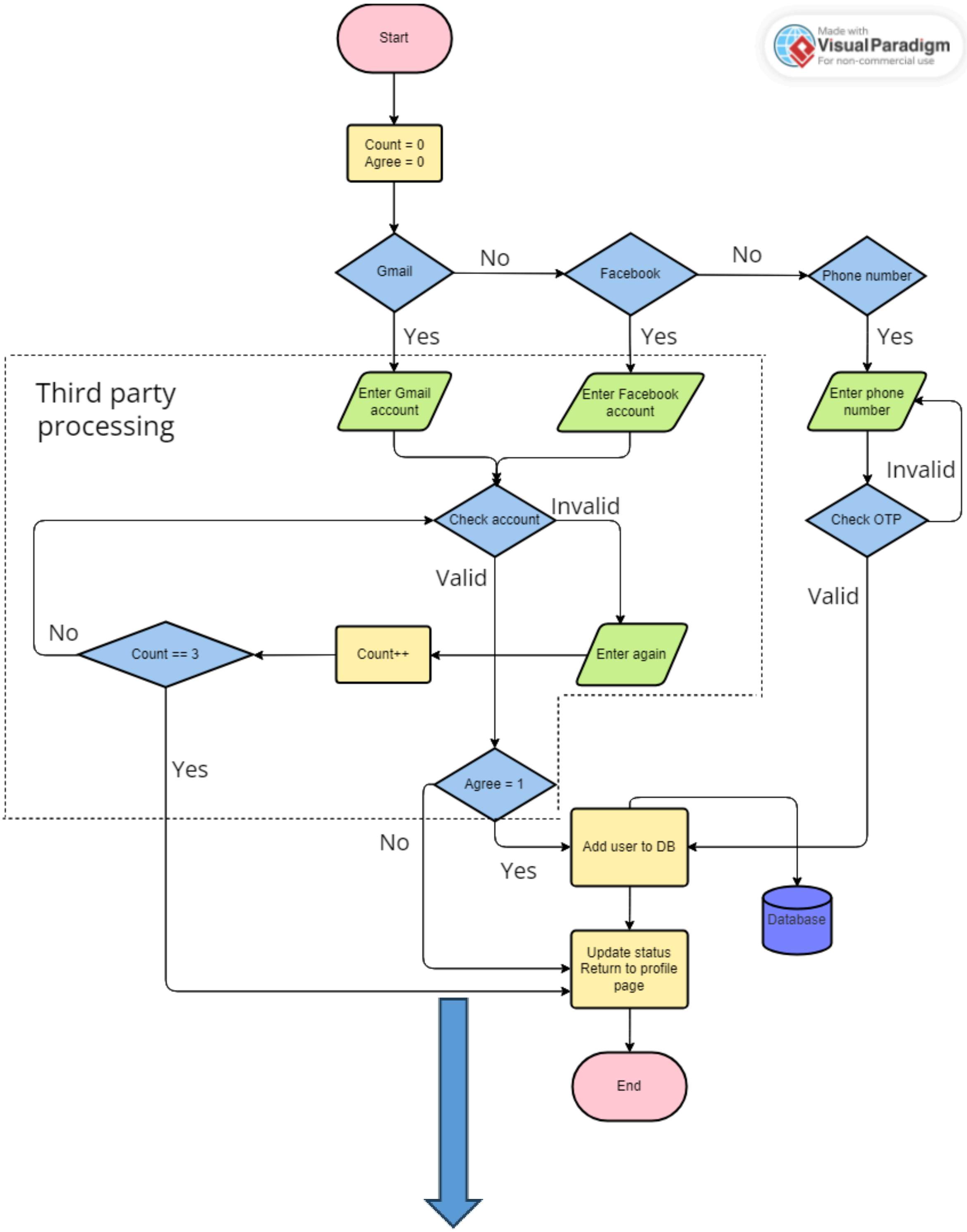
	Name	Code	Data Type	Length	Precision	P	F	M
1	AccountID	ACCOUNTID	char(7)	7		✓		✓
2	AName	ANAME	varchar(20)	20				✓
3	AGender	AGENDER	varchar(3)	3				✓
4	ADoB	ADOB	date					✓
5	APhoneNumber	APHONENUMB	char(10)	10				✓
6	AEmail	AEMAIL	varchar(20)	20				✓

3.2 Test 2 - Account Registration

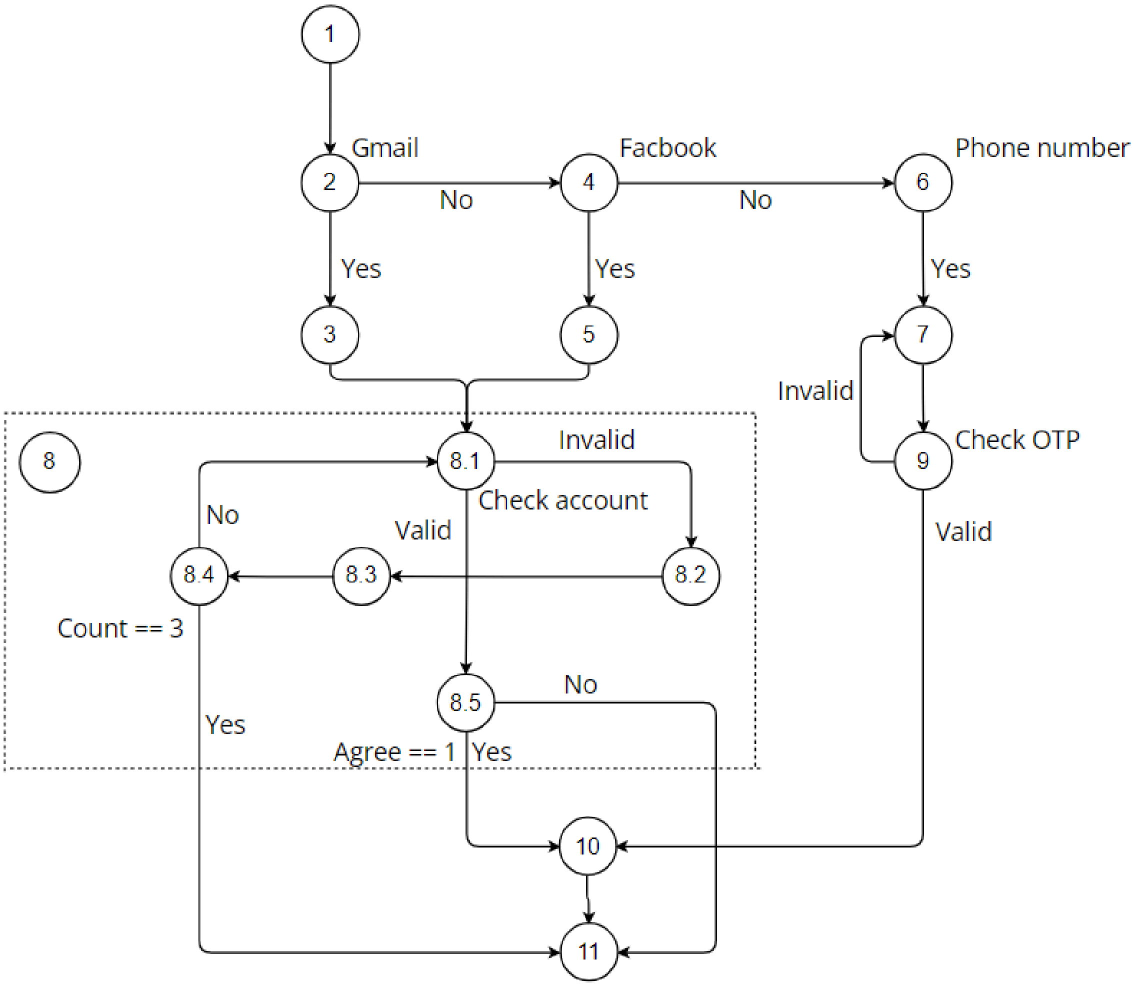
Test ID:	TC002
Test description:	Account Registration Testing
Created by:	Le Lu Huyen Tran
Date of creation:	26/4/2024
Date of review:	4/5/2024
Test priority:	High
Pre-requisite:	
Post-requisite:	

3.2.1 Drawing the flow graph

	Name	Code	Data Type	Length	Precision	P
1	AccountID	ACCOUNTID	char(7)	7		☑
2	AName	ANAME	varchar(20)	20		☐
3	AGender	AGENDER	varchar(3)	3		☐
4	ADoB	ADOB	date			☐
5	APhoneNumber	APHONENUMB	char(10)	10		☐
6	AEmail	AEMAIL	varchar(20)	20		☐



	Name	Code	Data Type	Length	Precision	P
1	AccountID	ACCOUNTID	char(7)	7		✓
2	AName	ANAME	varchar(20)	20		
3	AGender	AGENDER	varchar(3)	3		
4	ADoB	ADOB	date			
5	APhoneNumber	APHONENUMB	char(10)	10		
6	AEmail	AEMAIL	varchar(20)	20		



3.2.2 Determining a basis set of independent paths

Path 1: 8.1 - 8.2 - 8.3 - 8.4 - 10 - 11 (Invalid account more than 3 times)

Path 2: 8.1 - 8.5 - 11 (Valid account but user doesn't allow use their information)

Path 3: 8.1 - 8.2 - 8.3 - 8.4 - 8.1 - 8.5 - 10 - 11 (Invalid account less than 3 times but user doesn't allow use their information)

Path 4: 1 - 2 - 3 - 8 - 10 -11 (Sign up by Gmail account successfully)

Path 5: 1 - 2 - 4 - 5 - 10 - 11 (Sign up by Facebook account successfully)

Path 6: 1 - 2 - 4 - 6 - 7 - 9 - 10 - 11 (Sign up by Phone number successfully)

Path 7: 1 - 2 - 4 - 6 - 7 - 9 - 7 - 9 - 10 - 11 (Invalid OTP)

	Name	Code	Data Type	Length	Precision	P
1	AccountID	ACCOUNTID	char(7)	7		✓
2	AName	ANAME	varchar(20)	20		
3	AGender	AGENDER	varchar(3)	3		
4	ADoB	ADOB	date			
5	APhoneNumber	APHONENUMB	char(10)	10		
6	AEmail	AEMAIL	varchar(20)	20		

3.2.3 Preparing test cases

	Gmail account	Facebook account	Phone number
Valid	le123@gmail.com Pass: 12345678	Ht123 Pass: 12345678	0922331452
Invalid	abc345@asd Pass: 12345678	dfsf# Pass: 12345678	0123456783

# Test Case	Input Data				Expected Result	Actual Result	Status (Pass/ Fail)
	Gmail account	Facebook account	Phone number	Agree			
1	Invalid > 3	-	-	-	Return to profile page		
2	Valid	-	-	0	Return to profile page		
3	Invalid <= 3	-	-	0	Return to profile page		
4	Valid	-	-	1	Sign up successfully		
5	-	Valid	-	1	Sign up successfully		

-

	Name	Code	Data Type	Length	Precision	P
1	AccountID	ACCOUNTID	char(7)	7		☑
2	AName	ANAME	varchar(20)	20		☐
3	AGender	AGENDER	varchar(3)	3		☐
4	ADoB	ADOB	date			☐
5	APhoneNumber	APHONENUMB	char(10)	10		☐
6	AEmail	AEMAIL	varchar(20)	20		☐

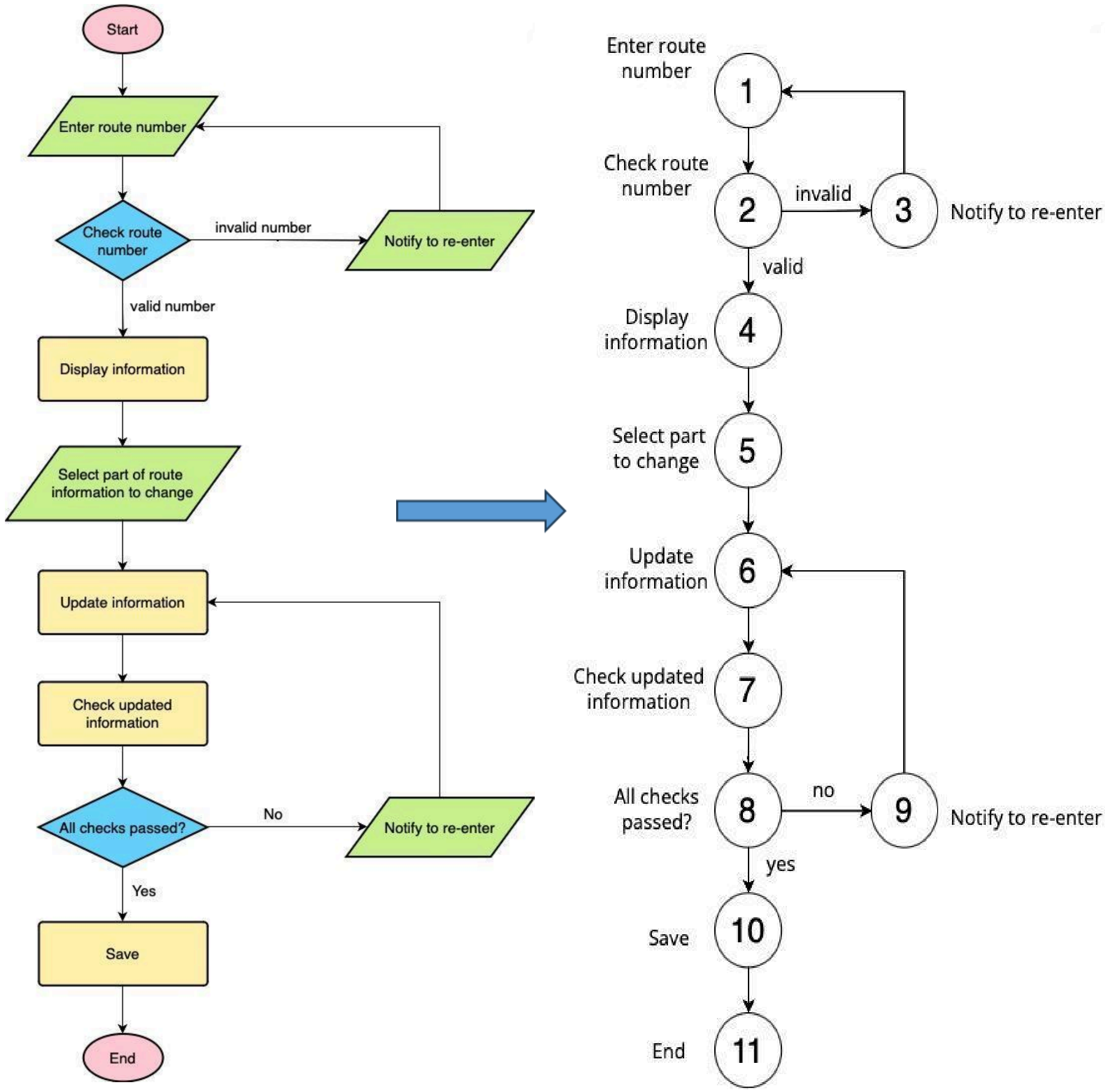
6	-	-	Valid	-	Sign up successfully		
7	-	-	Invalid	-	Please enter again		

3.3 Test 3 - Bus Route Editing

Test ID:	TC003
Test description:	Bus Route Editing Function Testing
Created by:	
Date of creation:	26/04/2024
Date of review:	28/04/2024
Test priority:	High
Pre-requisite:	
Post-requisite:	

3.3.1 Drawing the flow graph

	Name	Code	Data Type	Length	Precision	P
1	AccountID	ACCOUNTID	char(7)	7		☑
2	AName	ANAME	varchar(20)	20		☐
3	AGender	AGENDER	varchar(3)	3		☐
4	ADoB	ADOB	date			☐
5	APhoneNumber	APHONENUMB	char(10)	10		☐
6	AEmail	AEMAIL	varchar(20)	20		☐



3.3.2 Determining a basis set of independent paths

Path 1: 1 - 2 - 4 - 5 - 6 - 7 - 8 - 10 - 11 (Update successfully)

Path 2: 1- 2 - 3 - 1 - 2 - 4 - 5 - 6 - 7 - 8 - 10 - 11 (Invalid route number)

Path 3: 1 - 2 - 4 - 5 - 6 - 7 - 8 - 9 - 6 - 7 - 8 - 10 - 11 (Some invalidate updates)

3.3.3 Preparing test cases

	Name	Code	Data Type	Length	Precision	P
1	AccountID	ACCOUNTID	char(7)	7		☑
2	AName	ANAME	varchar(20)	20		☐
3	AGender	AGENDER	varchar(3)	3		☐
4	ADoB	ADOB	date			☐
5	APhoneNumber	APHONENUMB	char(10)	10		☐
6	AEmail	AEMAIL	varchar(20)	20		☐

# Test Case	Input Data								Expected Result	Actual Result	Status (Pass/Fail)
	Route number	Updated information									
		Route number	Number of seats	Route name	Ticket price	Operation time	Trip time	Trip spacing			
1	01	-	50	-	7.000	-	10	-	Update Successfully		
2	P@2								Route number not found		
3	01	P@2	-	-	-	6:00 - 25:00	-	-	Invalidate updates		

	Name	Code	Data Type	Length	Precision	P	F	M
1	AccountID	ACCOUNTID	char(7)	7		✓		✓
2	AName	ANAME	varchar(20)	20				✓
3	AGender	AGENDER	varchar(3)	3				✓
4	ADoB	ADOB	date					✓
5	APhoneNumber	APHONENUMB	char(10)	10				✓
6	AEmail	AEMAIL	varchar(20)	20				✓

3.4 Test 4 - Bus Routes Searching

Test ID:	TC004
Test description:	Bus Routes Searching Function Testing
Created by:	
Date of creation:	26/4/2024
Date of review:	30/4/2024
Test priority:	High
Pre-requisite:	
Post-requisite:	

3.4.1 Drawing the flow graph

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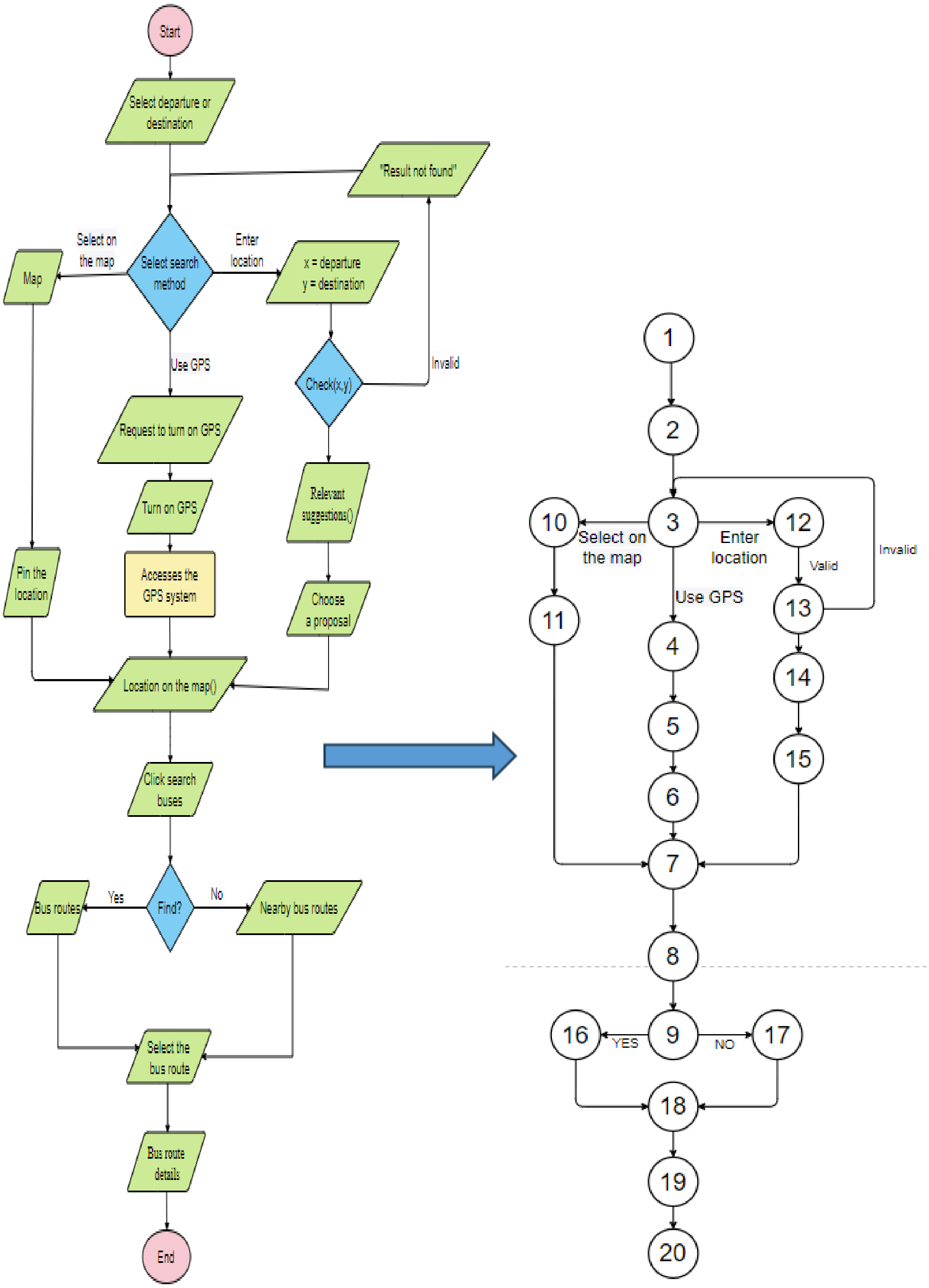
	Name	Code	Data Type	Length	Precision	P	F
1	AccountID	ACCOUNTID	char(7)	7		✓	
2	AName	ANAME	varchar(20)	20			
3	AGender	AGENDER	varchar(3)	3			
4	ADoB	ADOB	date				
5	APhoneNumber	APHONENUMB	char(10)	10			
6	AEmail	AEMAIL	varchar(20)	20			



3.4.2 Determining a basis set of independent paths

	Name	Code	Data Type	Length	Precision	P	F
1	AccountID	ACCOUNTID	char(7)	7		✓	
2	AName	ANAME	varchar(20)	20			
3	AGender	AGENDER	varchar(3)	3			
4	ADoB	ADOB	date				
5	APhoneNumber	APHONENUMB	char(10)	10			
6	AEmail	AEMAIL	varchar(20)	20			

Path 1: 1 - 2 - 3 - 4 - 5 - 6 - 7 - 8 - 9 - 16 - 18 - 19 - 20 (select on the map - bus routes)
Path 2: 1 - 2 - 3 - 4 - 5 - 6 - 7 - 8 - 9 - 17 - 18 - 19 - 20 (select on the map - nearby bus routes)
Path 3: 1 - 2 - 3 - 10 - 11 - 7 - 8 - 9 - 16 - 18 - 19 - 20 (use GPS-bus routes)
Path 4: 1 - 2 - 3 - 12 - 13 - 14 - 15 - 7 - 8 - 9 - 16 - 18 - 19 - 20 (enter location - valid - bus routes)
Path 5: 1 - 2 - 3 - 12 - 13 - 3 - 12 - 13 - 14 - 15 - 7 - 8 - 9 - 16 - 18 - 19 - 20 (enter location - invalid - bus routes)

3.4.3 Preparing test case

# Test Case	Input Data				Expected Result	Actual Result	Status (Pass/Fail)
	Method	GPS	Correct	Suitable routes			
1	select on the map	NO	-	YES	bus routes		
2	select on the map	NO	-	NO	nearby bus routes		
3	use GPS	YES	-	YES	bus routes		
4	enter location	NO	valid	YES	bus routes		
5	enter location	NO	invalid	YES	bus routes		

3.5 Test 5 - Online Ticket Booking

Test ID:	TC005
Test description:	Online Ticket Booking Function Testing

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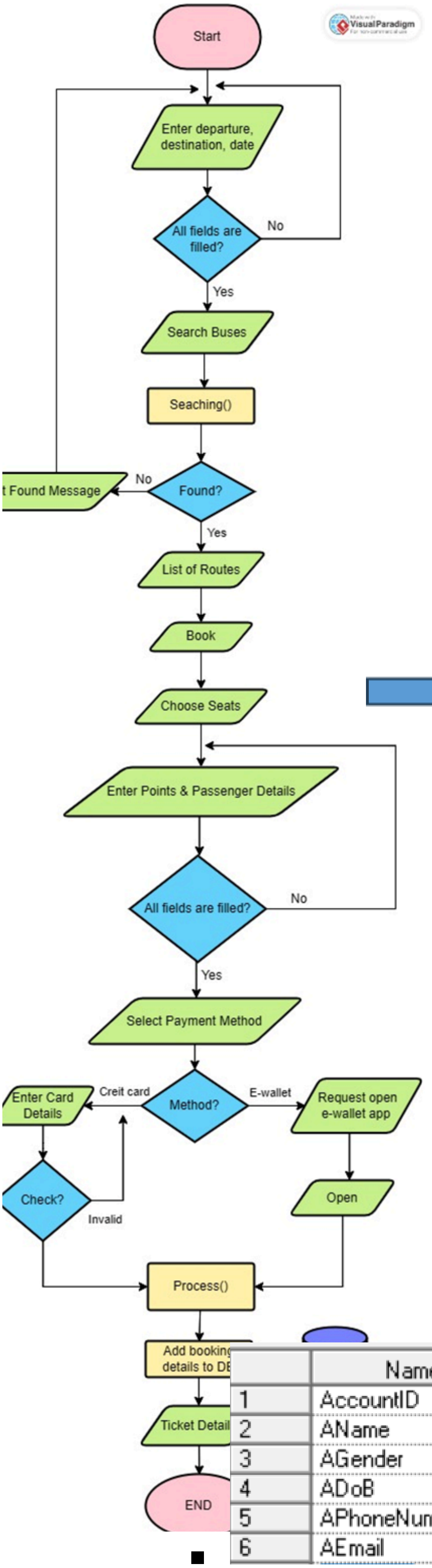
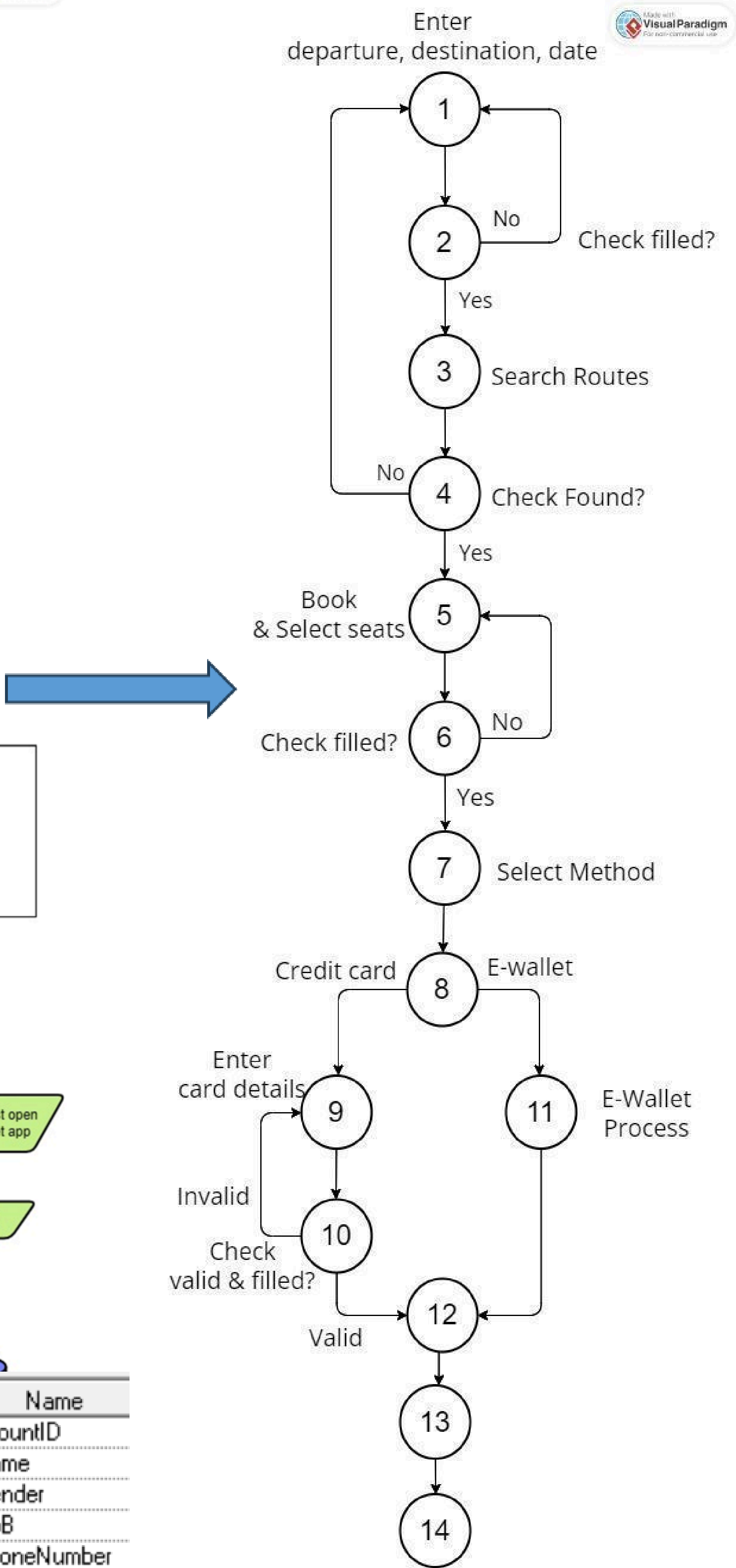
	Name	Code	Data Type	Length	Precision	P	F
1	AccountID	ACCOUNTID	char(7)	7		☑	
2	AName	ANAME	varchar(20)	20		☐	
3	AGender	AGENDER	varchar(3)	3		☐	
4	ADoB	ADOB	date			☐	
5	APhoneNumber	APHONENUMB	char(10)	10		☐	
6	AEmail	AEMAIL	varchar(20)	20		☐	

Created by:	
Date of creation:	26/04/2024
Date of review:	28/04/2024
Test priority:	High
Pre-requisite:	
Post-requisite:	

3.5.1 Drawing the flow graph

	Name	Code	Data Type	Length	Precision	P	F
1	AccountID	ACCOUNTID	char(7)	7		✓	
2	AName	ANAME	varchar(20)	20			
3	AGender	AGENDER	varchar(3)	3			
4	ADoB	ADOB	date				
5	APhoneNumber	APHONENUMB	char(10)	10			
6	AEmail	AEMAIL	varchar(20)	20			





	Name			
1	AccountID			
2	AName			
3	AGender			
4	ADoB			
5	APhoneNumber			
6	AEmail	AEMAIL	varchar(20)	20

3.5.2 Determining a basis set of independent paths

Path 1: 1-2-3-4-5-6-7-8-9-10-12-13-14 (Pass all branching nodes + Credit card)

Path 2: 1-2-3-4-5-6-7-8-11-12-13-14 (Pass all branching nodes + E-wallet)

Path 3: 1-2-1-2-3-4-5-6-7-8-9-10-12-13-14 (Destination, departure or date are not filled)

Path 4: 1-2-3-4-1-2-3-4-5-6-7-8-9-10-12-13-14 (Not found any routes)

Path 5: 1-2-3-4-5-6-5-6-7-8-9-10-12-13-14 (Points or Passenger Details are not filled)

Path 6: 1-2-3-4-5-6-7-8-9-10-11-10-12-13-14 (Invalid card details or details are not filled)

3.5.3 Preparing test cases

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	Name	Code	Data Type	Length	Precision	P	F
1	AccountID	ACCOUNTID	char(7)	7		✓	
2	AName	ANAME	varchar(20)	20			
3	AGender	AGENDER	varchar(3)	3			
4	ADoB	ADOB	date				
5	APhoneNumber	APHONENUMB	char(10)	10			
6	AEmail	AEMAIL	varchar(20)	20			

# Test Case	Input Data													Expected Result	Acutal Result	Status (Pass/Fail)
	Departure	Destination	Date	Seats	Boarding Point	Drop Point	Name	Email	Phone Number	Method	Card's Name	Card No	Card Exp Date			
1	Can Tho	Sai Gon	30/04/2024	C3,D3	Ben xe Can Tho	Ben xe Mien Tay	Nguyen Van A	abc@gmail.com	035920xxxx	Credit card	Nguyen Van A	9704 xxxx xxxx	04/2026	Payment Successful		
2	Can Tho	Sai Gon	30/04/2024	C3,D3	Ben xe Can Tho	Ben xe Mien Tay	Nguyen Van A	abc@gmail.com	035920xxxx	E-Wallet				Payment Successful		
3	Can Tho	null												Please enter again		

	Name	Code	Data Type	Length	Precision	P	F	M
1	AccountID	ACCOUNTID	char(7)	7		✓		✓
2	AName	ANAME	varchar(20)	20				✓
3	AGender	AGENDER	varchar(3)	3				✓
4	ADoB	ADOB	date					✓
5	APhoneNumber	APHONENUMB	char(10)	10				✓
6	AEmail	AEMAIL	varchar(20)	20				✓

4	Can Tho	Sai Gon	30/04/2024											Not Found Any Routes		
5	Can Tho	Sai Gon	30/04/2024	C3,D3	Ben xe Can Tho	Ben xe Mien Tay	Nguyen Van A	abc@gmail.com	035920xxxx	E-Wallet				Please enter passenger details again		
6	Can Tho	Sai Gon	30/04/2024	C3,D3	Ben xe Can Tho	Ben xe Mien Tay	Nguyen Van A	abc@gmail.com	035920xxxx	Credit card	Nguyen Van A	null	null	Please enter card details again		

	Name	Code	Data Type	Length	Precision	P	F	M
1	AccountID	ACCOUNTID	char(7)	7		✓		✓
2	AName	ANAME	varchar(20)	20				✓
3	AGender	AGENDER	varchar(3)	3				✓
4	ADoB	ADOB	date					✓
5	APhoneNumber	APHONENUMB	char(10)	10				✓
6	AEmail	AEMAIL	varchar(20)	20				✓